

EXAMPLE 8 A Rational Function with Degree of Numerator Greater than Degree of Denominator

Find the asymptotes of the graph of

$$f(x) = \frac{x^2 - 3}{2x - 4}.$$

Solution We are interested in the behavior as $x \rightarrow \pm\infty$ and also as $x \rightarrow 2$, where the denominator is zero. We divide $(2x - 4)$ into $(x^2 - 3)$:

$$\begin{array}{r} \frac{x}{2} + 1 \\ 2x - 4 \overline{) x^2 - 3} \\ \underline{x^2 - 2x} \\ 2x - 3 \\ \underline{2x - 4} \\ 1 \end{array}$$

This tells us that

$$f(x) = \frac{x^2 - 3}{2x - 4} = \underbrace{\frac{x}{2} + 1}_{\text{linear}} + \underbrace{\frac{1}{2x - 4}}_{\text{remainder}}.$$

Since $\lim_{x \rightarrow 2^+} f(x) = \infty$ and $\lim_{x \rightarrow 2^-} f(x) = -\infty$, the line $x = 2$ is a two-sided vertical asymptote. As $x \rightarrow \pm\infty$, the remainder approaches 0 and $f(x) \rightarrow (x/2) + 1$. The line $y = (x/2) + 1$ is an oblique asymptote both to the right and to the left (Figure 2.47). ■

Notice in Example 8, that if the degree of the numerator in a rational function is greater than the degree of the denominator, then the limit is $+\infty$ or $-\infty$, depending on the signs assumed by the numerator and denominator as $|x|$ becomes large.

Dominant Terms

Of all the observations we can make quickly about the function

$$f(x) = \frac{x^2 - 3}{2x - 4}$$

in Example 8, probably the most useful is that

$$f(x) = \frac{x}{2} + 1 + \frac{1}{2x - 4}.$$

This tells us immediately that

$$f(x) \approx \frac{x}{2} + 1 \quad \text{For } x \text{ numerically large}$$

$$f(x) \approx \frac{1}{2x - 4} \quad \text{For } x \text{ near } 2$$

If we want to know how f behaves, this is the way to find out. It behaves like $y = (x/2) + 1$ when x is numerically large and the contribution of $1/(2x - 4)$ to the total

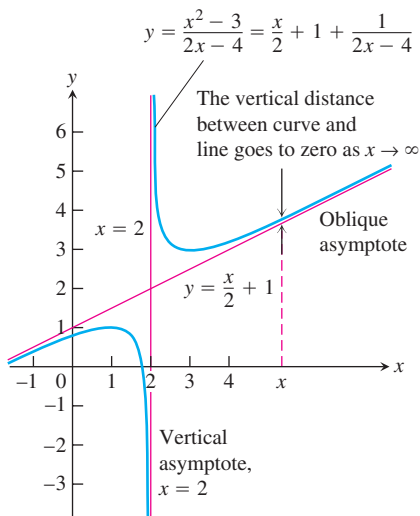


FIGURE 2.47 The graph of $f(x) = (x^2 - 3)/(2x - 4)$ has a vertical asymptote and an oblique asymptote (Example 8).

value of f is insignificant. It behaves like $1/(2x - 4)$ when x is so close to 2 that $1/(2x - 4)$ makes the dominant contribution.

We say that $(x/2) + 1$ **dominates** when x is numerically large, and we say that $1/(2x - 4)$ dominates when x is near 2. **Dominant terms** like these are the key to predicting a function's behavior. Here's another example.

EXAMPLE 9 Two Graphs Appearing Identical on a Large Scale

Let $f(x) = 3x^4 - 2x^3 + 3x^2 - 5x + 6$ and $g(x) = 3x^4$. Show that although f and g are quite different for numerically small values of x , they are virtually identical for $|x|$ very large.

Solution The graphs of f and g behave quite differently near the origin (Figure 2.48a), but appear as virtually identical on a larger scale (Figure 2.48b).

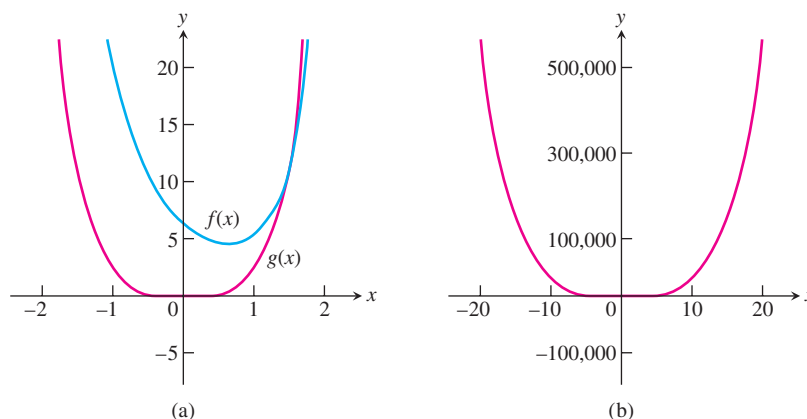


FIGURE 2.48 The graphs of f and g , (a) are distinct for $|x|$ small, and (b) nearly identical for $|x|$ large (Example 9).

We can test that the term $3x^4$ in f , represented graphically by g , dominates the polynomial f for numerically large values of x by examining the ratio of the two functions as $x \rightarrow \pm\infty$. We find that

$$\begin{aligned} \lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow \pm\infty} \frac{3x^4 - 2x^3 + 3x^2 - 5x + 6}{3x^4} \\ &= \lim_{x \rightarrow \pm\infty} \left(1 - \frac{2}{3x} + \frac{1}{x^2} - \frac{5}{3x^3} + \frac{2}{x^4} \right) \\ &= 1, \end{aligned}$$

so that f and g are nearly identical for $|x|$ large. ■

EXERCISES 2.5

Infinite Limits

Find the limits in Exercises 1–12.

1. $\lim_{x \rightarrow 0^+} \frac{1}{3x}$
2. $\lim_{x \rightarrow 0^-} \frac{5}{2x}$
3. $\lim_{x \rightarrow 2^-} \frac{3}{x-2}$
4. $\lim_{x \rightarrow 3^+} \frac{1}{x-3}$
5. $\lim_{x \rightarrow -8^+} \frac{2x}{x+8}$
6. $\lim_{x \rightarrow -5^-} \frac{3x}{2x+10}$
7. $\lim_{x \rightarrow 7} \frac{4}{(x-7)^2}$
8. $\lim_{x \rightarrow 0} \frac{-1}{x^2(x+1)}$
9. a. $\lim_{x \rightarrow 0^+} \frac{2}{3x^{1/3}}$
- b. $\lim_{x \rightarrow 0^-} \frac{2}{3x^{1/3}}$
10. a. $\lim_{x \rightarrow 0^+} \frac{2}{x^{1/5}}$
- b. $\lim_{x \rightarrow 0^-} \frac{2}{x^{1/5}}$
11. $\lim_{x \rightarrow 0} \frac{4}{x^{2/5}}$
12. $\lim_{x \rightarrow 0} \frac{1}{x^{2/3}}$

Find the limits in Exercises 13–16.

13. $\lim_{x \rightarrow (\pi/2)^-} \tan x$
14. $\lim_{x \rightarrow (-\pi/2)^+} \sec x$
15. $\lim_{\theta \rightarrow 0^-} (1 + \csc \theta)$
16. $\lim_{\theta \rightarrow 0} (2 - \cot \theta)$

Additional Calculations

Find the limits in Exercises 17–22.

17. $\lim_{x \rightarrow 4} \frac{1}{x^2 - 4}$ as
 - a. $x \rightarrow 2^+$
 - b. $x \rightarrow 2^-$
 - c. $x \rightarrow -2^+$
 - d. $x \rightarrow -2^-$
18. $\lim_{x \rightarrow 1} \frac{x}{x^2 - 1}$ as
 - a. $x \rightarrow 1^+$
 - b. $x \rightarrow 1^-$
 - c. $x \rightarrow -1^+$
 - d. $x \rightarrow -1^-$
19. $\lim_{x \rightarrow 0} \left(\frac{x^2}{2} - \frac{1}{x} \right)$ as
 - a. $x \rightarrow 0^+$
 - b. $x \rightarrow 0^-$
 - c. $x \rightarrow \sqrt[3]{2}$
 - d. $x \rightarrow -1$
20. $\lim_{x \rightarrow 4} \frac{x^2 - 1}{2x + 4}$ as
 - a. $x \rightarrow -2^+$
 - b. $x \rightarrow -2^-$
 - c. $x \rightarrow 1^+$
 - d. $x \rightarrow 0^-$

21. $\lim_{x \rightarrow 0} \frac{x^2 - 3x + 2}{x^3 - 2x^2}$ as
 - a. $x \rightarrow 0^+$
 - b. $x \rightarrow 2^+$
 - c. $x \rightarrow 2^-$
 - d. $x \rightarrow 2$
 - e. What, if anything, can be said about the limit as $x \rightarrow 0$?
22. $\lim_{x \rightarrow 0} \frac{x^2 - 3x + 2}{x^3 - 4x}$ as
 - a. $x \rightarrow 2^+$
 - b. $x \rightarrow -2^+$
 - c. $x \rightarrow 0^-$
 - d. $x \rightarrow 1^+$
 - e. What, if anything, can be said about the limit as $x \rightarrow 0$?

Find the limits in Exercises 23–26.

23. $\lim_{t \rightarrow 0} \left(2 - \frac{3}{t^{1/3}} \right)$ as
 - a. $t \rightarrow 0^+$
 - b. $t \rightarrow 0^-$
24. $\lim_{t \rightarrow 0} \left(\frac{1}{t^{3/5}} + 7 \right)$ as
 - a. $t \rightarrow 0^+$
 - b. $t \rightarrow 0^-$
25. $\lim_{x \rightarrow 1} \left(\frac{1}{x^{2/3}} + \frac{2}{(x-1)^{2/3}} \right)$ as
 - a. $x \rightarrow 0^+$
 - b. $x \rightarrow 0^-$
 - c. $x \rightarrow 1^+$
 - d. $x \rightarrow 1^-$
26. $\lim_{x \rightarrow 1} \left(\frac{1}{x^{1/3}} - \frac{1}{(x-1)^{4/3}} \right)$ as
 - a. $x \rightarrow 0^+$
 - b. $x \rightarrow 0^-$
 - c. $x \rightarrow 1^+$
 - d. $x \rightarrow 1^-$

Graphing Rational Functions

Graph the rational functions in Exercises 27–38. Include the graphs and equations of the asymptotes and dominant terms.

27. $y = \frac{1}{x-1}$
28. $y = \frac{1}{x+1}$
29. $y = \frac{1}{2x+4}$
30. $y = \frac{-3}{x-3}$
31. $y = \frac{x+3}{x+2}$
32. $y = \frac{2x}{x+1}$
33. $y = \frac{x^2}{x-1}$
34. $y = \frac{x^2+1}{x-1}$
35. $y = \frac{x^2-4}{x-1}$
36. $y = \frac{x^2-1}{2x+4}$
37. $y = \frac{x^2-1}{x}$
38. $y = \frac{x^3+1}{x^2}$

Inventing Graphs from Values and Limits

In Exercises 39–42, sketch the graph of a function $y = f(x)$ that satisfies the given conditions. No formulas are required—just label the coordinate axes and sketch an appropriate graph. (The answers are not unique, so your graphs may not be exactly like those in the answer section.)

39. $f(0) = 0$, $f(1) = 2$, $f(-1) = -2$, $\lim_{x \rightarrow -\infty} f(x) = -1$, and

$$\lim_{x \rightarrow \infty} f(x) = 1$$

40. $f(0) = 0$, $\lim_{x \rightarrow \pm\infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = 2$, and

$$\lim_{x \rightarrow 0^-} f(x) = -2$$

41. $f(0) = 0$, $\lim_{x \rightarrow \pm\infty} f(x) = 0$, $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) = \infty$,
 $\lim_{x \rightarrow 1^+} f(x) = -\infty$, and $\lim_{x \rightarrow -1^-} f(x) = -\infty$

42. $f(2) = 1$, $f(-1) = 0$, $\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$,
 $\lim_{x \rightarrow 0^-} f(x) = -\infty$, and $\lim_{x \rightarrow -\infty} f(x) = 1$

Inventing Functions

In Exercises 43–46, find a function that satisfies the given conditions and sketch its graph. (The answers here are not unique. Any function that satisfies the conditions is acceptable. Feel free to use formulas defined in pieces if that will help.)

43. $\lim_{x \rightarrow \pm\infty} f(x) = 0$, $\lim_{x \rightarrow 2^-} f(x) = \infty$, and $\lim_{x \rightarrow 2^+} f(x) = \infty$

44. $\lim_{x \rightarrow \pm\infty} g(x) = 0$, $\lim_{x \rightarrow 3^-} g(x) = -\infty$, and $\lim_{x \rightarrow 3^+} g(x) = \infty$

45. $\lim_{x \rightarrow -\infty} h(x) = -1$, $\lim_{x \rightarrow \infty} h(x) = 1$, $\lim_{x \rightarrow 0^-} h(x) = -1$, and
 $\lim_{x \rightarrow 0^+} h(x) = 1$

46. $\lim_{x \rightarrow \pm\infty} k(x) = 1$, $\lim_{x \rightarrow 1^-} k(x) = \infty$, and $\lim_{x \rightarrow 1^+} k(x) = -\infty$

The Formal Definition of Infinite Limit

Use formal definitions to prove the limit statements in Exercises 47–50.

47. $\lim_{x \rightarrow 0} \frac{-1}{x^2} = -\infty$

48. $\lim_{x \rightarrow 0} \frac{1}{|x|} = \infty$

49. $\lim_{x \rightarrow 3} \frac{-2}{(x-3)^2} = -\infty$

50. $\lim_{x \rightarrow -5} \frac{1}{(x+5)^2} = \infty$

Formal Definitions of Infinite One-Sided Limits

51. Here is the definition of **infinite right-hand limit**.

We say that $f(x)$ approaches infinity as x approaches x_0 from the right, and write

$$\lim_{x \rightarrow x_0^+} f(x) = \infty,$$

if, for every positive real number B , there exists a corresponding number $\delta > 0$ such that for all x

$$x_0 < x < x_0 + \delta \quad \Rightarrow \quad f(x) > B.$$

Modify the definition to cover the following cases.

a. $\lim_{x \rightarrow x_0^-} f(x) = \infty$

b. $\lim_{x \rightarrow x_0^+} f(x) = -\infty$

c. $\lim_{x \rightarrow x_0^-} f(x) = -\infty$

Use the formal definitions from Exercise 51 to prove the limit statements in Exercises 52–56.

52. $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$

53. $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$

54. $\lim_{x \rightarrow 2^-} \frac{1}{x-2} = -\infty$

55. $\lim_{x \rightarrow 2^+} \frac{1}{x-2} = \infty$

56. $\lim_{x \rightarrow 1^-} \frac{1}{1-x^2} = \infty$

Graphing Terms

Each of the functions in Exercises 57–60 is given as the sum or difference of two terms. First graph the terms (with the same set of axes). Then, using these graphs as guides, sketch in the graph of the function.

57. $y = \sec x + \frac{1}{x}$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

58. $y = \sec x - \frac{1}{x^2}$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

59. $y = \tan x + \frac{1}{x^2}$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

60. $y = \frac{1}{x} - \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

Grapher Explorations—Comparing Graphs with Formulas

Graph the curves in Exercises 61–64. Explain the relation between the curve's formula and what you see.

61. $y = \frac{x}{\sqrt{4-x^2}}$

62. $y = \frac{-1}{\sqrt{4-x^2}}$

63. $y = x^{2/3} + \frac{1}{x^{1/3}}$

64. $y = \sin\left(\frac{\pi}{x^2 + 1}\right)$

2.6 Continuity

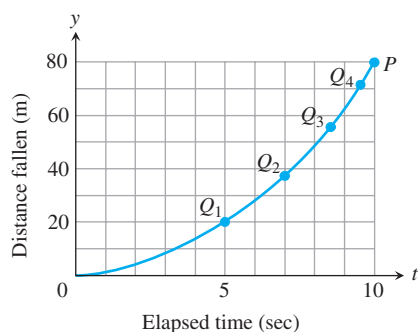


FIGURE 2.49 Connecting plotted points by an unbroken curve from experimental data $Q_1, Q_2, Q_3, \dots$ for a falling object.

When we plot function values generated in a laboratory or collected in the field, we often connect the plotted points with an unbroken curve to show what the function's values are likely to have been at the times we did not measure (Figure 2.49). In doing so, we are assuming that we are working with a *continuous function*, so its outputs vary continuously with the inputs and do not jump from one value to another without taking on the values in between. The limit of a continuous function as x approaches c can be found simply by calculating the value of the function at c . (We found this to be true for polynomials in Section 2.2.)

Any function $y = f(x)$ whose graph can be sketched over its domain in one continuous motion without lifting the pencil is an example of a continuous function. In this section we investigate more precisely what it means for a function to be continuous. We also study the properties of continuous functions, and see that many of the function types presented in Section 1.4 are continuous.

Continuity at a Point

To understand continuity, we need to consider a function like the one in Figure 2.50 whose limits we investigated in Example 2, Section 2.4.

EXAMPLE 1 Investigating Continuity

Find the points at which the function f in Figure 2.50 is continuous and the points at which f is discontinuous.

Solution The function f is continuous at every point in its domain $[0, 4]$ except at $x = 1$, $x = 2$, and $x = 4$. At these points, there are breaks in the graph. Note the relationship between the limit of f and the value of f at each point of the function's domain.

Points at which f is continuous:

$$\text{At } x = 0, \quad \lim_{x \rightarrow 0^+} f(x) = f(0).$$

$$\text{At } x = 3, \quad \lim_{x \rightarrow 3} f(x) = f(3).$$

$$\text{At } 0 < c < 4, c \neq 1, 2, \quad \lim_{x \rightarrow c} f(x) = f(c).$$

Points at which f is discontinuous:

$$\text{At } x = 1, \quad \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

$$\text{At } x = 2, \quad \lim_{x \rightarrow 2} f(x) = 1, \text{ but } 1 \neq f(2).$$

$$\text{At } x = 4, \quad \lim_{x \rightarrow 4^-} f(x) = 1, \text{ but } 1 \neq f(4).$$

$$\text{At } c < 0, c > 4, \quad \text{these points are not in the domain of } f.$$

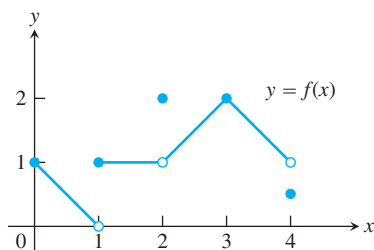


FIGURE 2.50 The function is continuous on $[0, 4]$ except at $x = 1$, $x = 2$, and $x = 4$ (Example 1).

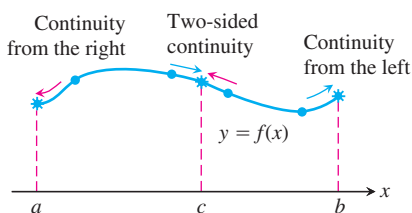


FIGURE 2.51 Continuity at points a , b , and c .

To define continuity at a point in a function's domain, we need to define continuity at an interior point (which involves a two-sided limit) and continuity at an endpoint (which involves a one-sided limit) (Figure 2.51).

DEFINITION Continuous at a Point

Interior point: A function $y = f(x)$ is **continuous at an interior point** c of its domain if

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Endpoint: A function $y = f(x)$ is **continuous at a left endpoint** a or is **continuous at a right endpoint** b of its domain if

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{or} \quad \lim_{x \rightarrow b^-} f(x) = f(b), \quad \text{respectively.}$$

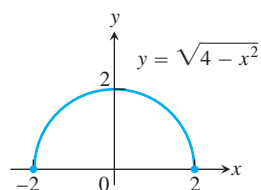


FIGURE 2.52 A function that is continuous at every domain point (Example 2).

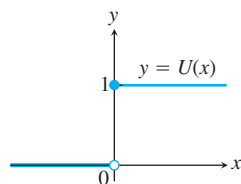


FIGURE 2.53 A function that is right-continuous, but not left-continuous, at the origin. It has a jump discontinuity there (Example 3).

If a function f is not continuous at a point c , we say that f is **discontinuous** at c and c is a **point of discontinuity** of f . Note that c need not be in the domain of f .

A function f is **right-continuous (continuous from the right)** at a point $x = c$ in its domain if $\lim_{x \rightarrow c^+} f(x) = f(c)$. It is **left-continuous (continuous from the left)** at c if $\lim_{x \rightarrow c^-} f(x) = f(c)$. Thus, a function is continuous at a left endpoint a of its domain if it is right-continuous at a and continuous at a right endpoint b of its domain if it is left-continuous at b . A function is continuous at an interior point c of its domain if and only if it is both right-continuous and left-continuous at c (Figure 2.51).

EXAMPLE 2 A Function Continuous Throughout Its Domain

The function $f(x) = \sqrt{4 - x^2}$ is continuous at every point of its domain, $[-2, 2]$ (Figure 2.52), including $x = -2$, where f is right-continuous, and $x = 2$, where f is left-continuous. ■

EXAMPLE 3 The Unit Step Function Has a Jump Discontinuity

The unit step function $U(x)$, graphed in Figure 2.53, is right-continuous at $x = 0$, but is neither left-continuous nor continuous there. It has a jump discontinuity at $x = 0$. ■

We summarize continuity at a point in the form of a test.

Continuity Test

A function $f(x)$ is continuous at $x = c$ if and only if it meets the following three conditions.

1. $f(c)$ exists (c lies in the domain of f)
2. $\lim_{x \rightarrow c} f(x)$ exists (f has a limit as $x \rightarrow c$)
3. $\lim_{x \rightarrow c} f(x) = f(c)$ (the limit equals the function value)

For one-sided continuity and continuity at an endpoint, the limits in parts 2 and 3 of the test should be replaced by the appropriate one-sided limits.

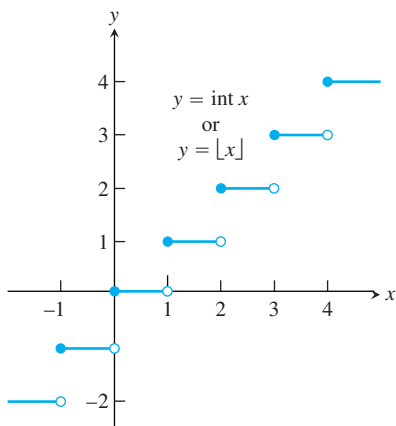


FIGURE 2.54 The greatest integer function is continuous at every noninteger point. It is right-continuous, but not left-continuous, at every integer point (Example 4).

EXAMPLE 4 The Greatest Integer Function

The function $y = \lfloor x \rfloor$ or $y = \text{int } x$, introduced in Chapter 1, is graphed in Figure 2.54. It is discontinuous at every integer because the limit does not exist at any integer n :

$$\lim_{x \rightarrow n^-} \text{int } x = n - 1 \quad \text{and} \quad \lim_{x \rightarrow n^+} \text{int } x = n$$

so the left-hand and right-hand limits are not equal as $x \rightarrow n$. Since $\text{int } n = n$, the greatest integer function is right-continuous at every integer n (but not left-continuous).

The greatest integer function is continuous at every real number other than the integers. For example,

$$\lim_{x \rightarrow 1.5} \text{int } x = 1 = \text{int } 1.5.$$

In general, if $n - 1 < c < n$, n an integer, then

$$\lim_{x \rightarrow c} \text{int } x = n - 1 = \text{int } c.$$

Figure 2.55 is a catalog of discontinuity types. The function in Figure 2.55a is continuous at $x = 0$. The function in Figure 2.55b would be continuous if it had $f(0) = 1$. The function in Figure 2.55c would be continuous if $f(0)$ were 1 instead of 2. The discontinuities in Figure 2.55b and c are **removable**. Each function has a limit as $x \rightarrow 0$, and we can remove the discontinuity by setting $f(0)$ equal to this limit.

The discontinuities in Figure 2.55d through f are more serious: $\lim_{x \rightarrow 0} f(x)$ does not exist, and there is no way to improve the situation by changing f at 0. The step function in Figure 2.55d has a **jump discontinuity**: The one-sided limits exist but have different values. The function $f(x) = 1/x^2$ in Figure 2.55e has an **infinite discontinuity**. The function in Figure 2.55f has an **oscillating discontinuity**: It oscillates too much to have a limit as $x \rightarrow 0$.

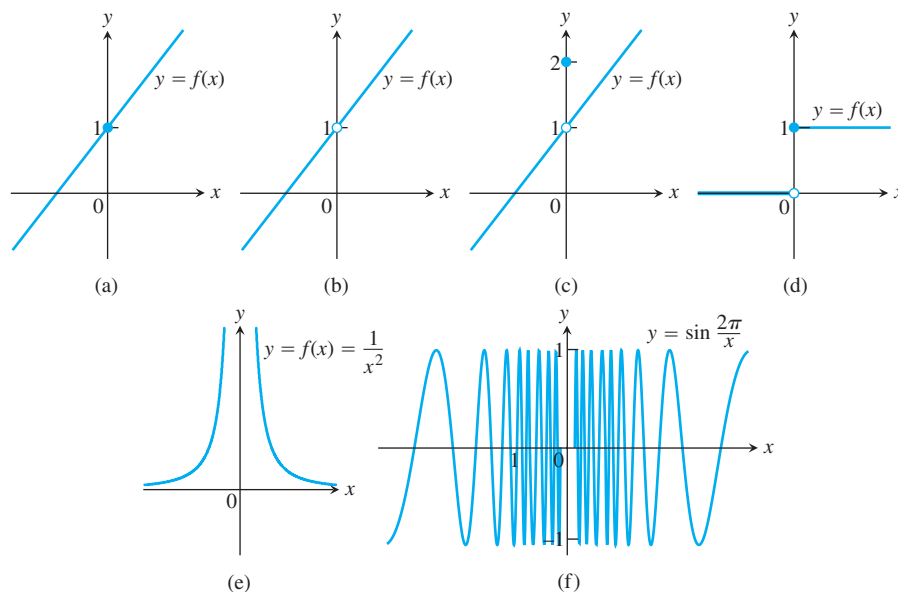


FIGURE 2.55 The function in (a) is continuous at $x = 0$; the functions in (b) through (f) are not.

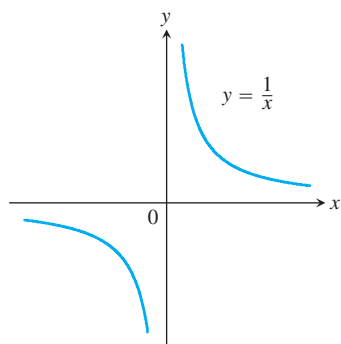


FIGURE 2.56 The function $y = 1/x$ is continuous at every value of x except $x = 0$. It has a point of discontinuity at $x = 0$ (Example 5).

Continuous Functions

A function is **continuous on an interval** if and only if it is continuous at every point of the interval. For example, the semicircle function graphed in Figure 2.52 is continuous on the interval $[-2, 2]$, which is its domain. A **continuous function** is one that is continuous at every point of its domain. A continuous function need not be continuous on every interval. For example, $y = 1/x$ is not continuous on $[-1, 1]$ (Figure 2.56), but it is continuous over its domain $(-\infty, 0) \cup (0, \infty)$.

EXAMPLE 5 Identifying Continuous Functions

- (a) The function $y = 1/x$ (Figure 2.56) is a continuous function because it is continuous at every point of its domain. It has a point of discontinuity at $x = 0$, however, because it is not defined there.
- (b) The identity function $f(x) = x$ and constant functions are continuous everywhere by Example 3, Section 2.3. ■

Algebraic combinations of continuous functions are continuous wherever they are defined.

THEOREM 9 Properties of Continuous Functions

If the functions f and g are continuous at $x = c$, then the following combinations are continuous at $x = c$.

- | | |
|------------------------|--|
| 1. Sums: | $f + g$ |
| 2. Differences: | $f - g$ |
| 3. Products: | $f \cdot g$ |
| 4. Constant multiples: | $k \cdot f$, for any number k |
| 5. Quotients: | f/g provided $g(c) \neq 0$ |
| 6. Powers: | $f^{r/s}$, provided it is defined on an open interval containing c , where r and s are integers |

Most of the results in Theorem 9 are easily proved from the limit rules in Theorem 1, Section 2.2. For instance, to prove the sum property we have

$$\begin{aligned}
 \lim_{x \rightarrow c} (f + g)(x) &= \lim_{x \rightarrow c} (f(x) + g(x)) \\
 &= \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x), && \text{Sum Rule, Theorem 1} \\
 &= f(c) + g(c) && \text{Continuity of } f, g \text{ at } c \\
 &= (f + g)(c).
 \end{aligned}$$

This shows that $f + g$ is continuous.

EXAMPLE 6 Polynomial and Rational Functions Are Continuous

- (a) Every polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ is continuous because $\lim_{x \rightarrow c} P(x) = P(c)$ by Theorem 2, Section 2.2.

- (b) If $P(x)$ and $Q(x)$ are polynomials, then the rational function $P(x)/Q(x)$ is continuous wherever it is defined ($Q(c) \neq 0$) by the Quotient Rule in Theorem 9.

EXAMPLE 7 Continuity of the Absolute Value Function

The function $f(x) = |x|$ is continuous at every value of x . If $x > 0$, we have $f(x) = x$, a polynomial. If $x < 0$, we have $f(x) = -x$, another polynomial. Finally, at the origin, $\lim_{x \rightarrow 0} |x| = 0 = |0|$. ■

The functions $y = \sin x$ and $y = \cos x$ are continuous at $x = 0$ by Example 6 of Section 2.2. Both functions are, in fact, continuous everywhere (see Exercise 62). It follows from Theorem 9 that all six trigonometric functions are then continuous wherever they are defined. For example, $y = \tan x$ is continuous on $\cdots \cup (-\pi/2, \pi/2) \cup (\pi/2, 3\pi/2) \cup \cdots$.

Composites

All composites of continuous functions are continuous. The idea is that if $f(x)$ is continuous at $x = c$ and $g(x)$ is continuous at $x = f(c)$, then $g \circ f$ is continuous at $x = c$ (Figure 2.57). In this case, the limit as $x \rightarrow c$ is $g(f(c))$.

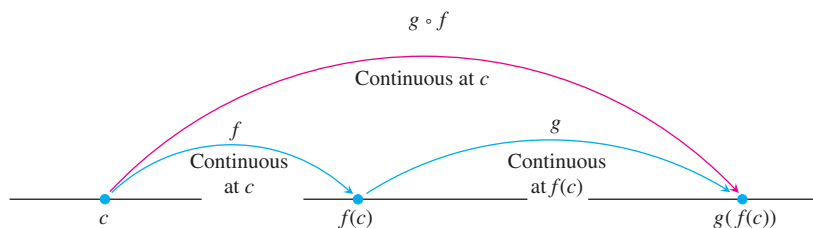


FIGURE 2.57 Composites of continuous functions are continuous.

THEOREM 10 Composite of Continuous Functions

If f is continuous at c and g is continuous at $f(c)$, then the composite $g \circ f$ is continuous at c .

Intuitively, Theorem 10 is reasonable because if x is close to c , then $f(x)$ is close to $f(c)$, and since g is continuous at $f(c)$, it follows that $g(f(x))$ is close to $g(f(c))$.

The continuity of composites holds for any finite number of functions. The only requirement is that each function be continuous where it is applied. For an outline of the proof of Theorem 10, see Exercise 6 in Appendix 2.

EXAMPLE 8 Applying Theorems 9 and 10

Show that the following functions are continuous everywhere on their respective domains.

(a) $y = \sqrt{x^2 - 2x - 5}$

(b) $y = \frac{x^{2/3}}{1 + x^4}$

(c) $y = \left| \frac{x - 2}{x^2 - 2} \right|$

(d) $y = \left| \frac{x \sin x}{x^2 + 2} \right|$

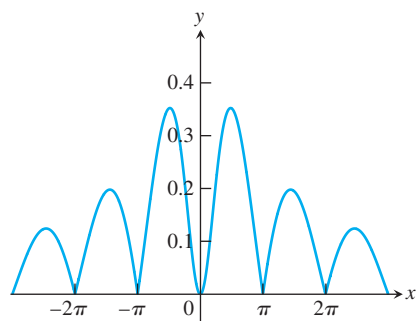


FIGURE 2.58 The graph suggests that $y = |(x \sin x)/(x^2 + 2)|$ is continuous (Example 8d).

Solution

- (a) The square root function is continuous on $[0, \infty)$ because it is a rational power of the continuous identity function $f(x) = x$ (Part 6, Theorem 9). The given function is then the composite of the polynomial $f(x) = x^2 - 2x - 5$ with the square root function $g(t) = \sqrt{t}$.
- (b) The numerator is a rational power of the identity function; the denominator is an everywhere-positive polynomial. Therefore, the quotient is continuous.
- (c) The quotient $(x - 2)/(x^2 - 2)$ is continuous for all $x \neq \pm\sqrt{2}$, and the function is the composition of this quotient with the continuous absolute value function (Example 7).
- (d) Because the sine function is everywhere-continuous (Exercise 62), the numerator term $x \sin x$ is the product of continuous functions, and the denominator term $x^2 + 2$ is an everywhere-positive polynomial. The given function is the composite of a quotient of continuous functions with the continuous absolute value function (Figure 2.58).

Continuous Extension to a Point

The function $y = (\sin x)/x$ is continuous at every point except $x = 0$. In this it is like the function $y = 1/x$. But $y = (\sin x)/x$ is different from $y = 1/x$ in that it has a finite limit as $x \rightarrow 0$ (Theorem 7). It is therefore possible to extend the function's domain to include the point $x = 0$ in such a way that the extended function is continuous at $x = 0$. We define

$$F(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0. \end{cases}$$

The function $F(x)$ is continuous at $x = 0$ because

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = F(0)$$

(Figure 2.59).

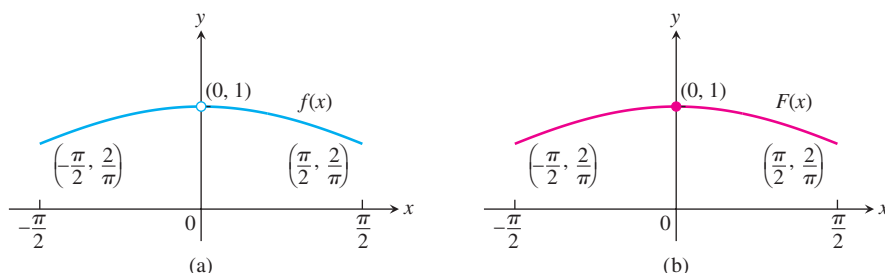


FIGURE 2.59 The graph (a) of $f(x) = (\sin x)/x$ for $-\pi/2 \leq x \leq \pi/2$ does not include the point $(0, 1)$ because the function is not defined at $x = 0$. (b) We can remove the discontinuity from the graph by defining the new function $F(x)$ with $F(0) = 1$ and $F(x) = f(x)$ everywhere else. Note that $F(0) = \lim_{x \rightarrow 0} f(x)$.

More generally, a function (such as a rational function) may have a limit even at a point where it is not defined. If $f(c)$ is not defined, but $\lim_{x \rightarrow c} f(x) = L$ exists, we can define a new function $F(x)$ by the rule

$$F(x) = \begin{cases} f(x), & \text{if } x \text{ is in the domain of } f \\ L, & \text{if } x = c. \end{cases}$$

The function F is continuous at $x = c$. It is called the **continuous extension** of f to $x = c$. For rational functions f , continuous extensions are usually found by canceling common factors.

EXAMPLE 9 A Continuous Extension

Show that

$$f(x) = \frac{x^2 + x - 6}{x^2 - 4}$$

has a continuous extension to $x = 2$, and find that extension.

Solution Although $f(2)$ is not defined, if $x \neq 2$ we have

$$f(x) = \frac{x^2 + x - 6}{x^2 - 4} = \frac{(x - 2)(x + 3)}{(x - 2)(x + 2)} = \frac{x + 3}{x + 2}.$$

The new function

$$F(x) = \frac{x + 3}{x + 2}$$

is equal to $f(x)$ for $x \neq 2$, but is continuous at $x = 2$, having there the value of $5/4$. Thus F is the continuous extension of f to $x = 2$, and

$$\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4} = \lim_{x \rightarrow 2} f(x) = \frac{5}{4}.$$

The graph of f is shown in Figure 2.60. The continuous extension F has the same graph except with no hole at $(2, 5/4)$. Effectively, F is the function f with its point of discontinuity at $x = 2$ removed. ■

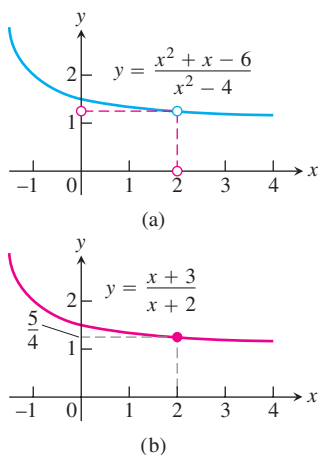


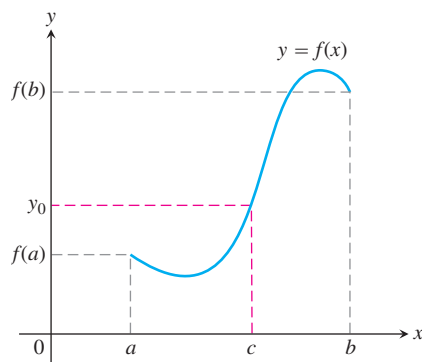
FIGURE 2.60 (a) The graph of $f(x)$ and (b) the graph of its continuous extension $F(x)$ (Example 9).

Intermediate Value Theorem for Continuous Functions

Functions that are continuous on intervals have properties that make them particularly useful in mathematics and its applications. One of these is the *Intermediate Value Property*. A function is said to have the **Intermediate Value Property** if whenever it takes on two values, it also takes on all the values in between.

THEOREM 11 The Intermediate Value Theorem for Continuous Functions

A function $y = f(x)$ that is continuous on a closed interval $[a, b]$ takes on every value between $f(a)$ and $f(b)$. In other words, if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.



Geometrically, the Intermediate Value Theorem says that any horizontal line $y = y_0$ crossing the y -axis between the numbers $f(a)$ and $f(b)$ will cross the curve $y = f(x)$ at least once over the interval $[a, b]$.

The proof of the Intermediate Value Theorem depends on the completeness property of the real number system and can be found in more advanced texts.

The continuity of f on the interval is essential to Theorem 11. If f is discontinuous at even one point of the interval, the theorem's conclusion may fail, as it does for the function graphed in Figure 2.61.

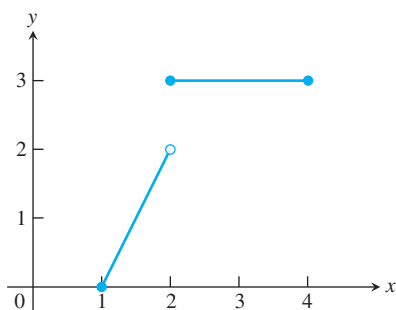


FIGURE 2.61 The function

$$f(x) = \begin{cases} 2x - 2, & 1 \leq x < 2 \\ 3, & 2 \leq x \leq 4 \end{cases}$$

does not take on all values between $f(1) = 0$ and $f(4) = 3$; it misses all the values between 2 and 3.

A Consequence for Graphing: Connectivity Theorem 11 is the reason the graph of a function continuous on an interval cannot have any breaks over the interval. It will be **connected**, a single, unbroken curve, like the graph of $\sin x$. It will not have jumps like the graph of the greatest integer function (Figure 2.54) or separate branches like the graph of $1/x$ (Figure 2.56).

A Consequence for Root Finding We call a solution of the equation $f(x) = 0$ a **root** of the equation or **zero** of the function f . The Intermediate Value Theorem tells us that if f is continuous, then any interval on which f changes sign contains a zero of the function.

In practical terms, when we see the graph of a continuous function cross the horizontal axis on a computer screen, we know it is not stepping across. There really is a point where the function's value is zero. This consequence leads to a procedure for estimating the zeros of any continuous function we can graph:

1. Graph the function over a large interval to see roughly where the zeros are.
2. Zoom in on each zero to estimate its x -coordinate value.

You can practice this procedure on your graphing calculator or computer in some of the exercises. Figure 2.62 shows a typical sequence of steps in a graphical solution of the equation $x^3 - x - 1 = 0$.

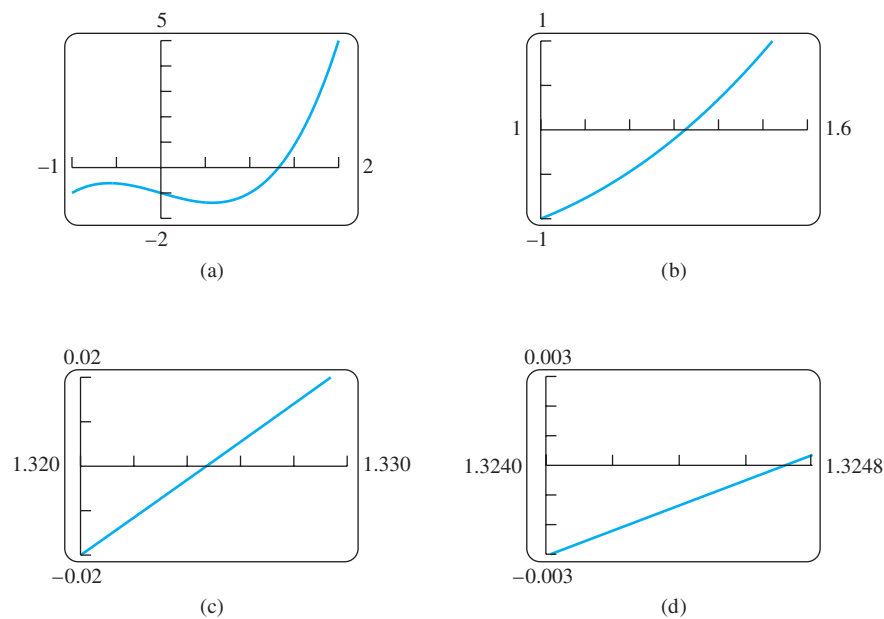


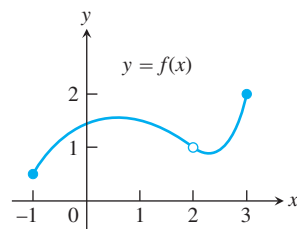
FIGURE 2.62 Zooming in on a zero of the function $f(x) = x^3 - x - 1$. The zero is near $x = 1.3247$.

EXERCISES 2.6

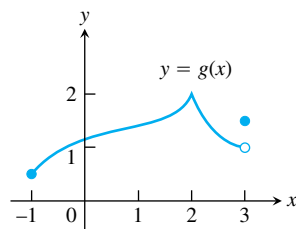
Continuity from Graphs

In Exercises 1–4, say whether the function graphed is continuous on $[-1, 3]$. If not, where does it fail to be continuous and why?

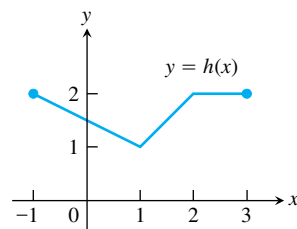
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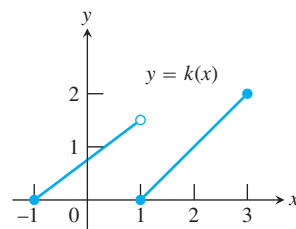
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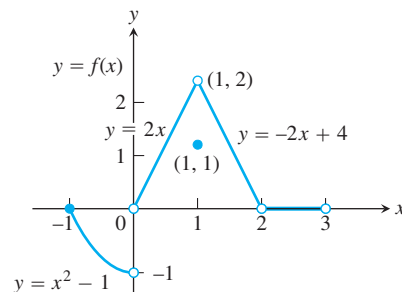
4.



Exercises 5–10 are about the function

$$f(x) = \begin{cases} x^2 - 1, & -1 \leq x < 0 \\ 2x, & 0 < x < 1 \\ 1, & x = 1 \\ -2x + 4, & 1 < x < 2 \\ 0, & 2 < x < 3 \end{cases}$$

graphed in the accompanying figure.



The graph for Exercises 5–10.

5. a. Does $f(-1)$ exist?
b. Does $\lim_{x \rightarrow -1^+} f(x)$ exist?
c. Does $\lim_{x \rightarrow -1^+} f(x) = f(-1)$?
d. Is f continuous at $x = -1$?
6. a. Does $f(1)$ exist?
b. Does $\lim_{x \rightarrow 1} f(x)$ exist?
c. Does $\lim_{x \rightarrow 1} f(x) = f(1)$?
d. Is f continuous at $x = 1$?
7. a. Is f defined at $x = 2$? (Look at the definition of f .)
b. Is f continuous at $x = 2$?
8. At what values of x is f continuous?
9. What value should be assigned to $f(2)$ to make the extended function continuous at $x = 2$?
10. To what new value should $f(1)$ be changed to remove the discontinuity?

Applying the Continuity Test

At which points do the functions in Exercises 11 and 12 fail to be continuous? At which points, if any, are the discontinuities removable? Not removable? Give reasons for your answers.

11. Exercise 1, Section 2.4 12. Exercise 2, Section 2.4

At what points are the functions in Exercises 13–28 continuous?

13. $y = \frac{1}{x-2} - 3x$
14. $y = \frac{1}{(x+2)^2} + 4$
15. $y = \frac{x+1}{x^2-4x+3}$
16. $y = \frac{x+3}{x^2-3x-10}$
17. $y = |x-1| + \sin x$
18. $y = \frac{1}{|x|+1} - \frac{x^2}{2}$
19. $y = \frac{\cos x}{x}$
20. $y = \frac{x+2}{\cos x}$
21. $y = \csc 2x$
22. $y = \tan \frac{\pi x}{2}$
23. $y = \frac{x \tan x}{x^2+1}$
24. $y = \frac{\sqrt{x^4+1}}{1+\sin^2 x}$
25. $y = \sqrt{2x+3}$
26. $y = \sqrt[4]{3x-1}$
27. $y = (2x-1)^{1/3}$
28. $y = (2-x)^{1/5}$

Composite Functions

Find the limits in Exercises 29–34. Are the functions continuous at the point being approached?

29. $\lim_{x \rightarrow \pi} \sin(x - \sin x)$
30. $\lim_{t \rightarrow 0} \sin\left(\frac{\pi}{2} \cos(\tan t)\right)$
31. $\lim_{y \rightarrow 1} \sec(y \sec^2 y - \tan^2 y - 1)$
32. $\lim_{x \rightarrow 0} \tan\left(\frac{\pi}{4} \cos(\sin x^{1/3})\right)$

33. $\lim_{t \rightarrow 0} \cos\left(\frac{\pi}{\sqrt{19-3 \sec 2t}}\right)$
34. $\lim_{x \rightarrow \pi/6} \sqrt{\csc^2 x + 5\sqrt{3} \tan x}$

Continuous Extensions

35. Define $g(3)$ in a way that extends $g(x) = (x^2 - 9)/(x - 3)$ to be continuous at $x = 3$.
36. Define $h(2)$ in a way that extends $h(t) = (t^2 + 3t - 10)/(t - 2)$ to be continuous at $t = 2$.
37. Define $f(1)$ in a way that extends $f(s) = (s^3 - 1)/(s^2 - 1)$ to be continuous at $s = 1$.
38. Define $g(4)$ in a way that extends $g(x) = (x^2 - 16)/(x^2 - 3x - 4)$ to be continuous at $x = 4$.
39. For what value of a is

$$f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$$

continuous at every x ?

40. For what value of b is

$$g(x) = \begin{cases} x, & x < -2 \\ bx^2, & x \geq -2 \end{cases}$$

continuous at every x ?

T In Exercises 41–44, graph the function f to see whether it appears to have a continuous extension to the origin. If it does, use Trace and Zoom to find a good candidate for the extended function's value at $x = 0$. If the function does not appear to have a continuous extension, can it be extended to be continuous at the origin from the right or from the left? If so, what do you think the extended function's value(s) should be?

41. $f(x) = \frac{10^x - 1}{x}$
42. $f(x) = \frac{10^{|x|} - 1}{x}$
43. $f(x) = \frac{\sin x}{|x|}$
44. $f(x) = (1 + 2x)^{1/x}$

Theory and Examples

45. A continuous function $y = f(x)$ is known to be negative at $x = 0$ and positive at $x = 1$. Why does the equation $f(x) = 0$ have at least one solution between $x = 0$ and $x = 1$? Illustrate with a sketch.
46. Explain why the equation $\cos x = x$ has at least one solution.
47. **Roots of a cubic** Show that the equation $x^3 - 15x + 1 = 0$ has three solutions in the interval $[-4, 4]$.
48. **A function value** Show that the function $F(x) = (x - a)^2 \cdot (x - b)^2 + x$ takes on the value $(a + b)/2$ for some value of x .
49. **Solving an equation** If $f(x) = x^3 - 8x + 10$, show that there are values c for which $f(c)$ equals (a) π ; (b) $-\sqrt{3}$; (c) 5,000,000.

50. Explain why the following five statements ask for the same information.
- Find the roots of $f(x) = x^3 - 3x - 1$.
 - Find the x -coordinates of the points where the curve $y = x^3$ crosses the line $y = 3x + 1$.
 - Find all the values of x for which $x^3 - 3x = 1$.
 - Find the x -coordinates of the points where the cubic curve $y = x^3 - 3x$ crosses the line $y = 1$.
 - Solve the equation $x^3 - 3x - 1 = 0$.
51. **Removable discontinuity** Give an example of a function $f(x)$ that is continuous for all values of x except $x = 2$, where it has a removable discontinuity. Explain how you know that f is discontinuous at $x = 2$, and how you know the discontinuity is removable.
52. **Nonremovable discontinuity** Give an example of a function $g(x)$ that is continuous for all values of x except $x = -1$, where it has a nonremovable discontinuity. Explain how you know that g is discontinuous there and why the discontinuity is not removable.
53. **A function discontinuous at every point**
- Use the fact that every nonempty interval of real numbers contains both rational and irrational numbers to show that the function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$$
 is discontinuous at every point.
 - Is f right-continuous or left-continuous at any point?
54. If functions $f(x)$ and $g(x)$ are continuous for $0 \leq x \leq 1$, could $f(x)/g(x)$ possibly be discontinuous at a point of $[0, 1]$? Give reasons for your answer.
55. If the product function $h(x) = f(x) \cdot g(x)$ is continuous at $x = 0$, must $f(x)$ and $g(x)$ be continuous at $x = 0$? Give reasons for your answer.
56. **Discontinuous composite of continuous functions** Give an example of functions f and g , both continuous at $x = 0$, for which the composite $f \circ g$ is discontinuous at $x = 0$. Does this contradict Theorem 10? Give reasons for your answer.
57. **Never-zero continuous functions** Is it true that a continuous function that is never zero on an interval never changes sign on that interval? Give reasons for your answer.

58. **Stretching a rubber band** Is it true that if you stretch a rubber band by moving one end to the right and the other to the left, some point of the band will end up in its original position? Give reasons for your answer.

59. **A fixed point theorem** Suppose that a function f is continuous on the closed interval $[0, 1]$ and that $0 \leq f(x) \leq 1$ for every x in $[0, 1]$. Show that there must exist a number c in $[0, 1]$ such that $f(c) = c$ (c is called a **fixed point** of f).

60. **The sign-preserving property of continuous functions** Let f be defined on an interval (a, b) and suppose that $f(c) \neq 0$ at some c where f is continuous. Show that there is an interval $(c - \delta, c + \delta)$ about c where f has the same sign as $f(c)$. Notice how remarkable this conclusion is. Although f is defined throughout (a, b) , it is not required to be continuous at any point except c . That and the condition $f(c) \neq 0$ are enough to make f different from zero (positive or negative) throughout an entire interval.

61. Prove that f is continuous at c if and only if

$$\lim_{h \rightarrow 0} f(c + h) = f(c).$$

62. Use Exercise 61 together with the identities

$$\sin(h + c) = \sin h \cos c + \cos h \sin c,$$

$$\cos(h + c) = \cos h \cos c - \sin h \sin c$$

to prove that $f(x) = \sin x$ and $g(x) = \cos x$ are continuous at every point $x = c$.

Solving Equations Graphically

T Use a graphing calculator or computer grapher to solve the equations in Exercises 63–70.

63. $x^3 - 3x - 1 = 0$

64. $2x^3 - 2x^2 - 2x + 1 = 0$

65. $x(x - 1)^2 = 1$ (one root)

66. $x^x = 2$

67. $\sqrt{x} + \sqrt{1 + x} = 4$

68. $x^3 - 15x + 1 = 0$ (three roots)

69. $\cos x = x$ (one root). Make sure you are using radian mode.

70. $2 \sin x = x$ (three roots). Make sure you are using radian mode.

2.7

Tangents and Derivatives

This section continues the discussion of secants and tangents begun in Section 2.1. We calculate limits of secant slopes to find tangents to curves.

What Is a Tangent to a Curve?

For circles, tangency is straightforward. A line L is tangent to a circle at a point P if L passes through P perpendicular to the radius at P (Figure 2.63). Such a line just *touches*

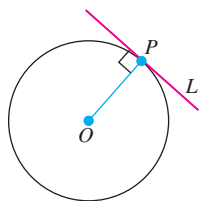


FIGURE 2.63 L is tangent to the circle at P if it passes through P perpendicular to radius OP .

the circle. But what does it mean to say that a line L is tangent to some other curve C at a point P ? Generalizing from the geometry of the circle, we might say that it means one of the following:

1. L passes through P perpendicular to the line from P to the center of C .
2. L passes through only one point of C , namely P .
3. L passes through P and lies on one side of C only.

Although these statements are valid if C is a circle, none of them works consistently for more general curves. Most curves do not have centers, and a line we may want to call tangent may intersect C at other points or cross C at the point of tangency (Figure 2.64).

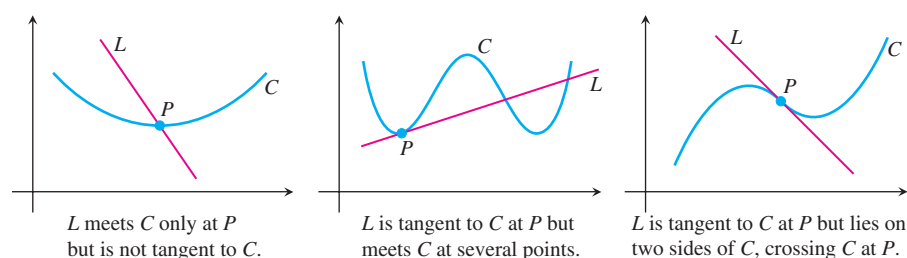


FIGURE 2.64 Exploding myths about tangent lines.

HISTORICAL BIOGRAPHY

Pierre de Fermat
(1601–1665)

To define tangency for general curves, we need a *dynamic* approach that takes into account the behavior of the secants through P and nearby points Q as Q moves toward P along the curve (Figure 2.65). It goes like this:

1. We start with what we *can* calculate, namely the slope of the secant PQ .
2. Investigate the limit of the secant slope as Q approaches P along the curve.
3. If the limit exists, take it to be the slope of the curve at P and define the tangent to the curve at P to be the line through P with this slope.

This approach is what we were doing in the falling-rock and fruit fly examples in Section 2.1.

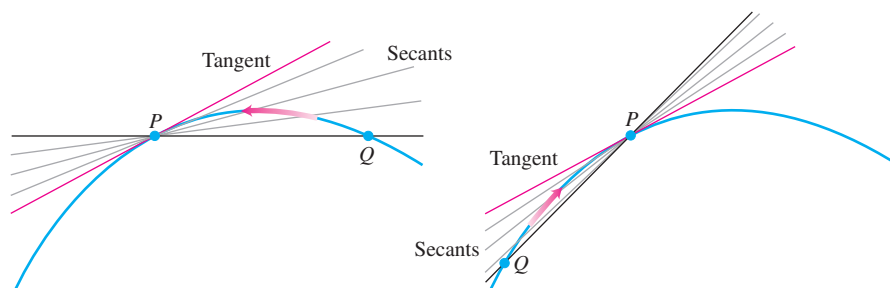


FIGURE 2.65 The dynamic approach to tangency. The tangent to the curve at P is the line through P whose slope is the limit of the secant slopes as $Q \rightarrow P$ from either side.

EXAMPLE 1 Tangent Line to a Parabola

Find the slope of the parabola $y = x^2$ at the point $P(2, 4)$. Write an equation for the tangent to the parabola at this point.

Solution We begin with a secant line through $P(2, 4)$ and $Q(2 + h, (2 + h)^2)$ nearby. We then write an expression for the slope of the secant PQ and investigate what happens to the slope as Q approaches P along the curve:

$$\begin{aligned}\text{Secant slope} &= \frac{\Delta y}{\Delta x} = \frac{(2 + h)^2 - 2^2}{h} = \frac{h^2 + 4h + 4 - 4}{h} \\ &= \frac{h^2 + 4h}{h} = h + 4.\end{aligned}$$

If $h > 0$, then Q lies above and to the right of P , as in Figure 2.66. If $h < 0$, then Q lies to the left of P (not shown). In either case, as Q approaches P along the curve, h approaches zero and the secant slope approaches 4:

$$\lim_{h \rightarrow 0} (h + 4) = 4.$$

We take 4 to be the parabola's slope at P .

The tangent to the parabola at P is the line through P with slope 4:

$$y = 4 + 4(x - 2) \quad \text{Point-slope equation}$$

$$y = 4x - 4.$$

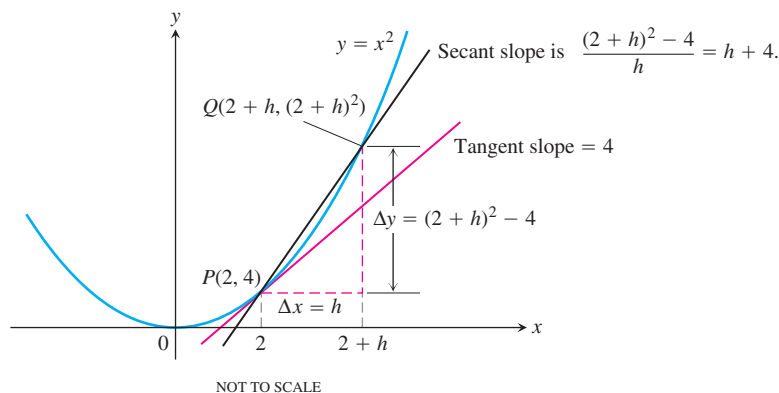


FIGURE 2.66 Finding the slope of the parabola $y = x^2$ at the point $P(2, 4)$ (Example 1).

Finding a Tangent to the Graph of a Function

The problem of finding a tangent to a curve was the dominant mathematical problem of the early seventeenth century. In optics, the tangent determined the angle at which a ray of light entered a curved lens. In mechanics, the tangent determined the direction of a body's motion at every point along its path. In geometry, the tangents to two curves at a point of intersection determined the angles at which they intersected. To find a tangent to an arbitrary curve $y = f(x)$ at a point $P(x_0, f(x_0))$, we use the same dynamic procedure. We calculate the slope of the secant through P and a point $Q(x_0 + h, f(x_0 + h))$. We then investigate the limit of the slope as $h \rightarrow 0$ (Figure 2.67). If the limit exists, we call it the slope of the curve at P and define the tangent at P to be the line through P having this slope.

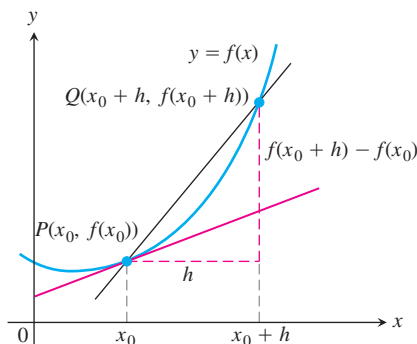


FIGURE 2.67 The slope of the tangent line at P is $\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$.

DEFINITIONS Slope, Tangent Line

The **slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$m = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (\text{provided the limit exists}).$$

The **tangent line** to the curve at P is the line through P with this slope.

Whenever we make a new definition, we try it on familiar objects to be sure it is consistent with results we expect in familiar cases. Example 2 shows that the new definition of slope agrees with the old definition from Section 1.2 when we apply it to nonvertical lines.

EXAMPLE 2 Testing the Definition

Show that the line $y = mx + b$ is its own tangent at any point $(x_0, mx_0 + b)$.

Solution We let $f(x) = mx + b$ and organize the work into three steps.

1. Find $f(x_0)$ and $f(x_0 + h)$.

$$f(x_0) = mx_0 + b$$

$$f(x_0 + h) = m(x_0 + h) + b = mx_0 + mh + b$$

2. Find the slope $\lim_{h \rightarrow 0} (f(x_0 + h) - f(x_0))/h$.

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{(mx_0 + mh + b) - (mx_0 + b)}{h} \\ &= \lim_{h \rightarrow 0} \frac{mh}{h} = m \end{aligned}$$

3. Find the tangent line using the point-slope equation. The tangent line at the point $(x_0, mx_0 + b)$ is

$$y = (mx_0 + b) + m(x - x_0)$$

$$y = mx_0 + b + mx - mx_0$$

$$y = mx + b. \quad \blacksquare$$

Let's summarize the steps in Example 2.

Finding the Tangent to the Curve $y = f(x)$ at (x_0, y_0)

1. Calculate $f(x_0)$ and $f(x_0 + h)$.
2. Calculate the slope

$$m = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

3. If the limit exists, find the tangent line as

$$y = y_0 + m(x - x_0).$$

EXAMPLE 3 Slope and Tangent to $y = 1/x$, $x \neq 0$

- (a) Find the slope of the curve $y = 1/x$ at $x = a \neq 0$.
 (b) Where does the slope equal $-1/4$?
 (c) What happens to the tangent to the curve at the point $(a, 1/a)$ as a changes?

Solution

- (a) Here $f(x) = 1/x$. The slope at $(a, 1/a)$ is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{h} \frac{a - (a+h)}{a(a+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{ha(a+h)} \\ &= \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = -\frac{1}{a^2}. \end{aligned}$$

Notice how we had to keep writing “ $\lim_{h \rightarrow 0}$ ” before each fraction until the stage where we could evaluate the limit by substituting $h = 0$. The number a may be positive or negative, but not 0.

- (b) The slope of $y = 1/x$ at the point where $x = a$ is $-1/a^2$. It will be $-1/4$ provided that

$$-\frac{1}{a^2} = -\frac{1}{4}.$$

This equation is equivalent to $a^2 = 4$, so $a = 2$ or $a = -2$. The curve has slope $-1/4$ at the two points $(2, 1/2)$ and $(-2, -1/2)$ (Figure 2.68).

- (c) Notice that the slope $-1/a^2$ is always negative if $a \neq 0$. As $a \rightarrow 0^+$, the slope approaches $-\infty$ and the tangent becomes increasingly steep (Figure 2.69). We see this situation again as $a \rightarrow 0^-$. As a moves away from the origin in either direction, the slope approaches 0^- and the tangent levels off.

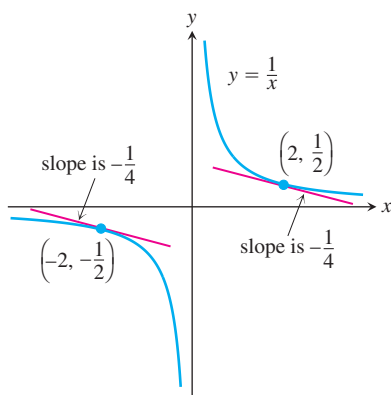


FIGURE 2.68 The two tangent lines to $y = 1/x$ having slope $-1/4$ (Example 3).

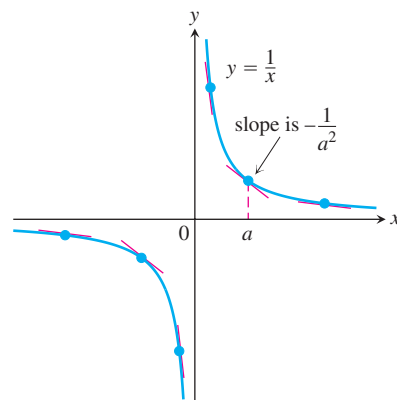


FIGURE 2.69 The tangent slopes, steep near the origin, become more gradual as the point of tangency moves away.

Rates of Change: Derivative at a Point

The expression

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

is called the **difference quotient of f at x_0 with increment h** . If the difference quotient has a limit as h approaches zero, that limit is called the **derivative of f at x_0** . If we interpret the difference quotient as a secant slope, the derivative gives the slope of the curve and tangent at the point where $x = x_0$. If we interpret the difference quotient as an average rate of change, as we did in Section 2.1, the derivative gives the function's rate of change with respect to x at the point $x = x_0$. The derivative is one of the two most important mathematical objects considered in calculus. We begin a thorough study of it in Chapter 3. The other important object is the integral, and we initiate its study in Chapter 5.

EXAMPLE 4 Instantaneous Speed (Continuation of Section 2.1, Examples 1 and 2)

In Examples 1 and 2 in Section 2.1, we studied the speed of a rock falling freely from rest near the surface of the earth. We knew that the rock fell $y = 16t^2$ feet during the first t sec, and we used a sequence of average rates over increasingly short intervals to estimate the rock's speed at the instant $t = 1$. Exactly what *was* the rock's speed at this time?

Solution We let $f(t) = 16t^2$. The average speed of the rock over the interval between $t = 1$ and $t = 1 + h$ seconds was

$$\frac{f(1 + h) - f(1)}{h} = \frac{16(1 + h)^2 - 16(1)^2}{h} = \frac{16(h^2 + 2h)}{h} = 16(h + 2).$$

The rock's speed at the instant $t = 1$ was

$$\lim_{h \rightarrow 0} 16(h + 2) = 16(0 + 2) = 32 \text{ ft/sec}.$$

Our original estimate of 32 ft/sec was right. ■

Summary

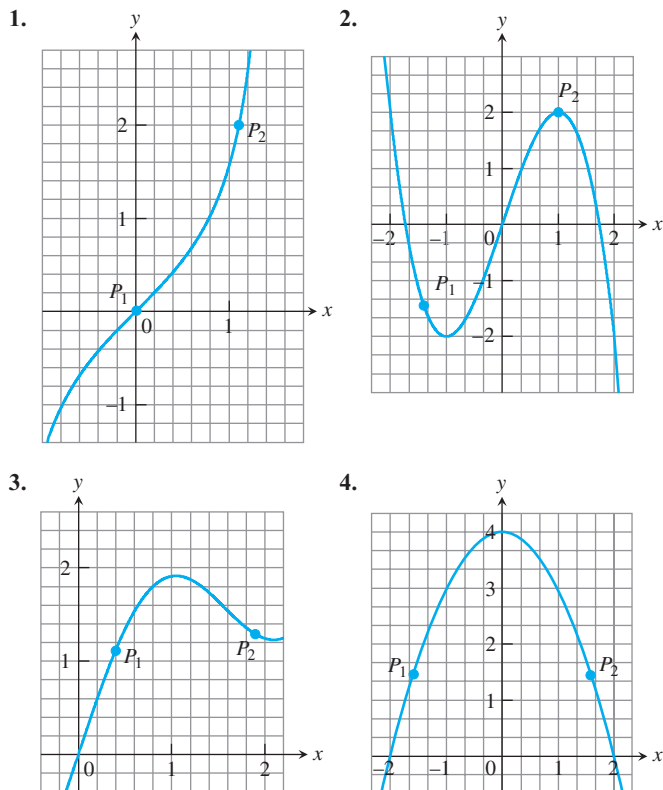
We have been discussing slopes of curves, lines tangent to a curve, the rate of change of a function, the limit of the difference quotient, and the derivative of a function at a point. All of these ideas refer to the same thing, summarized here:

1. The slope of $y = f(x)$ at $x = x_0$
2. The slope of the tangent to the curve $y = f(x)$ at $x = x_0$
3. The rate of change of $f(x)$ with respect to x at $x = x_0$
4. The derivative of f at $x = x_0$
5. The limit of the difference quotient, $\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$

EXERCISES 2.7

Slopes and Tangent Lines

In Exercises 1–4, use the grid and a straight edge to make a rough estimate of the slope of the curve (in y -units per x -unit) at the points P_1 and P_2 . Graphs can shift during a press run, so your estimates may be somewhat different from those in the back of the book.



In Exercises 5–10, find an equation for the tangent to the curve at the given point. Then sketch the curve and tangent together.

5. $y = 4 - x^2$, $(-1, 3)$
6. $y = (x - 1)^2 + 1$, $(1, 1)$
7. $y = 2\sqrt{x}$, $(1, 2)$
8. $y = \frac{1}{x^2}$, $(-1, 1)$
9. $y = x^3$, $(-2, -8)$
10. $y = \frac{1}{x^3}$, $(-2, -\frac{1}{8})$

In Exercises 11–18, find the slope of the function's graph at the given point. Then find an equation for the line tangent to the graph there.

11. $f(x) = x^2 + 1$, $(2, 5)$
12. $f(x) = x - 2x^2$, $(1, -1)$
13. $g(x) = \frac{x}{x-2}$, $(3, 3)$
14. $g(x) = \frac{8}{x^2}$, $(2, 2)$
15. $h(t) = t^3$, $(2, 8)$
16. $h(t) = t^3 + 3t$, $(1, 4)$
17. $f(x) = \sqrt{x}$, $(4, 2)$
18. $f(x) = \sqrt{x+1}$, $(8, 3)$

In Exercises 19–22, find the slope of the curve at the point indicated.

19. $y = 5x^2$, $x = -1$
20. $y = 1 - x^2$, $x = 2$
21. $y = \frac{1}{x-1}$, $x = 3$
22. $y = \frac{x-1}{x+1}$, $x = 0$

Tangent Lines with Specified Slopes

At what points do the graphs of the functions in Exercises 23 and 24 have horizontal tangents?

23. $f(x) = x^2 + 4x - 1$
24. $g(x) = x^3 - 3x$
25. Find equations of all lines having slope -1 that are tangent to the curve $y = 1/(x-1)$.
26. Find an equation of the straight line having slope $1/4$ that is tangent to the curve $y = \sqrt{x}$.

Rates of Change

27. **Object dropped from a tower** An object is dropped from the top of a 100-m-high tower. Its height above ground after t sec is $100 - 4.9t^2$ m. How fast is it falling 2 sec after it is dropped?
28. **Speed of a rocket** At t sec after liftoff, the height of a rocket is $3t^2$ ft. How fast is the rocket climbing 10 sec after liftoff?
29. **Circle's changing area** What is the rate of change of the area of a circle ($A = \pi r^2$) with respect to the radius when the radius is $r = 3$?
30. **Ball's changing volume** What is the rate of change of the volume of a ball ($V = (4/3)\pi r^3$) with respect to the radius when the radius is $r = 2$?

Testing for Tangents

31. Does the graph of

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

have a tangent at the origin? Give reasons for your answer.

32. Does the graph of

$$g(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

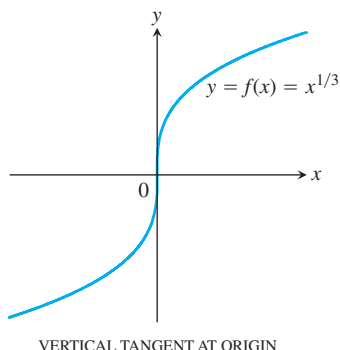
have a tangent at the origin? Give reasons for your answer.

Vertical Tangents

We say that the curve $y = f(x)$ has a **vertical tangent** at the point where $x = x_0$ if $\lim_{h \rightarrow 0} (f(x_0 + h) - f(x_0))/h = \infty$ or $-\infty$.

Vertical tangent at $x = 0$ (see accompanying figure):

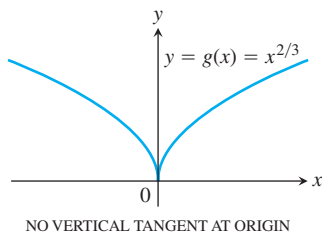
$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} &= \lim_{h \rightarrow 0} \frac{h^{1/3} - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h^{2/3}} = \infty \end{aligned}$$



No vertical tangent at $x = 0$ (see next figure):

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{g(0+h) - g(0)}{h} &= \lim_{h \rightarrow 0} \frac{h^{2/3} - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h^{1/3}}\end{aligned}$$

does not exist, because the limit is ∞ from the right and $-\infty$ from the left.



33. Does the graph of

$$f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

have a vertical tangent at the origin? Give reasons for your answer.

34. Does the graph of

$$U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

have a vertical tangent at the point $(0, 1)$? Give reasons for your answer.

- T** a. Graph the curves in Exercises 35–44. Where do the graphs appear to have vertical tangents?
b. Confirm your findings in part (a) with limit calculations. But before you do, read the introduction to Exercises 33 and 34.

35. $y = x^{2/5}$

36. $y = x^{4/5}$

37. $y = x^{1/5}$

38. $y = x^{3/5}$

39. $y = 4x^{2/5} - 2x$

40. $y = x^{5/3} - 5x^{2/3}$

41. $y = x^{2/3} - (x - 1)^{1/3}$

42. $y = x^{1/3} + (x - 1)^{1/3}$

43. $y = \begin{cases} -\sqrt{|x|}, & x \leq 0 \\ \sqrt{x}, & x > 0 \end{cases}$

44. $y = \sqrt{|4 - x|}$

COMPUTER EXPLORATIONS

Graphing Secant and Tangent Lines

Use a CAS to perform the following steps for the functions in Exercises 45–48.

- a. Plot $y = f(x)$ over the interval $(x_0 - 1/2) \leq x \leq (x_0 + 3)$.
b. Holding x_0 fixed, the difference quotient

$$q(h) = \frac{f(x_0 + h) - f(x_0)}{h}$$

at x_0 becomes a function of the step size h . Enter this function into your CAS workspace.

- c. Find the limit of q as $h \rightarrow 0$.
d. Define the secant lines $y = f(x_0) + q \cdot (x - x_0)$ for $h = 3, 2$, and 1 . Graph them together with f and the tangent line over the interval in part (a).

45. $f(x) = x^3 + 2x, \quad x_0 = 0$ 46. $f(x) = x + \frac{5}{x}, \quad x_0 = 1$

47. $f(x) = x + \sin(2x), \quad x_0 = \pi/2$

48. $f(x) = \cos x + 4 \sin(2x), \quad x_0 = \pi$

Chapter 2

Additional and Advanced Exercises

- T 1. Assigning a value to 0^0** The rules of exponents (see Appendix 9) tell us that $a^0 = 1$ if a is any number different from zero. They also tell us that $0^n = 0$ if n is any positive number.

If we tried to extend these rules to include the case 0^0 , we would get conflicting results. The first rule would say $0^0 = 1$, whereas the second would say $0^0 = 0$.

We are not dealing with a question of right or wrong here. Neither rule applies as it stands, so there is no contradiction. We could, in fact, define 0^0 to have any value we wanted as long as we could persuade others to agree.

What value would you like 0^0 to have? Here is an example that might help you to decide. (See Exercise 2 below for another example.)

- Calculate x^x for $x = 0.1, 0.01, 0.001$, and so on as far as your calculator can go. Record the values you get. What pattern do you see?
- Graph the function $y = x^x$ for $0 < x \leq 1$. Even though the function is not defined for $x \leq 0$, the graph will approach the y -axis from the right. Toward what y -value does it seem to be headed? Zoom in to further support your idea.

- T 2. A reason you might want 0^0 to be something other than 0 or 1** As the number x increases through positive values, the numbers $1/x$ and $1/(\ln x)$ both approach zero. What happens to the number

$$f(x) = \left(\frac{1}{x}\right)^{1/(\ln x)}$$

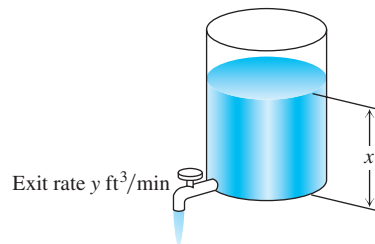
as x increases? Here are two ways to find out.

- Evaluate f for $x = 10, 100, 1000$, and so on as far as your calculator can reasonably go. What pattern do you see?
 - Graph f in a variety of graphing windows, including windows that contain the origin. What do you see? Trace the y -values along the graph. What do you find?
- 3. Lorentz contraction** In relativity theory, the length of an object, say a rocket, appears to an observer to depend on the speed at which the object is traveling with respect to the observer. If the observer measures the rocket's length as L_0 at rest, then at speed v the length will appear to be

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}.$$

This equation is the Lorentz contraction formula. Here, c is the speed of light in a vacuum, about 3×10^8 m/sec. What happens to L as v increases? Find $\lim_{v \rightarrow c^-} L$. Why was the left-hand limit needed?

- 4. Controlling the flow from a draining tank** Torricelli's law says that if you drain a tank like the one in the figure shown, the rate y at which water runs out is a constant times the square root of the water's depth x . The constant depends on the size and shape of the exit valve.



Suppose that $y = \sqrt{x}/2$ for a certain tank. You are trying to maintain a fairly constant exit rate by adding water to the tank with a hose from time to time. How deep must you keep the water if you want to maintain the exit rate

- within $0.2 \text{ ft}^3/\text{min}$ of the rate $y_0 = 1 \text{ ft}^3/\text{min}$?
 - within $0.1 \text{ ft}^3/\text{min}$ of the rate $y_0 = 1 \text{ ft}^3/\text{min}$?
- 5. Thermal expansion in precise equipment** As you may know, most metals expand when heated and contract when cooled. The dimensions of a piece of laboratory equipment are sometimes so critical that the shop where the equipment is made must be held at the same temperature as the laboratory where the equipment is to be used. A typical aluminum bar that is 10 cm wide at 70°F will be

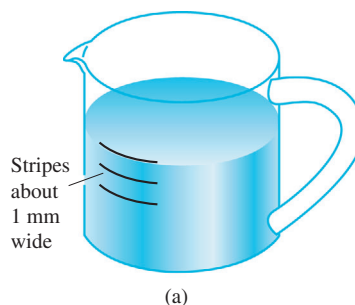
$$y = 10 + (t - 70) \times 10^{-4}$$

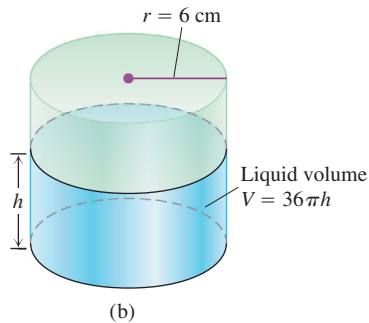
centimeters wide at a nearby temperature t . Suppose that you are using a bar like this in a gravity wave detector, where its width must stay within 0.0005 cm of the ideal 10 cm. How close to $t_0 = 70^\circ\text{F}$ must you maintain the temperature to ensure that this tolerance is not exceeded?

- 6. Stripes on a measuring cup** The interior of a typical 1-L measuring cup is a right circular cylinder of radius 6 cm (see accompanying figure). The volume of water we put in the cup is therefore a function of the level h to which the cup is filled, the formula being

$$V = \pi 6^2 h = 36\pi h.$$

How closely must we measure h to measure out 1 L of water (1000 cm^3) with an error of no more than 1% (10 cm^3)?





A 1-L measuring cup (a), modeled as a right circular cylinder (b) of radius $r = 6$ cm

Precise Definition of Limit

In Exercises 7–10, use the formal definition of limit to prove that the function is continuous at x_0 .

7. $f(x) = x^2 - 7$, $x_0 = 1$ 8. $g(x) = 1/(2x)$, $x_0 = 1/4$
 9. $h(x) = \sqrt{2x - 3}$, $x_0 = 2$ 10. $F(x) = \sqrt{9 - x}$, $x_0 = 5$

11. **Uniqueness of limits** Show that a function cannot have two different limits at the same point. That is, if $\lim_{x \rightarrow x_0} f(x) = L_1$ and $\lim_{x \rightarrow x_0} f(x) = L_2$, then $L_1 = L_2$.

12. Prove the limit Constant Multiple Rule:

$$\lim_{x \rightarrow c} kf(x) = k \lim_{x \rightarrow c} f(x) \quad \text{for any constant } k.$$

13. **One-sided limits** If $\lim_{x \rightarrow 0^+} f(x) = A$ and $\lim_{x \rightarrow 0^-} f(x) = B$, find

- a. $\lim_{x \rightarrow 0^+} f(x^3 - x)$ b. $\lim_{x \rightarrow 0^-} f(x^3 - x)$
 c. $\lim_{x \rightarrow 0^+} f(x^2 - x^4)$ d. $\lim_{x \rightarrow 0^-} f(x^2 - x^4)$

14. **Limits and continuity** Which of the following statements are true, and which are false? If true, say why; if false, give a counterexample (that is, an example confirming the falsehood).

- a. If $\lim_{x \rightarrow a} f(x)$ exists but $\lim_{x \rightarrow a} g(x)$ does not exist, then $\lim_{x \rightarrow a} (f(x) + g(x))$ does not exist.
 b. If neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exists, then $\lim_{x \rightarrow a} (f(x) + g(x))$ does not exist.
 c. If f is continuous at x , then so is $|f|$.
 d. If $|f|$ is continuous at a , then so is f .

In Exercises 15 and 16, use the formal definition of limit to prove that the function has a continuous extension to the given value of x .

15. $f(x) = \frac{x^2 - 1}{x + 1}$, $x = -1$ 16. $g(x) = \frac{x^2 - 2x - 3}{2x - 6}$, $x = 3$

17. **A function continuous at only one point** Let

$$f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

- a. Show that f is continuous at $x = 0$.
 b. Use the fact that every nonempty open interval of real numbers contains both rational and irrational numbers to show that f is not continuous at any nonzero value of x .

18. **The Dirichlet ruler function** If x is a rational number, then x can be written in a unique way as a quotient of integers m/n where $n > 0$ and m and n have no common factors greater than 1. (We say that such a fraction is in *lowest terms*. For example, $6/4$ written in lowest terms is $3/2$.) Let $f(x)$ be defined for all x in the interval $[0, 1]$ by

$$f(x) = \begin{cases} 1/n, & \text{if } x = m/n \text{ is a rational number in lowest terms} \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

For instance, $f(0) = f(1) = 1$, $f(1/2) = 1/2$, $f(1/3) = f(2/3) = 1/3$, $f(1/4) = f(3/4) = 1/4$, and so on.

- a. Show that f is discontinuous at every rational number in $[0, 1]$.
 b. Show that f is continuous at every irrational number in $[0, 1]$. (Hint: If ϵ is a given positive number, show that there are only finitely many rational numbers r in $[0, 1]$ such that $f(r) \geq \epsilon$.)
 c. Sketch the graph of f . Why do you think f is called the “ruler function”?

19. **Antipodal points** Is there any reason to believe that there is always a pair of antipodal (diametrically opposite) points on Earth’s equator where the temperatures are the same? Explain.

20. If $\lim_{x \rightarrow c} (f(x) + g(x)) = 3$ and $\lim_{x \rightarrow c} (f(x) - g(x)) = -1$, find $\lim_{x \rightarrow c} f(x)g(x)$.

21. **Roots of a quadratic equation that is almost linear** The equation $ax^2 + 2x - 1 = 0$, where a is a constant, has two roots if $a > -1$ and $a \neq 0$, one positive and one negative:

$$r_+(a) = \frac{-1 + \sqrt{1+a}}{a}, \quad r_-(a) = \frac{-1 - \sqrt{1+a}}{a}.$$

- a. What happens to $r_+(a)$ as $a \rightarrow 0$? As $a \rightarrow -1^+$?
 b. What happens to $r_-(a)$ as $a \rightarrow 0$? As $a \rightarrow -1^+$?
 c. Support your conclusions by graphing $r_+(a)$ and $r_-(a)$ as functions of a . Describe what you see.
 d. For added support, graph $f(x) = ax^2 + 2x - 1$ simultaneously for $a = 1, 0.5, 0.2, 0.1$, and 0.05 .

22. **Root of an equation** Show that the equation $x + 2 \cos x = 0$ has at least one solution.

23. **Bounded functions** A real-valued function f is **bounded from above** on a set D if there exists a number N such that $f(x) \leq N$ for all x in D . We call N , when it exists, an **upper bound** for f on D and say that f is bounded from above by N . In a similar manner, we say that f is **bounded from below** on D if there exists a number M such that $f(x) \geq M$ for all x in D . We call M , when it exists, a **lower bound** for f on D and say that f is bounded from below by M . We say that f is **bounded** on D if it is bounded from both above and below.

- a. Show that f is bounded on D if and only if there exists a number B such that $|f(x)| \leq B$ for all x in D .
 b. Suppose that f is bounded from above by N . Show that if $\lim_{x \rightarrow x_0} f(x) = L$, then $L \leq N$.
 c. Suppose that f is bounded from below by M . Show that if $\lim_{x \rightarrow x_0} f(x) = L$, then $L \geq M$.

24. Max $\{a, b\}$ and min $\{a, b\}$ **a.** Show that the expression

$$\max \{a, b\} = \frac{a + b}{2} + \frac{|a - b|}{2}$$

equals a if $a \geq b$ and equals b if $b \geq a$. In other words, $\max \{a, b\}$ gives the larger of the two numbers a and b .

b. Find a similar expression for $\min \{a, b\}$, the smaller of a and b .**Generalized Limits Involving $\frac{\sin \theta}{\theta}$**

The formula $\lim_{\theta \rightarrow 0} (\sin \theta)/\theta = 1$ can be generalized. If $\lim_{x \rightarrow c} f(x) = 0$ and $f(x)$ is never zero in an open interval containing the point $x = c$, except possibly c itself, then

$$\lim_{x \rightarrow c} \frac{\sin f(x)}{f(x)} = 1.$$

Here are several examples.

a. $\lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} = 1.$

b. $\lim_{x \rightarrow 0} \frac{\sin x^2}{x} = \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} \lim_{x \rightarrow 0} \frac{x^2}{x} = 1 \cdot 0 = 0.$

c. $\lim_{x \rightarrow -1} \frac{\sin(x^2 - x - 2)}{x + 1} = \lim_{x \rightarrow -1} \frac{\sin(x^2 - x - 2)}{(x^2 - x - 2)}.$

$$\lim_{x \rightarrow -1} \frac{(x^2 - x - 2)}{x + 1} = 1 \cdot \lim_{x \rightarrow -1} \frac{(x + 1)(x - 2)}{x + 1} = -3.$$

d. $\lim_{x \rightarrow 1} \frac{\sin(1 - \sqrt{x})}{x - 1} = \lim_{x \rightarrow 1} \frac{\sin(1 - \sqrt{x})}{1 - \sqrt{x}} \frac{1 - \sqrt{x}}{x - 1} =$

$$1 \cdot \lim_{x \rightarrow 1} \frac{(1 - \sqrt{x})(1 + \sqrt{x})}{(x - 1)(1 + \sqrt{x})} = \lim_{x \rightarrow 1} \frac{1 - x}{(x - 1)(1 + \sqrt{x})} = -\frac{1}{2}.$$

Find the limits in Exercises 25–30.

25. $\lim_{x \rightarrow 0} \frac{\sin(1 - \cos x)}{x}$

26. $\lim_{x \rightarrow 0^+} \frac{\sin x}{\sin \sqrt{x}}$

27. $\lim_{x \rightarrow 0} \frac{\sin(\sin x)}{x}$

28. $\lim_{x \rightarrow 0} \frac{\sin(x^2 + x)}{x}$

29. $\lim_{x \rightarrow 2} \frac{\sin(x^2 - 4)}{x - 2}$

30. $\lim_{x \rightarrow 9} \frac{\sin(\sqrt{x} - 3)}{x - 9}$

Chapter 2

Practice Exercises

Limits and Continuity

1. Graph the function

$$f(x) = \begin{cases} 1, & x \leq -1 \\ -x, & -1 < x < 0 \\ 1, & x = 0 \\ -x, & 0 < x < 1 \\ 1, & x \geq 1. \end{cases}$$

Then discuss, in detail, limits, one-sided limits, continuity, and one-sided continuity of f at $x = -1, 0$, and 1 . Are any of the discontinuities removable? Explain.

2. Repeat the instructions of Exercise 1 for

$$f(x) = \begin{cases} 0, & x \leq -1 \\ 1/x, & 0 < |x| < 1 \\ 0, & x = 1 \\ 1, & x > 1. \end{cases}$$

3. Suppose that
- $f(t)$
- and
- $g(t)$
- are defined for all
- t
- and that
- $\lim_{t \rightarrow t_0} f(t) = -7$
- and
- $\lim_{t \rightarrow t_0} g(t) = 0$
- . Find the limit as
- $t \rightarrow t_0$
- of the following functions.

a. $3f(t)$

b. $(f(t))^2$

c. $f(t) \cdot g(t)$

d. $\frac{f(t)}{g(t) - 7}$

e. $\cos(g(t))$

f. $|f(t)|$

g. $f(t) + g(t)$

h. $1/f(t)$

4. Suppose that $f(x)$ and $g(x)$ are defined for all x and that $\lim_{x \rightarrow 0} f(x) = 1/2$ and $\lim_{x \rightarrow 0} g(x) = \sqrt{2}$. Find the limits as $x \rightarrow 0$ of the following functions.

- | | |
|------------------|--------------------------------------|
| a. $-g(x)$ | b. $g(x) \cdot f(x)$ |
| c. $f(x) + g(x)$ | d. $1/f(x)$ |
| e. $x + f(x)$ | f. $\frac{f(x) \cdot \cos x}{x - 1}$ |

In Exercises 5 and 6, find the value that $\lim_{x \rightarrow 0} g(x)$ must have if the given limit statements hold.

5. $\lim_{x \rightarrow 0} \left(\frac{4 - g(x)}{x} \right) = 1$ 6. $\lim_{x \rightarrow -4} \left(x \lim_{x \rightarrow 0} g(x) \right) = 2$

7. On what intervals are the following functions continuous?

- | | |
|----------------------|----------------------|
| a. $f(x) = x^{1/3}$ | b. $g(x) = x^{3/4}$ |
| c. $h(x) = x^{-2/3}$ | d. $k(x) = x^{-1/6}$ |

8. On what intervals are the following functions continuous?

- | | |
|------------------------------------|------------------------------|
| a. $f(x) = \tan x$ | b. $g(x) = \csc x$ |
| c. $h(x) = \frac{\cos x}{x - \pi}$ | d. $k(x) = \frac{\sin x}{x}$ |

Finding Limits

In Exercises 9–16, find the limit or explain why it does not exist.

9. $\lim_{x \rightarrow 0} \frac{x^2 - 4x + 4}{x^3 + 5x^2 - 14x}$
- | | |
|-------------------------|-------------------------|
| a. as $x \rightarrow 0$ | b. as $x \rightarrow 2$ |
|-------------------------|-------------------------|
10. $\lim_{x \rightarrow 0} \frac{x^2 + x}{x^5 + 2x^4 + x^3}$
- | | |
|-------------------------|--------------------------|
| a. as $x \rightarrow 0$ | b. as $x \rightarrow -1$ |
|-------------------------|--------------------------|
11. $\lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{1 - x}$
12. $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^4 - a^4}$
13. $\lim_{h \rightarrow 0} \frac{(x + h)^2 - x^2}{h}$
14. $\lim_{x \rightarrow 0} \frac{(x + h)^2 - x^2}{h}$
15. $\lim_{x \rightarrow 0} \frac{\frac{1}{2+x} - \frac{1}{2}}{x}$
16. $\lim_{x \rightarrow 0} \frac{(2+x)^3 - 8}{x}$

In Exercises 17–20, find the limit of $g(x)$ as x approaches the indicated value.

17. $\lim_{x \rightarrow 0^+} (4g(x))^{1/3} = 2$
18. $\lim_{x \rightarrow \sqrt{5}} \frac{1}{x + g(x)} = 2$
19. $\lim_{x \rightarrow 1} \frac{3x^2 + 1}{g(x)} = \infty$
20. $\lim_{x \rightarrow -2} \frac{5 - x^2}{\sqrt{g(x)}} = 0$

Limits at Infinity

Find the limits in Exercises 21–30.

21. $\lim_{x \rightarrow \infty} \frac{2x + 3}{5x + 7}$
22. $\lim_{x \rightarrow -\infty} \frac{2x^2 + 3}{5x^2 + 7}$

23. $\lim_{x \rightarrow -\infty} \frac{x^2 - 4x + 8}{3x^3}$
24. $\lim_{x \rightarrow \infty} \frac{1}{x^2 - 7x + 1}$
25. $\lim_{x \rightarrow -\infty} \frac{x^2 - 7x}{x + 1}$
26. $\lim_{x \rightarrow \infty} \frac{x^4 + x^3}{12x^3 + 128}$
27. $\lim_{x \rightarrow \infty} \frac{\sin x}{\lfloor x \rfloor}$ (If you have a grapher, try graphing the function for $-5 \leq x \leq 5$.)
28. $\lim_{\theta \rightarrow \infty} \frac{\cos \theta - 1}{\theta}$ (If you have a grapher, try graphing $f(x) = x(\cos(1/x) - 1)$ near the origin to “see” the limit at infinity.)
29. $\lim_{x \rightarrow \infty} \frac{x + \sin x + 2\sqrt{x}}{x + \sin x}$
30. $\lim_{x \rightarrow \infty} \frac{x^{2/3} + x^{-1}}{x^{2/3} + \cos^2 x}$

Continuous Extension

31. Can $f(x) = x(x^2 - 1)/|x^2 - 1|$ be extended to be continuous at $x = 1$ or -1 ? Give reasons for your answers. (Graph the function—you will find the graph interesting.)
32. Explain why the function $f(x) = \sin(1/x)$ has no continuous extension to $x = 0$.

T In Exercises 33–36, graph the function to see whether it appears to have a continuous extension to the given point a . If it does, use Trace and Zoom to find a good candidate for the extended function's value at a . If the function does not appear to have a continuous extension, can it be extended to be continuous from the right or left? If so, what do you think the extended function's value should be?

33. $f(x) = \frac{x - 1}{x - \sqrt[4]{x}}$, $a = 1$
34. $g(\theta) = \frac{5 \cos \theta}{4\theta - 2\pi}$, $a = \pi/2$
35. $h(t) = (1 + |t|)^{1/t}$, $a = 0$
36. $k(x) = \frac{x}{1 - 2|x|}$, $a = 0$

Roots

- T** 37. Let $f(x) = x^3 - x - 1$.
- Show that f has a zero between -1 and 2 .
 - Solve the equation $f(x) = 0$ graphically with an error of magnitude at most 10^{-8} .
 - It can be shown that the exact value of the solution in part (b) is

$$\left(\frac{1}{2} + \frac{\sqrt{69}}{18} \right)^{1/3} + \left(\frac{1}{2} - \frac{\sqrt{69}}{18} \right)^{1/3}$$

Evaluate this exact answer and compare it with the value you found in part (b).

- T** 38. Let $f(\theta) = \theta^3 - 2\theta + 2$.
- Show that f has a zero between -2 and 0 .
 - Solve the equation $f(\theta) = 0$ graphically with an error of magnitude at most 10^{-4} .
 - It can be shown that the exact value of the solution in part (b) is

$$\left(\sqrt{\frac{19}{27}} - 1 \right)^{1/3} - \left(\sqrt{\frac{19}{27}} + 1 \right)^{1/3}$$

Evaluate this exact answer and compare it with the value you found in part (b).

Chapter 2

Questions to Guide Your Review

1. What is the average rate of change of the function $g(t)$ over the interval from $t = a$ to $t = b$? How is it related to a secant line?
2. What limit must be calculated to find the rate of change of a function $g(t)$ at $t = t_0$?
3. What is an informal or intuitive definition of the limit

$$\lim_{x \rightarrow x_0} f(x) = L?$$

Why is the definition “informal”? Give examples.

4. Does the existence and value of the limit of a function $f(x)$ as x approaches x_0 ever depend on what happens at $x = x_0$? Explain and give examples.
5. What function behaviors might occur for which the limit may fail to exist? Give examples.
6. What theorems are available for calculating limits? Give examples of how the theorems are used.

7. How are one-sided limits related to limits? How can this relationship sometimes be used to calculate a limit or prove it does not exist? Give examples.
8. What is the value of $\lim_{\theta \rightarrow 0} ((\sin \theta)/\theta)$? Does it matter whether θ is measured in degrees or radians? Explain.
9. What exactly does $\lim_{x \rightarrow x_0} f(x) = L$ mean? Give an example in which you find a $\delta > 0$ for a given f , L , x_0 , and $\epsilon > 0$ in the precise definition of limit.
10. Give precise definitions of the following statements.
 - a. $\lim_{x \rightarrow 2^-} f(x) = 5$
 - b. $\lim_{x \rightarrow 2^+} f(x) = 5$
 - c. $\lim_{x \rightarrow 2} f(x) = \infty$
 - d. $\lim_{x \rightarrow 2} f(x) = -\infty$
11. What exactly do $\lim_{x \rightarrow \infty} f(x) = L$ and $\lim_{x \rightarrow -\infty} f(x) = L$ mean? Give examples.
12. What are $\lim_{x \rightarrow \pm\infty} k$ (k a constant) and $\lim_{x \rightarrow \pm\infty} (1/x)$? How do you extend these results to other functions? Give examples.
13. How do you find the limit of a rational function as $x \rightarrow \pm\infty$? Give examples.
14. What are horizontal, vertical, and oblique asymptotes? Give examples.
15. What conditions must be satisfied by a function if it is to be continuous at an interior point of its domain? At an endpoint?
16. How can looking at the graph of a function help you tell where the function is continuous?
17. What does it mean for a function to be right-continuous at a point? Left-continuous? How are continuity and one-sided continuity related?
18. What can be said about the continuity of polynomials? Of rational functions? Of trigonometric functions? Of rational powers and al-

gebraic combinations of functions? Of composites of functions? Of absolute values of functions?

19. Under what circumstances can you extend a function $f(x)$ to be continuous at a point $x = c$? Give an example.
20. What does it mean for a function to be continuous on an interval?
21. What does it mean for a function to be continuous? Give examples to illustrate the fact that a function that is not continuous on its entire domain may still be continuous on selected intervals within the domain.
22. What are the basic types of discontinuity? Give an example of each. What is a removable discontinuity? Give an example.
23. What does it mean for a function to have the Intermediate Value Property? What conditions guarantee that a function has this property over an interval? What are the consequences for graphing and solving the equation $f(x) = 0$?
24. It is often said that a function is continuous if you can draw its graph without having to lift your pen from the paper. Why is that?
25. What does it mean for a line to be tangent to a curve C at a point P ?
26. What is the significance of the formula

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}?$$

Interpret the formula geometrically and physically.

27. How do you find the tangent to the curve $y = f(x)$ at a point (x_0, y_0) on the curve?
28. How does the slope of the curve $y = f(x)$ at $x = x_0$ relate to the function's rate of change with respect to x at $x = x_0$? To the derivative of f at x_0 ?

Chapter 2 Technology Application Projects

Mathematica-Maple Module

Take It to the Limit

Part I

Part II (Zero Raised to the Power Zero: What Does it Mean?)

Part III (One-Sided Limits)

Visualize and interpret the limit concept through graphical and numerical explorations.

Part IV (What a Difference a Power Makes)

See how sensitive limits can be with various powers of x .

Mathematica-Maple Module

Going to Infinity

Part I (Exploring Function Behavior as $x \rightarrow \infty$ or $x \rightarrow -\infty$)

This module provides four examples to explore the behavior of a function as $x \rightarrow \infty$ or $x \rightarrow -\infty$.

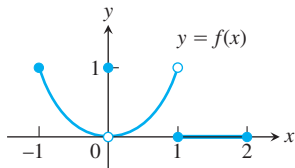
Part II (Rates of Growth)

Observe graphs that *appear* to be continuous, yet the function is not continuous. Several issues of continuity are explored to obtain results that you may find surprising.

EXERCISES 2.4

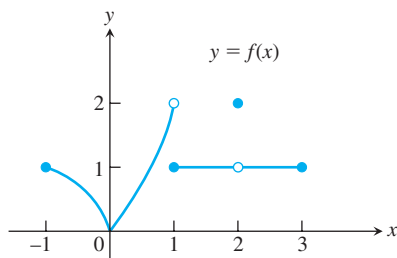
Finding Limits Graphically

1. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?



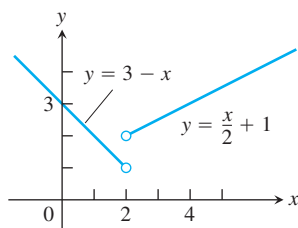
- | | |
|---|--|
| a. $\lim_{x \rightarrow -1^+} f(x) = 1$ | b. $\lim_{x \rightarrow 0^-} f(x) = 0$ |
| c. $\lim_{x \rightarrow 0^-} f(x) = 1$ | d. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$ |
| e. $\lim_{x \rightarrow 0} f(x)$ exists | f. $\lim_{x \rightarrow 0} f(x) = 0$ |
| g. $\lim_{x \rightarrow 0} f(x) = 1$ | h. $\lim_{x \rightarrow 1} f(x) = 1$ |
| i. $\lim_{x \rightarrow 1} f(x) = 0$ | j. $\lim_{x \rightarrow 2^-} f(x) = 2$ |
| k. $\lim_{x \rightarrow -1^-} f(x)$ does not exist. | l. $\lim_{x \rightarrow 2^+} f(x) = 0$ |

2. Which of the following statements about the function $y = f(x)$ graphed here are true, and which are false?



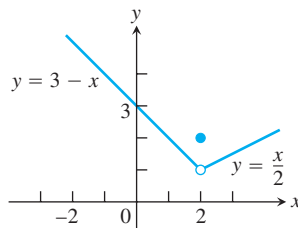
- a. $\lim_{x \rightarrow -1^+} f(x) = 1$
b. $\lim_{x \rightarrow 2} f(x)$ does not exist.
c. $\lim_{x \rightarrow 2} f(x) = 2$
d. $\lim_{x \rightarrow 1^-} f(x) = 2$
e. $\lim_{x \rightarrow 1^+} f(x) = 1$
f. $\lim_{x \rightarrow 1} f(x)$ does not exist.
g. $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
h. $\lim_{x \rightarrow c} f(x)$ exists at every c in the open interval $(-1, 1)$.
i. $\lim_{x \rightarrow c} f(x)$ exists at every c in the open interval $(1, 3)$.
j. $\lim_{x \rightarrow -1^-} f(x) = 0$
k. $\lim_{x \rightarrow 3^+} f(x)$ does not exist.

3. Let $f(x) = \begin{cases} 3 - x, & x < 2 \\ \frac{x}{2} + 1, & x > 2. \end{cases}$



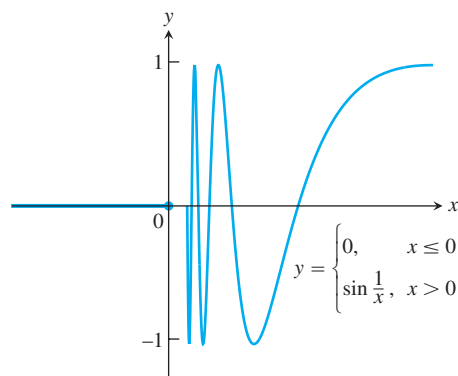
- a. Find $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$.
b. Does $\lim_{x \rightarrow 2} f(x)$ exist? If so, what is it? If not, why not?
c. Find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$.
d. Does $\lim_{x \rightarrow 4} f(x)$ exist? If so, what is it? If not, why not?

4. Let $f(x) = \begin{cases} 3 - x, & x < 2 \\ 2, & x = 2 \\ \frac{x}{2}, & x > 2. \end{cases}$



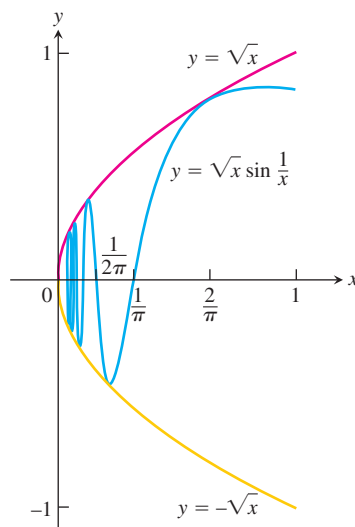
- a. Find $\lim_{x \rightarrow 2^+} f(x)$, $\lim_{x \rightarrow 2^-} f(x)$, and $f(2)$.
b. Does $\lim_{x \rightarrow 2} f(x)$ exist? If so, what is it? If not, why not?
c. Find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$.
d. Does $\lim_{x \rightarrow -1} f(x)$ exist? If so, what is it? If not, why not?

5. Let $f(x) = \begin{cases} 0, & x \leq 0 \\ \sin \frac{1}{x}, & x > 0. \end{cases}$



- a. Does $\lim_{x \rightarrow 0^+} f(x)$ exist? If so, what is it? If not, why not?
b. Does $\lim_{x \rightarrow 0^-} f(x)$ exist? If so, what is it? If not, why not?
c. Does $\lim_{x \rightarrow 0} f(x)$ exist? If so, what is it? If not, why not?

6. Let $g(x) = \sqrt{x} \sin(1/x)$.



- a. Does $\lim_{x \rightarrow 0^+} g(x)$ exist? If so, what is it? If not, why not?
b. Does $\lim_{x \rightarrow 0^-} g(x)$ exist? If so, what is it? If not, why not?
c. Does $\lim_{x \rightarrow 0} g(x)$ exist? If so, what is it? If not, why not?

7. a. Graph $f(x) = \begin{cases} x^3, & x \neq 1 \\ 0, & x = 1. \end{cases}$
 b. Find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$.
 c. Does $\lim_{x \rightarrow 1} f(x)$ exist? If so, what is it? If not, why not?
8. a. Graph $f(x) = \begin{cases} 1 - x^2, & x \neq 1 \\ 2, & x = 1. \end{cases}$
 b. Find $\lim_{x \rightarrow 1^+} f(x)$ and $\lim_{x \rightarrow 1^-} f(x)$.
 c. Does $\lim_{x \rightarrow 1} f(x)$ exist? If so, what is it? If not, why not?

Graph the functions in Exercises 9 and 10. Then answer these questions.

- a. What are the domain and range of f ?
 b. At what points c , if any, does $\lim_{x \rightarrow c} f(x)$ exist?
 c. At what points does only the left-hand limit exist?
 d. At what points does only the right-hand limit exist?
9. $f(x) = \begin{cases} \sqrt{1 - x^2}, & 0 \leq x < 1 \\ 1, & 1 \leq x < 2 \\ 2, & x = 2 \end{cases}$
10. $f(x) = \begin{cases} x, & -1 \leq x < 0, \text{ or } 0 < x \leq 1 \\ 1, & x = 0 \\ 0, & x < -1, \text{ or } x > 1 \end{cases}$

Finding One-Sided Limits Algebraically

Find the limits in Exercises 11–18.

11. $\lim_{x \rightarrow -0.5^-} \sqrt{\frac{x+2}{x+1}}$ 12. $\lim_{x \rightarrow 1^+} \sqrt{\frac{x-1}{x+2}}$
13. $\lim_{x \rightarrow -2^+} \left(\frac{x}{x+1} \right) \left(\frac{2x+5}{x^2+x} \right)$
14. $\lim_{x \rightarrow 1^-} \left(\frac{1}{x+1} \right) \left(\frac{x+6}{x} \right) \left(\frac{3-x}{7} \right)$
15. $\lim_{h \rightarrow 0^+} \frac{\sqrt{h^2 + 4h + 5} - \sqrt{5}}{h}$
16. $\lim_{h \rightarrow 0^-} \frac{\sqrt{6 - \sqrt{5h^2 + 11h + 6}}}{h}$
17. a. $\lim_{x \rightarrow -2^+} (x+3) \frac{|x+2|}{x+2}$ b. $\lim_{x \rightarrow -2^-} (x+3) \frac{|x+2|}{x+2}$
18. a. $\lim_{x \rightarrow 1^+} \frac{\sqrt{2x(x-1)}}{|x-1|}$ b. $\lim_{x \rightarrow 1^-} \frac{\sqrt{2x(x-1)}}{|x-1|}$

Use the graph of the greatest integer function $y = \lfloor x \rfloor$ (sometimes written $y = \text{int } x$), Figure 1.31 in Section 1.3, to help you find the limits in Exercises 19 and 20.

19. a. $\lim_{\theta \rightarrow 3^+} \frac{\lfloor \theta \rfloor}{\theta}$ b. $\lim_{\theta \rightarrow 3^-} \frac{\lfloor \theta \rfloor}{\theta}$
20. a. $\lim_{t \rightarrow 4^+} (t - \lfloor t \rfloor)$ b. $\lim_{t \rightarrow 4^-} (t - \lfloor t \rfloor)$

Using $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Find the limits in Exercises 21–36.

21. $\lim_{\theta \rightarrow 0} \frac{\sin \sqrt{2\theta}}{\sqrt{2\theta}}$ 22. $\lim_{t \rightarrow 0} \frac{\sin kt}{t} \quad (k \text{ constant})$
23. $\lim_{y \rightarrow 0} \frac{\sin 3y}{4y}$ 24. $\lim_{h \rightarrow 0^+} \frac{h}{\sin 3h}$
25. $\lim_{x \rightarrow 0} \frac{\tan 2x}{x}$ 26. $\lim_{t \rightarrow 0} \frac{2t}{\tan t}$
27. $\lim_{x \rightarrow 0} \frac{x \csc 2x}{\cos 5x}$ 28. $\lim_{x \rightarrow 0} 6x^2 (\cot x) (\csc 2x)$
29. $\lim_{x \rightarrow 0} \frac{x + x \cos x}{\sin x \cos x}$ 30. $\lim_{x \rightarrow 0} \frac{x^2 - x + \sin x}{2x}$
31. $\lim_{t \rightarrow 0} \frac{\sin(1 - \cos t)}{1 - \cos t}$ 32. $\lim_{h \rightarrow 0} \frac{\sin(\sin h)}{\sin h}$
33. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\sin 2\theta}$ 34. $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 4x}$
35. $\lim_{y \rightarrow 0} \frac{\tan 3x}{\sin 8x}$ 36. $\lim_{y \rightarrow 0} \frac{\sin 3y \cot 5y}{y \cot 4y}$

Calculating Limits as $x \rightarrow \pm \infty$

In Exercises 37–42, find the limit of each function (a) as $x \rightarrow \infty$ and (b) as $x \rightarrow -\infty$. (You may wish to visualize your answer with a graphing calculator or computer.)

37. $f(x) = \frac{2}{x} - 3$ 38. $f(x) = \pi - \frac{2}{x^2}$
39. $g(x) = \frac{1}{2 + (1/x)}$ 40. $g(x) = \frac{1}{8 - (5/x^2)}$
41. $h(x) = \frac{-5 + (7/x)}{3 - (1/x^2)}$ 42. $h(x) = \frac{3 - (2/x)}{4 + (\sqrt{2}/x^2)}$

Find the limits in Exercises 43–46.

43. $\lim_{x \rightarrow \infty} \frac{\sin 2x}{x}$ 44. $\lim_{\theta \rightarrow -\infty} \frac{\cos \theta}{3\theta}$
45. $\lim_{t \rightarrow -\infty} \frac{2 - t + \sin t}{t + \cos t}$ 46. $\lim_{r \rightarrow \infty} \frac{r + \sin r}{2r + 7 - 5 \sin r}$

Limits of Rational Functions

In Exercises 47–56, find the limit of each rational function (a) as $x \rightarrow \infty$ and (b) as $x \rightarrow -\infty$.

47. $f(x) = \frac{2x+3}{5x+7}$ 48. $f(x) = \frac{2x^3+7}{x^3-x^2+x+7}$
49. $f(x) = \frac{x+1}{x^2+3}$ 50. $f(x) = \frac{3x+7}{x^2-2}$
51. $h(x) = \frac{7x^3}{x^3-3x^2+6x}$ 52. $g(x) = \frac{1}{x^3-4x+1}$

$$53. g(x) = \frac{10x^5 + x^4 + 31}{x^6}$$

$$54. h(x) = \frac{9x^4 + x}{2x^4 + 5x^2 - x + 6}$$

$$55. h(x) = \frac{-2x^3 - 2x + 3}{3x^3 + 3x^2 - 5x}$$

$$56. h(x) = \frac{-x^4}{x^4 - 7x^3 + 7x^2 + 9}$$

Limits with Noninteger or Negative Powers

The process by which we determine limits of rational functions applies equally well to ratios containing noninteger or negative powers of x : divide numerator and denominator by the highest power of x in the denominator and proceed from there. Find the limits in Exercises 57–62.

$$57. \lim_{x \rightarrow \infty} \frac{2\sqrt{x} + x^{-1}}{3x - 7}$$

$$58. \lim_{x \rightarrow \infty} \frac{2 + \sqrt{x}}{2 - \sqrt{x}}$$

$$59. \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x} - \sqrt[5]{x}}{\sqrt[3]{x} + \sqrt[5]{x}}$$

$$60. \lim_{x \rightarrow \infty} \frac{x^{-1} + x^{-4}}{x^{-2} - x^{-3}}$$

$$61. \lim_{x \rightarrow \infty} \frac{2x^{5/3} - x^{1/3} + 7}{x^{8/5} + 3x + \sqrt{x}}$$

$$62. \lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x} - 5x + 3}{2x + x^{2/3} - 4}$$

Theory and Examples

63. Once you know $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ at an interior point of the domain of f , do you then know $\lim_{x \rightarrow a} f(x)$? Give reasons for your answer.
64. If you know that $\lim_{x \rightarrow c} f(x)$ exists, can you find its value by calculating $\lim_{x \rightarrow c^+} f(x)$? Give reasons for your answer.
65. Suppose that f is an odd function of x . Does knowing that $\lim_{x \rightarrow 0^+} f(x) = 3$ tell you anything about $\lim_{x \rightarrow 0^-} f(x)$? Give reasons for your answer.
66. Suppose that f is an even function of x . Does knowing that $\lim_{x \rightarrow 2^-} f(x) = 7$ tell you anything about either $\lim_{x \rightarrow -2^-} f(x)$ or $\lim_{x \rightarrow -2^+} f(x)$? Give reasons for your answer.
67. Suppose that $f(x)$ and $g(x)$ are polynomials in x and that $\lim_{x \rightarrow \infty} (f(x)/g(x)) = 2$. Can you conclude anything about $\lim_{x \rightarrow -\infty} (f(x)/g(x))$? Give reasons for your answer.
68. Suppose that $f(x)$ and $g(x)$ are polynomials in x . Can the graph of $f(x)/g(x)$ have an asymptote if $g(x)$ is never zero? Give reasons for your answer.
69. How many horizontal asymptotes can the graph of a given rational function have? Give reasons for your answer.
70. Find $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - \sqrt{x^2 - x})$.

Use the formal definitions of limits as $x \rightarrow \pm\infty$ to establish the limits in Exercises 71 and 72.

71. If f has the constant value $f(x) = k$, then $\lim_{x \rightarrow \infty} f(x) = k$.

72. If f has the constant value $f(x) = k$, then $\lim_{x \rightarrow -\infty} f(x) = k$.

Formal Definitions of One-Sided Limits

73. Given $\epsilon > 0$, find an interval $I = (5, 5 + \delta)$, $\delta > 0$, such that if x lies in I , then $\sqrt{x - 5} < \epsilon$. What limit is being verified and what is its value?

74. Given $\epsilon > 0$, find an interval $I = (4 - \delta, 4)$, $\delta > 0$, such that if x lies in I , then $\sqrt{4 - x} < \epsilon$. What limit is being verified and what is its value?

Use the definitions of right-hand and left-hand limits to prove the limit statements in Exercises 75 and 76.

$$75. \lim_{x \rightarrow 0^+} \frac{x}{|x|} = -1$$

$$76. \lim_{x \rightarrow 2^+} \frac{x - 2}{|x - 2|} = 1$$

77. **Greatest integer function** Find (a) $\lim_{x \rightarrow 400^+} \lfloor x \rfloor$ and (b) $\lim_{x \rightarrow 400^-} \lfloor x \rfloor$; then use limit definitions to verify your findings. (c) Based on your conclusions in parts (a) and (b), can anything be said about $\lim_{x \rightarrow 400} \lfloor x \rfloor$? Give reasons for your answers.

78. **One-sided limits** Let $f(x) = \begin{cases} x^2 \sin(1/x), & x < 0 \\ \sqrt{x}, & x > 0. \end{cases}$

Find (a) $\lim_{x \rightarrow 0^+} f(x)$ and (b) $\lim_{x \rightarrow 0^-} f(x)$; then use limit definitions to verify your findings. (c) Based on your conclusions in parts (a) and (b), can anything be said about $\lim_{x \rightarrow 0} f(x)$? Give reasons for your answer.

Grapher Explorations—“Seeing” Limits at Infinity

Sometimes a change of variable can change an unfamiliar expression into one whose limit we know how to find. For example,

$$\lim_{x \rightarrow \infty} \sin \frac{1}{x} = \lim_{\theta \rightarrow 0^+} \sin \theta \quad \text{Substitute } \theta = 1/x \\ = 0.$$

This suggests a creative way to “see” limits at infinity. Describe the procedure and use it to picture and determine limits in Exercises 79–84.

$$79. \lim_{x \rightarrow \pm\infty} x \sin \frac{1}{x}$$

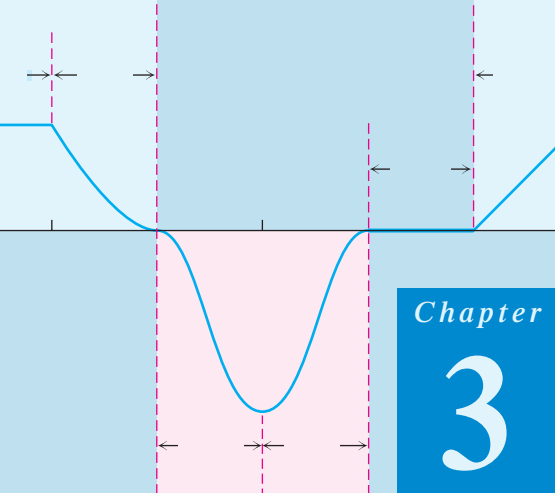
$$80. \lim_{x \rightarrow -\infty} \frac{\cos(1/x)}{1 + (1/x)}$$

$$81. \lim_{x \rightarrow \pm\infty} \frac{3x + 4}{2x - 5}$$

$$82. \lim_{x \rightarrow \infty} \left(\frac{1}{x} \right)^{1/x}$$

$$83. \lim_{x \rightarrow \pm\infty} \left(3 + \frac{2}{x} \right) \left(\cos \frac{1}{x} \right)$$

$$84. \lim_{x \rightarrow \infty} \left(\frac{3}{x^2} - \cos \frac{1}{x} \right) \left(1 + \sin \frac{1}{x} \right)$$



Chapter

3

DIFFERENTIATION

OVERVIEW In Chapter 2, we defined the slope of a curve at a point as the limit of secant slopes. This limit, called a derivative, measures the rate at which a function changes, and it is one of the most important ideas in calculus. Derivatives are used to calculate velocity and acceleration, to estimate the rate of spread of a disease, to set levels of production so as to maximize efficiency, to find the best dimensions of a cylindrical can, to find the age of a prehistoric artifact, and for many other applications. In this chapter, we develop techniques to calculate derivatives easily and learn how to use derivatives to approximate complicated functions.

3.1

The Derivative as a Function

HISTORICAL ESSAY

The Derivative

At the end of Chapter 2, we defined the slope of a curve $y = f(x)$ at the point where $x = x_0$ to be

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

We called this limit, when it existed, the derivative of f at x_0 . We now investigate the derivative as a *function* derived from f by considering the limit at each point of the domain of f .

DEFINITION Derivative Function

The **derivative** of the function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h},$$

provided the limit exists.

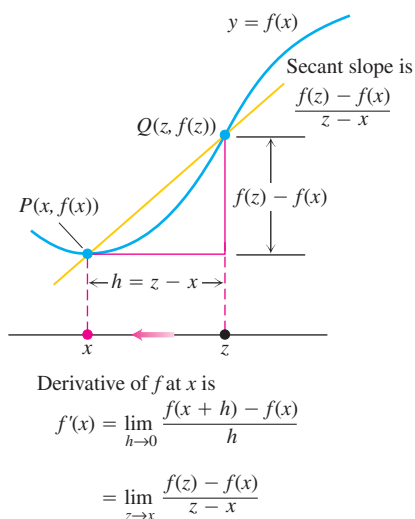


FIGURE 3.1 The way we write the difference quotient for the derivative of a function f depends on how we label the points involved.

We use the notation $f(x)$ rather than simply f in the definition to emphasize the independent variable x , which we are differentiating with respect to. The domain of f' is the set of points in the domain of f for which the limit exists, and the domain may be the same or smaller than the domain of f . If f' exists at a particular x , we say that f is **differentiable (has a derivative)** at x . If f' exists at every point in the domain of f , we call f **differentiable**.

If we write $z = x + h$, then $h = z - x$ and h approaches 0 if and only if z approaches x . Therefore, an equivalent definition of the derivative is as follows (see Figure 3.1).

Alternative Formula for the Derivative

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}.$$

Calculating Derivatives from the Definition

The process of calculating a derivative is called **differentiation**. To emphasize the idea that differentiation is an operation performed on a function $y = f(x)$, we use the notation

$$\frac{d}{dx} f(x)$$

as another way to denote the derivative $f'(x)$. Examples 2 and 3 of Section 2.7 illustrate the differentiation process for the functions $y = mx + b$ and $y = 1/x$. Example 2 shows that

$$\frac{d}{dx} (mx + b) = m.$$

For instance,

$$\frac{d}{dx} \left(\frac{3}{2}x - 4 \right) = \frac{3}{2}.$$

In Example 3, we see that

$$\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}.$$

Here are two more examples.

EXAMPLE 1 Applying the Definition

Differentiate $f(x) = \frac{x}{x-1}$.

Solution Here we have $f(x) = \frac{x}{x-1}$

and

$$\begin{aligned}
 f(x+h) &= \frac{(x+h)}{(x+h)-1}, \text{ so} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \frac{\frac{x+h}{x+h-1} - \frac{x}{x-1}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{(x+h)(x-1) - x(x+h-1)}{(x+h-1)(x-1)} \quad \frac{a}{b} - \frac{c}{d} = \frac{ad - cb}{bd} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{-h}{(x+h-1)(x-1)} \\
 &= \lim_{h \rightarrow 0} \frac{-1}{(x+h-1)(x-1)} = \frac{-1}{(x-1)^2}.
 \end{aligned}$$

EXAMPLE 2 Derivative of the Square Root Function

- (a) Find the derivative of $y = \sqrt{x}$ for $x > 0$.
 (b) Find the tangent line to the curve $y = \sqrt{x}$ at $x = 4$.

You will often need to know the derivative of $\sqrt{x}$ for $x > 0$:

$$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}.$$

Solution

- (a) We use the equivalent form to calculate f' :

$$\begin{aligned}
 f'(x) &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} \\
 &= \lim_{z \rightarrow x} \frac{\sqrt{z} - \sqrt{x}}{z - x} \\
 &= \lim_{z \rightarrow x} \frac{\sqrt{z} - \sqrt{x}}{(\sqrt{z} - \sqrt{x})(\sqrt{z} + \sqrt{x})} \\
 &= \lim_{z \rightarrow x} \frac{1}{\sqrt{z} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.
 \end{aligned}$$

- (b) The slope of the curve at $x = 4$ is

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

The tangent is the line through the point $(4, 2)$ with slope $1/4$ (Figure 3.2):

$$y = 2 + \frac{1}{4}(x - 4)$$

$$y = \frac{1}{4}x + 1.$$

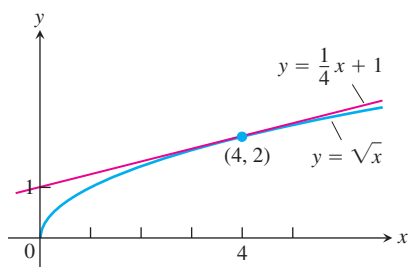


FIGURE 3.2 The curve $y = \sqrt{x}$ and its tangent at $(4, 2)$. The tangent's slope is found by evaluating the derivative at $x = 4$ (Example 2).

We consider the derivative of $y = \sqrt{x}$ when $x = 0$ in Example 6.

Notations

There are many ways to denote the derivative of a function $y = f(x)$, where the independent variable is x and the dependent variable is y . Some common alternative notations for the derivative are

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x).$$

The symbols d/dx and D indicate the operation of differentiation and are called **differentiation operators**. We read dy/dx as “the derivative of y with respect to x ,” and df/dx and $(d/dx)f(x)$ as “the derivative of f with respect to x .” The “prime” notations y' and f' come from notations that Newton used for derivatives. The d/dx notations are similar to those used by Leibniz. The symbol dy/dx should not be regarded as a ratio (until we introduce the idea of “differentials” in Section 3.8).

Be careful not to confuse the notation $D(f)$ as meaning the domain of the function f instead of the derivative function f' . The distinction should be clear from the context.

To indicate the value of a derivative at a specified number $x = a$, we use the notation

$$f'(a) = \left. \frac{dy}{dx} \right|_{x=a} = \left. \frac{df}{dx} \right|_{x=a} = \left. \frac{d}{dx} f(x) \right|_{x=a}.$$

For instance, in Example 2b we could write

$$f'(4) = \left. \frac{d}{dx} \sqrt{x} \right|_{x=4} = \left. \frac{1}{2\sqrt{x}} \right|_{x=4} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

To evaluate an expression, we sometimes use the right bracket $]$ in place of the vertical bar $|$.

Graphing the Derivative

We can often make a reasonable plot of the derivative of $y = f(x)$ by estimating the slopes on the graph of f . That is, we plot the points $(x, f'(x))$ in the xy -plane and connect them with a smooth curve, which represents $y = f'(x)$.

EXAMPLE 3 Graphing a Derivative

Graph the derivative of the function $y = f(x)$ in Figure 3.3a.

Solution We sketch the tangents to the graph of f at frequent intervals and use their slopes to estimate the values of $f'(x)$ at these points. We plot the corresponding $(x, f'(x))$ pairs and connect them with a smooth curve as sketched in Figure 3.3b. ■

What can we learn from the graph of $y = f'(x)$? At a glance we can see

1. where the rate of change of f is positive, negative, or zero;
2. the rough size of the growth rate at any x and its size in relation to the size of $f(x)$;
3. where the rate of change itself is increasing or decreasing.

Here's another example.

EXAMPLE 4 Concentration of Blood Sugar

On April 23, 1988, the human-powered airplane *Daedalus* flew a record-breaking 119 km from Crete to the island of Santorini in the Aegean Sea, southeast of mainland Greece. Dur-

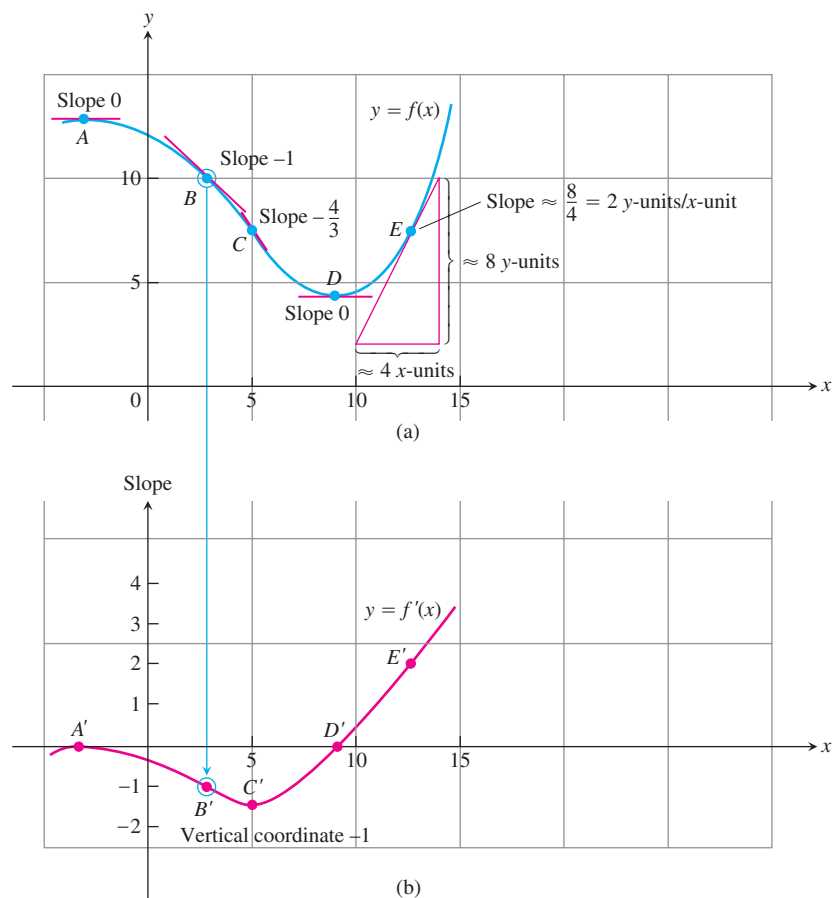


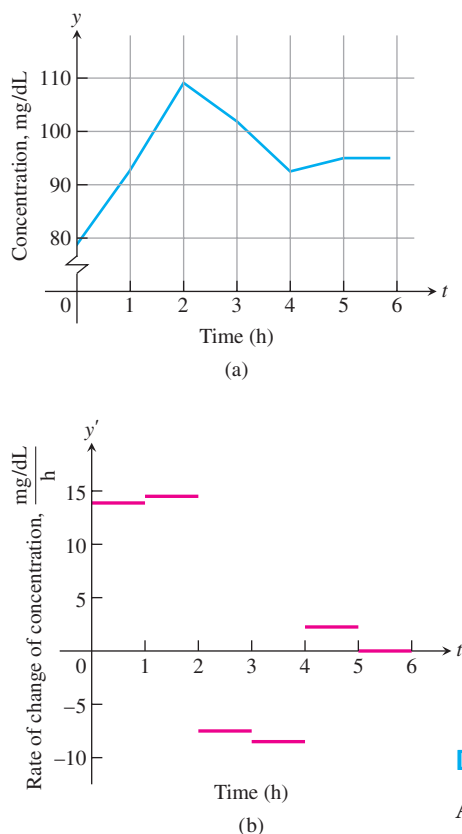
FIGURE 3.3 We made the graph of $y = f'(x)$ in (b) by plotting slopes from the graph of $y = f(x)$ in (a). The vertical coordinate of B' is the slope at B and so on. The graph of f' is a visual record of how the slope of f changes with x .

ing the 6-hour endurance tests before the flight, researchers monitored the prospective pilots' blood-sugar concentrations. The concentration graph for one of the athlete-pilots is shown in Figure 3.4a, where the concentration in milligrams/deciliter is plotted against time in hours.

The graph consists of line segments connecting data points. The constant slope of each segment gives an estimate of the derivative of the concentration between measurements. We calculated the slope of each segment from the coordinate grid and plotted the derivative as a step function in Figure 3.4b. To make the plot for the first hour, for instance, we observed that the concentration increased from about 79 mg/dL to 93 mg/dL. The net increase was $\Delta y = 93 - 79 = 14$ mg/dL. Dividing this by $\Delta t = 1$ hour gave the rate of change as

$$\frac{\Delta y}{\Delta t} = \frac{14}{1} = 14 \text{ mg/dL per hour.}$$

Notice that we can make no estimate of the concentration's rate of change at times $t = 1, 2, \dots, 5$, where the graph we have drawn for the concentration has a corner and no slope. The derivative step function is not defined at these times. ■



Daedalus's flight path on April 23, 1988

FIGURE 3.4 (a) Graph of the sugar concentration in the blood of a *Daedalus* pilot during a 6-hour preflight endurance test. (b) The derivative of the pilot's blood-sugar concentration shows how rapidly the concentration rose and fell during various portions of the test.

Differentiable on an Interval; One-Sided Derivatives

A function $y = f(x)$ is **differentiable** on an open interval (finite or infinite) if it has a derivative at each point of the interval. It is differentiable on a closed interval $[a, b]$ if it is differentiable on the interior (a, b) and if the limits

$$\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h} \quad \text{Right-hand derivative at } a$$

$$\lim_{h \rightarrow 0^-} \frac{f(b+h) - f(b)}{h} \quad \text{Left-hand derivative at } b$$

exist at the endpoints (Figure 3.5).

Right-hand and left-hand derivatives may be defined at any point of a function's domain. The usual relation between one-sided and two-sided limits holds for these derivatives. Because of Theorem 6, Section 2.4, a function has a derivative at a point if and only if it has left-hand and right-hand derivatives there, and these one-sided derivatives are equal.

EXAMPLE 5 $y = |x|$ Is Not Differentiable at the Origin

Show that the function $y = |x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$ but has no derivative at $x = 0$.

Solution To the right of the origin,

$$\frac{d}{dx}(|x|) = \frac{d}{dx}(x) = \frac{d}{dx}(1 \cdot x) = 1. \quad \frac{d}{dx}(mx + b) = m, |x| = x$$

To the left,

$$\frac{d}{dx}(|x|) = \frac{d}{dx}(-x) = \frac{d}{dx}(-1 \cdot x) = -1 \quad |x| = -x$$

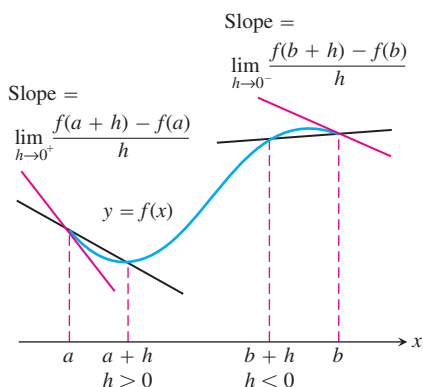


FIGURE 3.5 Derivatives at endpoints are one-sided limits.

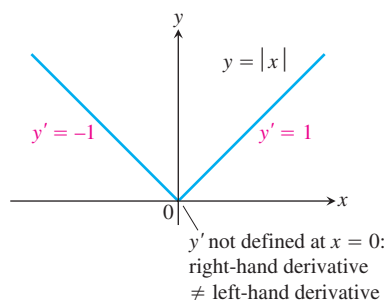


FIGURE 3.6 The function $y = |x|$ is not differentiable at the origin where the graph has a “corner.”

(Figure 3.6). There can be no derivative at the origin because the one-sided derivatives differ there:

$$\begin{aligned}\text{Right-hand derivative of } |x| \text{ at zero} &= \lim_{h \rightarrow 0^+} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{|h|}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{h}{h} \quad |h| = h \text{ when } h > 0. \\ &= \lim_{h \rightarrow 0^+} 1 = 1\end{aligned}$$

$$\begin{aligned}\text{Left-hand derivative of } |x| \text{ at zero} &= \lim_{h \rightarrow 0^-} \frac{|0 + h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{|h|}{h} \\ &= \lim_{h \rightarrow 0^-} \frac{-h}{h} \quad |h| = -h \text{ when } h < 0. \\ &= \lim_{h \rightarrow 0^-} -1 = -1.\end{aligned}$$

EXAMPLE 6 $y = \sqrt{x}$ Is Not Differentiable at $x = 0$

In Example 2 we found that for $x > 0$,

$$\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}.$$

We apply the definition to examine if the derivative exists at $x = 0$:

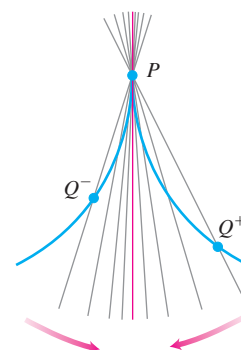
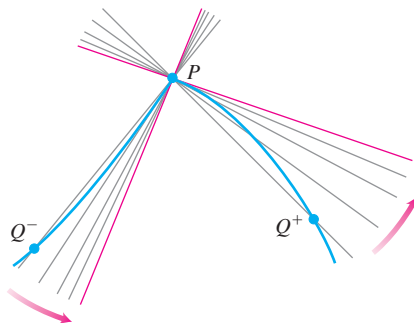
$$\lim_{h \rightarrow 0^+} \frac{\sqrt{0 + h} - \sqrt{0}}{h} = \lim_{h \rightarrow 0^+} \frac{1}{\sqrt{h}} = \infty.$$

Since the (right-hand) limit is not finite, there is no derivative at $x = 0$. Since the slopes of the secant lines joining the origin to the points $(h, \sqrt{h})$ on a graph of $y = \sqrt{x}$ approach ∞ , the graph has a *vertical tangent* at the origin.

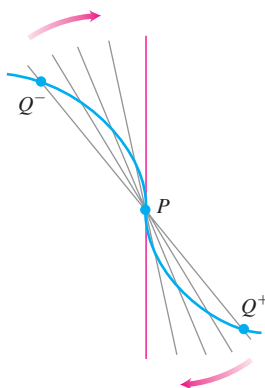
When Does a Function Not Have a Derivative at a Point?

A function has a derivative at a point x_0 if the slopes of the secant lines through $P(x_0, f(x_0))$ and a nearby point Q on the graph approach a limit as Q approaches P . Whenever the secants fail to take up a limiting position or become vertical as Q approaches P , the derivative does not exist. Thus differentiability is a “smoothness” condition on the graph of f . A function whose graph is otherwise smooth will fail to have a derivative at a point for several reasons, such as at points where the graph has

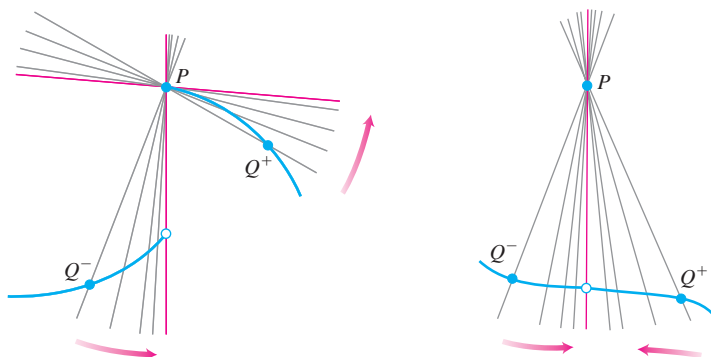
1. a *corner*, where the one-sided derivatives differ.
2. a *cusp*, where the slope of PQ approaches ∞ from one side and $-\infty$ from the other.



3. a *vertical tangent*, where the slope of PQ approaches ∞ from both sides or approaches $-\infty$ from both sides (here, $-\infty$).



4. a *discontinuity*.



Differentiable Functions Are Continuous

A function is continuous at every point where it has a derivative.

THEOREM 1 Differentiability Implies Continuity

If f has a derivative at $x = c$, then f is continuous at $x = c$.

Proof Given that $f'(c)$ exists, we must show that $\lim_{x \rightarrow c} f(x) = f(c)$, or equivalently, that $\lim_{h \rightarrow 0} f(c + h) = f(c)$. If $h \neq 0$, then

$$\begin{aligned} f(c + h) &= f(c) + (f(c + h) - f(c)) \\ &= f(c) + \frac{f(c + h) - f(c)}{h} \cdot h. \end{aligned}$$

Now take limits as $h \rightarrow 0$. By Theorem 1 of Section 2.2,

$$\begin{aligned}\lim_{h \rightarrow 0} f(c + h) &= \lim_{h \rightarrow 0} f(c) + \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} \cdot \lim_{h \rightarrow 0} h \\ &= f(c) + f'(c) \cdot 0 \\ &= f(c) + 0 \\ &= f(c).\end{aligned}$$

Similar arguments with one-sided limits show that if f has a derivative from one side (right or left) at $x = c$ then f is continuous from that side at $x = c$.

Theorem 1 on page 154 says that if a function has a discontinuity at a point (for instance, a jump discontinuity), then it cannot be differentiable there. The greatest integer function $y = \lfloor x \rfloor = \text{int } x$ fails to be differentiable at every integer $x = n$ (Example 4, Section 2.6).

CAUTION The converse of Theorem 1 is false. A function need not have a derivative at a point where it is continuous, as we saw in Example 5.

The Intermediate Value Property of Derivatives

Not every function can be some function's derivative, as we see from the following theorem.

THEOREM 2

If a and b are any two points in an interval on which f is differentiable, then f' takes on every value between $f'(a)$ and $f'(b)$.

Theorem 2 (which we will not prove) says that a function cannot be a derivative on an interval unless it has the Intermediate Value Property there. For example, the unit step function in Figure 3.7 cannot be the derivative of any real-valued function on the real line. In Chapter 5 we will see that every continuous function is a derivative of some function.

In Section 4.4, we invoke Theorem 2 to analyze what happens at a point on the graph of a twice-differentiable function where it changes its “bending” behavior.

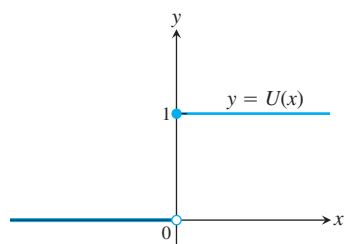


FIGURE 3.7 The unit step function does not have the Intermediate Value Property and cannot be the derivative of a function on the real line.

EXERCISES 3.1

Finding Derivative Functions and Values

Using the definition, calculate the derivatives of the functions in Exercises 1–6. Then find the values of the derivatives as specified.

1. $f(x) = 4 - x^2$; $f'(-3), f'(0), f'(1)$

2. $F(x) = (x - 1)^2 + 1$; $F'(-1), F'(0), F'(2)$

3. $g(t) = \frac{1}{t^2}$; $g'(-1), g'(2), g'(\sqrt{3})$

4. $k(z) = \frac{1-z}{2z}$; $k'(-1), k'(1), k'(\sqrt{2})$

5. $p(\theta) = \sqrt{3\theta}$; $p'(1), p'(3), p'(2/3)$

6. $r(s) = \sqrt{2s+1}$; $r'(0), r'(1), r'(1/2)$

In Exercises 7–12, find the indicated derivatives.

7. $\frac{dy}{dx}$ if $y = 2x^3$

8. $\frac{dr}{ds}$ if $r = \frac{s^3}{2} + 1$

9. $\frac{ds}{dt}$ if $s = \frac{t}{2t+1}$
10. $\frac{dv}{dt}$ if $v = t - \frac{1}{t}$
11. $\frac{dp}{dq}$ if $p = \frac{1}{\sqrt{q+1}}$
12. $\frac{dz}{dw}$ if $z = \frac{1}{\sqrt{3w-2}}$

Slopes and Tangent Lines

In Exercises 13–16, differentiate the functions and find the slope of the tangent line at the given value of the independent variable.

13. $f(x) = x + \frac{9}{x}$, $x = -3$

14. $k(x) = \frac{1}{2+x}$, $x = 2$

15. $s = t^3 - t^2$, $t = -1$

16. $y = (x+1)^3$, $x = -2$

In Exercises 17–18, differentiate the functions. Then find an equation of the tangent line at the indicated point on the graph of the function.

17. $y = f(x) = \frac{8}{\sqrt{x-2}}$, $(x, y) = (6, 4)$

18. $w = g(z) = 1 + \sqrt{4-z}$, $(z, w) = (3, 2)$

In Exercises 19–22, find the values of the derivatives.

19. $\left. \frac{ds}{dt} \right|_{t=-1}$ if $s = 1 - 3t^2$

20. $\left. \frac{dy}{dx} \right|_{x=\sqrt{3}}$ if $y = 1 - \frac{1}{x}$

21. $\left. \frac{dr}{d\theta} \right|_{\theta=0}$ if $r = \frac{2}{\sqrt{4-\theta}}$

22. $\left. \frac{dw}{dz} \right|_{z=4}$ if $w = z + \sqrt{z}$

Using the Alternative Formula for Derivatives

Use the formula

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

to find the derivative of the functions in Exercises 23–26.

23. $f(x) = \frac{1}{x+2}$

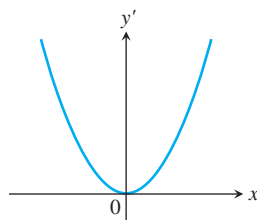
24. $f(x) = \frac{1}{(x-1)^2}$

25. $g(x) = \frac{x}{x-1}$

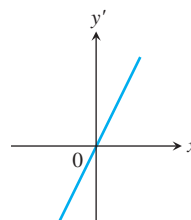
26. $g(x) = 1 + \sqrt{x}$

Graphs

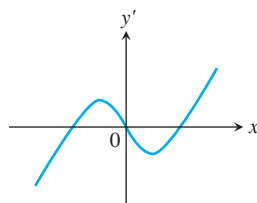
Match the functions graphed in Exercises 27–30 with the derivatives graphed in the accompanying figures (a)–(d).



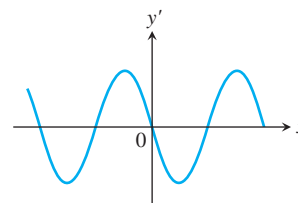
(a)



(b)

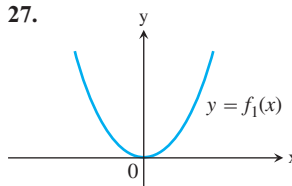


(c)

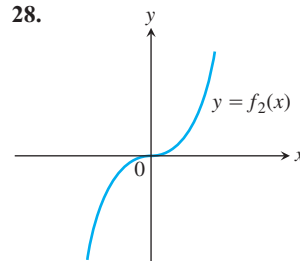


(d)

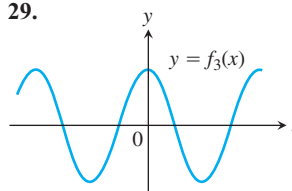
27.



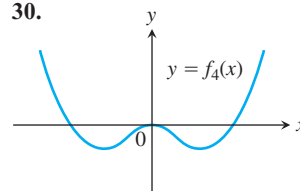
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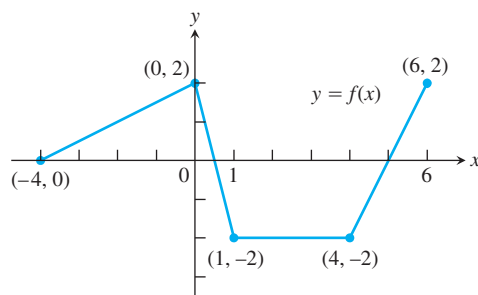
29.



30.



31. a. The graph in the accompanying figure is made of line segments joined end to end. At which points of the interval $[-4, 6]$ is f' not defined? Give reasons for your answer.

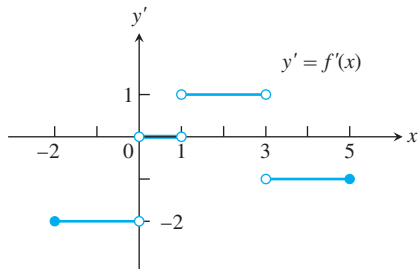


- b. Graph the derivative of f .

The graph should show a step function.

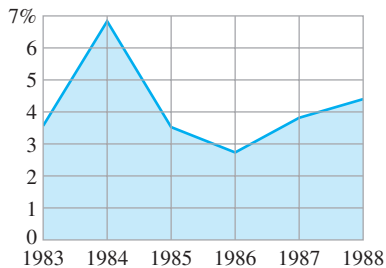
32. Recovering a function from its derivative

- a. Use the following information to graph the function f over the closed interval $[-2, 5]$.
- The graph of f is made of closed line segments joined end to end.
 - The graph starts at the point $(-2, 3)$.
 - The derivative of f is the step function in the figure shown here.

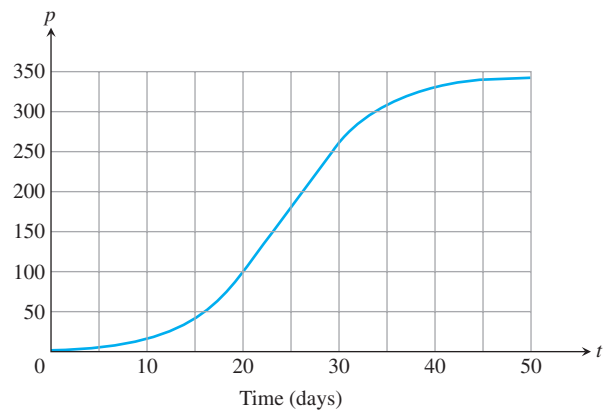


- b. Repeat part (a) assuming that the graph starts at $(-2, 0)$ instead of $(-2, 3)$.

33. **Growth in the economy** The graph in the accompanying figure shows the average annual percentage change $y = f(t)$ in the U.S. gross national product (GNP) for the years 1983–1988. Graph dy/dt (where defined). (Source: *Statistical Abstracts of the United States*, 110th Edition, U.S. Department of Commerce, p. 427.)



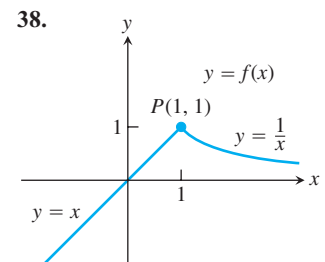
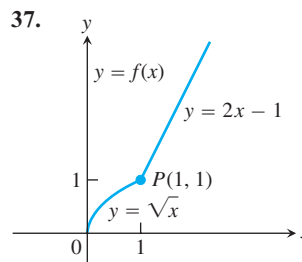
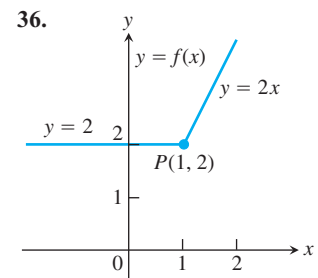
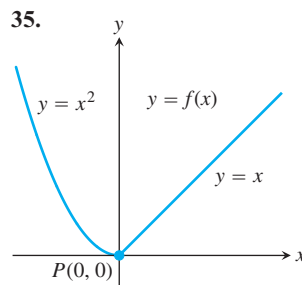
34. **Fruit flies** (Continuation of Example 3, Section 2.1.) Populations starting out in closed environments grow slowly at first, when there are relatively few members, then more rapidly as the number of reproducing individuals increases and resources are still abundant, then slowly again as the population reaches the carrying capacity of the environment.
- a. Use the graphical technique of Example 3 to graph the derivative of the fruit fly population introduced in Section 2.1. The graph of the population is reproduced here.



- b. During what days does the population seem to be increasing fastest? Slowest?

One-Sided Derivatives

Compare the right-hand and left-hand derivatives to show that the functions in Exercises 35–38 are not differentiable at the point P .



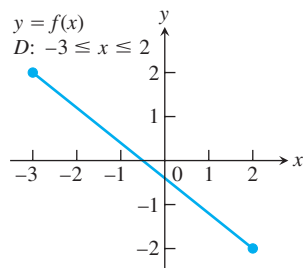
Differentiability and Continuity on an Interval

Each figure in Exercises 39–44 shows the graph of a function over a closed interval D . At what domain points does the function appear to be

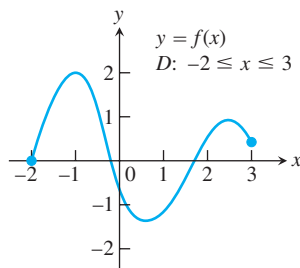
- differentiable?
- continuous but not differentiable?
- neither continuous nor differentiable?

Give reasons for your answers.

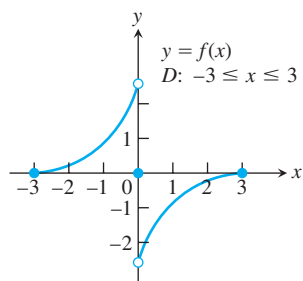
39.



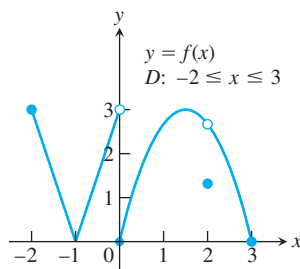
40.



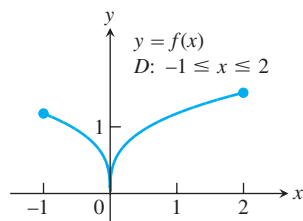
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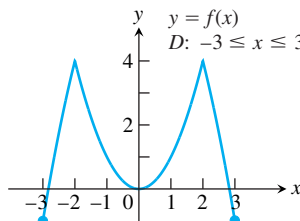
42.



43.



44.



Theory and Examples

In Exercises 45–48,

- Find the derivative $f'(x)$ of the given function $y = f(x)$.
 - Graph $y = f(x)$ and $y = f'(x)$ side by side using separate sets of coordinate axes, and answer the following questions.
 - For what values of x , if any, is f' positive? Zero? Negative?
 - Over what intervals of x -values, if any, does the function $y = f(x)$ increase as x increases? Decrease as x increases? How is this related to what you found in part (c)? (We will say more about this relationship in Chapter 4.)
- $y = -x^2$
 - $y = -1/x$
 - $y = x^3/3$
 - $y = x^4/4$
 - Does the curve $y = x^3$ ever have a negative slope? If so, where? Give reasons for your answer.
 - Does the curve $y = 2\sqrt{x}$ have any horizontal tangents? If so, where? Give reasons for your answer.

- Tangent to a parabola** Does the parabola $y = 2x^2 - 13x + 5$ have a tangent whose slope is -1 ? If so, find an equation for the line and the point of tangency. If not, why not?
- Tangent to $y = \sqrt{x}$** Does any tangent to the curve $y = \sqrt{x}$ cross the x -axis at $x = -1$? If so, find an equation for the line and the point of tangency. If not, why not?
- Greatest integer in x** Does any function differentiable on $(-\infty, \infty)$ have $y = \text{int } x$, the greatest integer in x (see Figure 2.55), as its derivative? Give reasons for your answer.
- Derivative of $y = |x|$** Graph the derivative of $f(x) = |x|$. Then graph $y = (|x| - 0)/(x - 0) = |x|/x$. What can you conclude?
- Derivative of $-f$** Does knowing that a function $f(x)$ is differentiable at $x = x_0$ tell you anything about the differentiability of the function $-f$ at $x = x_0$? Give reasons for your answer.
- Derivative of multiples** Does knowing that a function $g(t)$ is differentiable at $t = 7$ tell you anything about the differentiability of the function $3g$ at $t = 7$? Give reasons for your answer.
- Limit of a quotient** Suppose that functions $g(t)$ and $h(t)$ are defined for all values of t and $g(0) = h(0) = 0$. Can $\lim_{t \rightarrow 0} (g(t)/h(t))$ exist? If it does exist, must it equal zero? Give reasons for your answers.
- a. Let $f(x)$ be a function satisfying $|f(x)| \leq x^2$ for $-1 \leq x \leq 1$. Show that f is differentiable at $x = 0$ and find $f'(0)$.

b. Show that

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is differentiable at $x = 0$ and find $f'(0)$.

- T** Graph $y = 1/(2\sqrt{x})$ in a window that has $0 \leq x \leq 2$. Then, on the same screen, graph

$$y = \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

for $h = 1, 0.5, 0.1$. Then try $h = -1, -0.5, -0.1$. Explain what is going on.

- T** Graph $y = 3x^2$ in a window that has $-2 \leq x \leq 2, 0 \leq y \leq 3$. Then, on the same screen, graph

$$y = \frac{(x+h)^3 - x^3}{h}$$

for $h = 2, 1, 0.2$. Then try $h = -2, -1, -0.2$. Explain what is going on.

- T** **Weierstrass's nowhere differentiable continuous function** The sum of the first eight terms of the Weierstrass function $f(x) = \sum_{n=0}^{\infty} (2/3)^n \cos(9^n \pi x)$ is

$$g(x) = \cos(\pi x) + (2/3)^1 \cos(9\pi x) + (2/3)^2 \cos(9^2 \pi x) + (2/3)^3 \cos(9^3 \pi x) + \cdots + (2/3)^7 \cos(9^7 \pi x).$$

Graph this sum. Zoom in several times. How wiggly and bumpy is this graph? Specify a viewing window in which the displayed portion of the graph is smooth.

COMPUTER EXPLORATIONS

Use a CAS to perform the following steps for the functions in Exercises 62–67.

- a. Plot $y = f(x)$ to see that function's global behavior.
- b. Define the difference quotient q at a general point x , with general step size h .
- c. Take the limit as $h \rightarrow 0$. What formula does this give?
- d. Substitute the value $x = x_0$ and plot the function $y = f(x)$ together with its tangent line at that point.
- e. Substitute various values for x larger and smaller than x_0 into the formula obtained in part (c). Do the numbers make sense with your picture?

- f. Graph the formula obtained in part (c). What does it mean when its values are negative? Zero? Positive? Does this make sense with your plot from part (a)? Give reasons for your answer.

62. $f(x) = x^3 + x^2 - x, \quad x_0 = 1$

63. $f(x) = x^{1/3} + x^{2/3}, \quad x_0 = 1$

64. $f(x) = \frac{4x}{x^2 + 1}, \quad x_0 = 2$

65. $f(x) = \frac{x - 1}{3x^2 + 1}, \quad x_0 = -1$

66. $f(x) = \sin 2x, \quad x_0 = \pi/2$

67. $f(x) = x^2 \cos x, \quad x_0 = \pi/4$

3.2

Differentiation Rules

This section introduces a few rules that allow us to differentiate a great variety of functions. By proving these rules here, we can differentiate functions without having to apply the definition of the derivative each time.

Powers, Multiples, Sums, and Differences

The first rule of differentiation is that the derivative of every constant function is zero.

RULE 1 Derivative of a Constant Function

If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

EXAMPLE 1

If f has the constant value $f(x) = 8$, then

$$\frac{df}{dx} = \frac{d}{dx}(8) = 0.$$

Similarly,

$$\frac{d}{dx}\left(-\frac{\pi}{2}\right) = 0 \quad \text{and} \quad \frac{d}{dx}\left(\sqrt{3}\right) = 0. \quad \blacksquare$$

Proof of Rule 1 We apply the definition of derivative to $f(x) = c$, the function whose outputs have the constant value c (Figure 3.8). At every value of x , we find that

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0. \quad \blacksquare$$

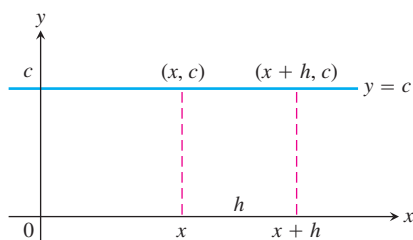


FIGURE 3.8 The rule $(d/dx)(c) = 0$ is another way to say that the values of constant functions never change and that the slope of a horizontal line is zero at every point.

The second rule tells how to differentiate x^n if n is a positive integer.

RULE 2 Power Rule for Positive Integers

If n is a positive integer, then

$$\frac{d}{dx} x^n = nx^{n-1}.$$

To apply the Power Rule, we subtract 1 from the original exponent (n) and multiply the result by n .

EXAMPLE 2 Interpreting Rule 2

f	x	x^2	x^3	x^4	$\dots$
f'	1	$2x$	$3x^2$	$4x^3$	$\dots$

HISTORICAL BIOGRAPHY

Richard Courant
(1888–1972)

First Proof of Rule 2 The formula

$$z^n - x^n = (z - x)(z^{n-1} + z^{n-2}x + \dots + zx^{n-2} + x^{n-1})$$

can be verified by multiplying out the right-hand side. Then from the alternative form for the definition of the derivative,

$$\begin{aligned} f'(x) &= \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x} = \lim_{z \rightarrow x} \frac{z^n - x^n}{z - x} \\ &= \lim_{z \rightarrow x} (z^{n-1} + z^{n-2}x + \dots + zx^{n-2} + x^{n-1}) \\ &= nx^{n-1} \end{aligned}$$

Second Proof of Rule 2 If $f(x) = x^n$, then $f(x + h) = (x + h)^n$. Since n is a positive integer, we can expand $(x + h)^n$ by the Binomial Theorem to get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x + h)^n - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left[x^n + nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + nxh^{n-1} + h^n \right] - x^n}{h} \\ &= \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \frac{n(n-1)}{2}x^{n-2}h^2 + \dots + nxh^{n-1} + h^n}{h} \\ &= \lim_{h \rightarrow 0} \left[nx^{n-1} + \frac{n(n-1)}{2}x^{n-2}h + \dots + nxh^{n-2} + h^{n-1} \right] \\ &= nx^{n-1} \end{aligned}$$

The third rule says that when a differentiable function is multiplied by a constant, its derivative is multiplied by the same constant.

RULE 3 Constant Multiple Rule

If u is a differentiable function of x , and c is a constant, then

$$\frac{d}{dx}(cu) = c \frac{du}{dx}.$$

In particular, if n is a positive integer, then

$$\frac{d}{dx}(cx^n) = cnx^{n-1}.$$

EXAMPLE 3

(a) The derivative formula

$$\frac{d}{dx}(3x^2) = 3 \cdot 2x = 6x$$

says that if we rescale the graph of $y = x^2$ by multiplying each y -coordinate by 3, then we multiply the slope at each point by 3 (Figure 3.9).

(b) A useful special case

The derivative of the negative of a differentiable function u is the negative of the function's derivative. Rule 3 with $c = -1$ gives

$$\frac{d}{dx}(-u) = \frac{d}{dx}(-1 \cdot u) = -1 \cdot \frac{d}{dx}(u) = -\frac{du}{dx}.$$

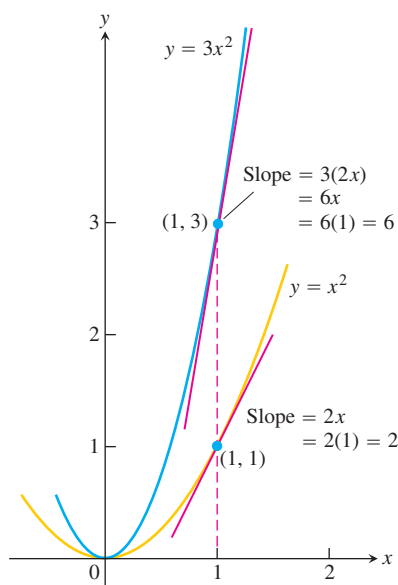


FIGURE 3.9 The graphs of $y = x^2$ and $y = 3x^2$. Tripling the y -coordinates triples the slope (Example 3).

Proof of Rule 3

$$\begin{aligned} \frac{d}{dx}cu &= \lim_{h \rightarrow 0} \frac{cu(x+h) - cu(x)}{h} && \text{Derivative definition with } f(x) = cu(x) \\ &= c \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h} && \text{Limit property} \\ &= c \frac{du}{dx} && u \text{ is differentiable.} \end{aligned}$$

The next rule says that the derivative of the sum of two differentiable functions is the sum of their derivatives.

Denoting Functions by u and v

The functions we are working with when we need a differentiation formula are likely to be denoted by letters like f and g . When we apply the formula, we do not want to find it using these same letters in some other way. To guard against this problem, we denote the functions in differentiation rules by letters like u and v that are not likely to be already in use.

RULE 4 Derivative Sum Rule

If u and v are differentiable functions of x , then their sum $u + v$ is differentiable at every point where u and v are both differentiable. At such points,

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}.$$

EXAMPLE 4 Derivative of a Sum

$$\begin{aligned}
 y &= x^4 + 12x \\
 \frac{dy}{dx} &= \frac{d}{dx}(x^4) + \frac{d}{dx}(12x) \\
 &= 4x^3 + 12
 \end{aligned}$$

Proof of Rule 4 We apply the definition of derivative to $f(x) = u(x) + v(x)$:

$$\begin{aligned}
 \frac{d}{dx}[u(x) + v(x)] &= \lim_{h \rightarrow 0} \frac{[u(x+h) + v(x+h)] - [u(x) + v(x)]}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{u(x+h) - u(x)}{h} + \frac{v(x+h) - v(x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h} + \lim_{h \rightarrow 0} \frac{v(x+h) - v(x)}{h} = \frac{du}{dx} + \frac{dv}{dx}.
 \end{aligned}$$

Combining the Sum Rule with the Constant Multiple Rule gives the **Difference Rule**, which says that the derivative of a *difference* of differentiable functions is the difference of their derivatives.

$$\frac{d}{dx}(u - v) = \frac{d}{dx}[u + (-1)v] = \frac{du}{dx} + (-1)\frac{dv}{dx} = \frac{du}{dx} - \frac{dv}{dx}$$

The Sum Rule also extends to sums of more than two functions, as long as there are only finitely many functions in the sum. If $u_1, u_2, \dots, u_n$ are differentiable at x , then so is $u_1 + u_2 + \dots + u_n$, and

$$\frac{d}{dx}(u_1 + u_2 + \dots + u_n) = \frac{du_1}{dx} + \frac{du_2}{dx} + \dots + \frac{du_n}{dx}.$$

EXAMPLE 5 Derivative of a Polynomial

$$\begin{aligned}
 y &= x^3 + \frac{4}{3}x^2 - 5x + 1 \\
 \frac{dy}{dx} &= \frac{d}{dx}x^3 + \frac{d}{dx}\left(\frac{4}{3}x^2\right) - \frac{d}{dx}(5x) + \frac{d}{dx}(1) \\
 &= 3x^2 + \frac{4}{3} \cdot 2x - 5 + 0 \\
 &= 3x^2 + \frac{8}{3}x - 5
 \end{aligned}$$

Notice that we can differentiate any polynomial term by term, the way we differentiated the polynomial in Example 5. All polynomials are differentiable everywhere.

Proof of the Sum Rule for Sums of More Than Two Functions We prove the statement

$$\frac{d}{dx}(u_1 + u_2 + \dots + u_n) = \frac{du_1}{dx} + \frac{du_2}{dx} + \dots + \frac{du_n}{dx}$$

by mathematical induction (see Appendix 1). The statement is true for $n = 2$, as was just proved. This is Step 1 of the induction proof.

Step 2 is to show that if the statement is true for any positive integer $n = k$, where $k \geq n_0 = 2$, then it is also true for $n = k + 1$. So suppose that

$$\frac{d}{dx}(u_1 + u_2 + \cdots + u_k) = \frac{du_1}{dx} + \frac{du_2}{dx} + \cdots + \frac{du_k}{dx}. \quad (1)$$

Then

$$\begin{aligned} & \frac{d}{dx} \underbrace{(u_1 + u_2 + \cdots + u_k)}_{\text{Call the function defined by this sum } u.} + \underbrace{u_{k+1}}_{\text{Call this function } v.} \\ &= \frac{d}{dx}(u_1 + u_2 + \cdots + u_k) + \frac{du_{k+1}}{dx} \quad \text{Rule 4 for } \frac{d}{dx}(u + v) \\ &= \frac{du_1}{dx} + \frac{du_2}{dx} + \cdots + \frac{du_k}{dx} + \frac{du_{k+1}}{dx}. \quad \text{Eq. (1)} \end{aligned}$$

With these steps verified, the mathematical induction principle now guarantees the Sum Rule for every integer $n \geq 2$. ■

EXAMPLE 6 Finding Horizontal Tangents

Does the curve $y = x^4 - 2x^2 + 2$ have any horizontal tangents? If so, where?

Solution The horizontal tangents, if any, occur where the slope dy/dx is zero. We have,

$$\frac{dy}{dx} = \frac{d}{dx}(x^4 - 2x^2 + 2) = 4x^3 - 4x.$$

Now solve the equation $\frac{dy}{dx} = 0$ for x :

$$\begin{aligned} 4x^3 - 4x &= 0 \\ 4x(x^2 - 1) &= 0 \\ x &= 0, 1, -1. \end{aligned}$$

The curve $y = x^4 - 2x^2 + 2$ has horizontal tangents at $x = 0, 1$, and -1 . The corresponding points on the curve are $(0, 2)$, $(1, 1)$ and $(-1, 1)$. See Figure 3.10. ■

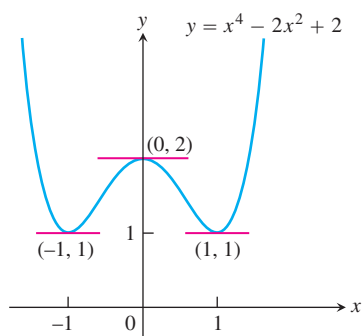


FIGURE 3.10 The curve $y = x^4 - 2x^2 + 2$ and its horizontal tangents (Example 6).

Products and Quotients

While the derivative of the sum of two functions is the sum of their derivatives, the derivative of the product of two functions is *not* the product of their derivatives. For instance,

$$\frac{d}{dx}(x \cdot x) = \frac{d}{dx}(x^2) = 2x, \quad \text{while} \quad \frac{d}{dx}(x) \cdot \frac{d}{dx}(x) = 1 \cdot 1 = 1.$$

The derivative of a product of two functions is the sum of *two* products, as we now explain.

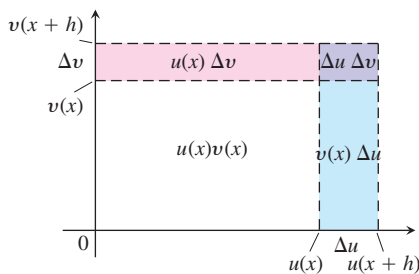
RULE 5 Derivative Product Rule

If u and v are differentiable at x , then so is their product uv , and

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

Picturing the Product Rule

If $u(x)$ and $v(x)$ are positive and increase when x increases, and if $h > 0$,



then the total shaded area in the picture is

$$\begin{aligned} & u(x+h)v(x+h) - u(x)v(x) \\ &= u(x+h)\Delta v + v(x+h)\Delta u - \Delta u\Delta v. \end{aligned}$$

Dividing both sides of this equation by h gives

$$\begin{aligned} & \frac{u(x+h)v(x+h) - u(x)v(x)}{h} \\ &= u(x+h)\frac{\Delta v}{h} + v(x+h)\frac{\Delta u}{h} - \Delta u\frac{\Delta v}{h}. \end{aligned}$$

As $h \rightarrow 0^+$,

$$\Delta u \cdot \frac{\Delta v}{h} \rightarrow 0 \cdot \frac{dv}{dx} = 0,$$

leaving

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

The derivative of the product uv is u times the derivative of v plus v times the derivative of u . In *prime notation*, $(uv)' = uv' + vu'$. In *function notation*,

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x).$$

EXAMPLE 7 Using the Product Rule

Find the derivative of

$$y = \frac{1}{x} \left(x^2 + \frac{1}{x} \right).$$

Solution We apply the Product Rule with $u = 1/x$ and $v = x^2 + (1/x)$:

$$\begin{aligned} \frac{d}{dx} \left[\frac{1}{x} \left(x^2 + \frac{1}{x} \right) \right] &= \frac{1}{x} \left(2x - \frac{1}{x^2} \right) + \left(x^2 + \frac{1}{x} \right) \left(-\frac{1}{x^2} \right) \\ &= 2 - \frac{1}{x^3} - 1 - \frac{1}{x^3} \\ &= 1 - \frac{2}{x^3}. \end{aligned}$$

$$\begin{aligned} \frac{d}{dx}(uv) &= u\frac{dv}{dx} + v\frac{du}{dx}, \text{ and} \\ \frac{d}{dx} \left(\frac{1}{x} \right) &= -\frac{1}{x^2} \text{ by} \\ &\text{Example 3, Section 2.7.} \end{aligned}$$

Proof of Rule 5

$$\frac{d}{dx}(uv) = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}$$

To change this fraction into an equivalent one that contains difference quotients for the derivatives of u and v , we subtract and add $u(x+h)v(x)$ in the numerator:

$$\begin{aligned} \frac{d}{dx}(uv) &= \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x+h)v(x) + u(x+h)v(x) - u(x)v(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[u(x+h) \frac{v(x+h) - v(x)}{h} + v(x) \frac{u(x+h) - u(x)}{h} \right] \\ &= \lim_{h \rightarrow 0} u(x+h) \cdot \lim_{h \rightarrow 0} \frac{v(x+h) - v(x)}{h} + v(x) \cdot \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h}. \end{aligned}$$

As h approaches zero, $u(x+h)$ approaches $u(x)$ because u , being differentiable at x , is continuous at x . The two fractions approach the values of dv/dx at x and du/dx at x . In short,

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

In the following example, we have only numerical values with which to work.

EXAMPLE 8 Derivative from Numerical Values

Let $y = uv$ be the product of the functions u and v . Find $y'(2)$ if

$$u(2) = 3, \quad u'(2) = -4, \quad v(2) = 1, \quad \text{and} \quad v'(2) = 2.$$

Solution From the Product Rule, in the form

$$y' = (uv)' = uv' + vu',$$

we have

$$\begin{aligned} y'(2) &= u(2)v'(2) + v(2)u'(2) \\ &= (3)(2) + (1)(-4) = 6 - 4 = 2. \end{aligned}$$

EXAMPLE 9 Differentiating a Product in Two Ways

Find the derivative of $y = (x^2 + 1)(x^3 + 3)$.

Solution

(a) From the Product Rule with $u = x^2 + 1$ and $v = x^3 + 3$, we find

$$\begin{aligned} \frac{d}{dx} [(x^2 + 1)(x^3 + 3)] &= (x^2 + 1)(3x^2) + (x^3 + 3)(2x) \\ &= 3x^4 + 3x^2 + 2x^4 + 6x \\ &= 5x^4 + 3x^2 + 6x. \end{aligned}$$

(b) This particular product can be differentiated as well (perhaps better) by multiplying out the original expression for y and differentiating the resulting polynomial:

$$\begin{aligned} y &= (x^2 + 1)(x^3 + 3) = x^5 + x^3 + 3x^2 + 3 \\ \frac{dy}{dx} &= 5x^4 + 3x^2 + 6x. \end{aligned}$$

This is in agreement with our first calculation.

Just as the derivative of the product of two differentiable functions is not the product of their derivatives, the derivative of the quotient of two functions is not the quotient of their derivatives. What happens instead is the Quotient Rule.

RULE 6 Derivative Quotient Rule

If u and v are differentiable at x and if $v(x) \neq 0$, then the quotient u/v is differentiable at x , and

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

In function notation,

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}.$$

EXAMPLE 10 Using the Quotient Rule

Find the derivative of

$$y = \frac{t^2 - 1}{t^2 + 1}.$$

Solution

We apply the Quotient Rule with $u = t^2 - 1$ and $v = t^2 + 1$:

$$\begin{aligned}\frac{dy}{dt} &= \frac{(t^2 + 1) \cdot 2t - (t^2 - 1) \cdot 2t}{(t^2 + 1)^2} & \frac{d}{dt} \left(\frac{u}{v} \right) &= \frac{v(du/dt) - u(dv/dt)}{v^2} \\ &= \frac{2t^3 + 2t - 2t^3 + 2t}{(t^2 + 1)^2} \\ &= \frac{4t}{(t^2 + 1)^2}.\end{aligned}$$

Proof of Rule 6

$$\begin{aligned}\frac{d}{dx} \left(\frac{u}{v} \right) &= \lim_{h \rightarrow 0} \frac{\frac{u(x+h)}{v(x+h)} - \frac{u(x)}{v(x)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{v(x)u(x+h) - u(x)v(x+h)}{hv(x+h)v(x)}\end{aligned}$$

To change the last fraction into an equivalent one that contains the difference quotients for the derivatives of u and v , we subtract and add $v(x)u(x)$ in the numerator. We then get

$$\begin{aligned}\frac{d}{dx} \left(\frac{u}{v} \right) &= \lim_{h \rightarrow 0} \frac{v(x)u(x+h) - v(x)u(x) + v(x)u(x) - u(x)v(x+h)}{hv(x+h)v(x)} \\ &= \lim_{h \rightarrow 0} \frac{v(x) \frac{u(x+h) - u(x)}{h} - u(x) \frac{v(x+h) - v(x)}{h}}{v(x+h)v(x)}.\end{aligned}$$

Taking the limit in the numerator and denominator now gives the Quotient Rule. ■

Negative Integer Powers of x

The Power Rule for negative integers is the same as the rule for positive integers.

RULE 7 Power Rule for Negative Integers

If n is a negative integer and $x \neq 0$, then

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

EXAMPLE 11

$$(a) \quad \frac{d}{dx} \left(\frac{1}{x} \right) = \frac{d}{dx}(x^{-1}) = (-1)x^{-2} = -\frac{1}{x^2}$$

Agrees with Example 3, Section 2.7

$$(b) \quad \frac{d}{dx} \left(\frac{4}{x^3} \right) = 4 \frac{d}{dx}(x^{-3}) = 4(-3)x^{-4} = -\frac{12}{x^4}$$

Proof of Rule 7 The proof uses the Quotient Rule. If n is a negative integer, then $n = -m$, where m is a positive integer. Hence, $x^n = x^{-m} = 1/x^m$, and

$$\begin{aligned}\frac{d}{dx}(x^n) &= \frac{d}{dx}\left(\frac{1}{x^m}\right) \\ &= \frac{x^m \cdot \frac{d}{dx}(1) - 1 \cdot \frac{d}{dx}(x^m)}{(x^m)^2} && \text{Quotient Rule with } u = 1 \text{ and } v = x^m \\ &= \frac{0 - mx^{m-1}}{x^{2m}} && \text{Since } m > 0, \frac{d}{dx}(x^m) = mx^{m-1} \\ &= -mx^{-m-1} \\ &= nx^{n-1}. && \text{Since } -m = n\end{aligned}$$

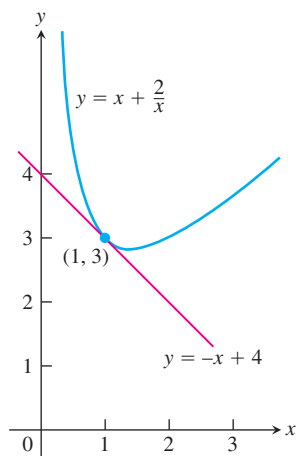


FIGURE 3.11 The tangent to the curve $y = x + (2/x)$ at $(1, 3)$ in Example 12. The curve has a third-quadrant portion not shown here. We see how to graph functions like this one in Chapter 4.

EXAMPLE 12 Tangent to a Curve

Find an equation for the tangent to the curve

$$y = x + \frac{2}{x}$$

at the point $(1, 3)$ (Figure 3.11).

Solution The slope of the curve is

$$\frac{dy}{dx} = \frac{d}{dx}(x) + 2 \frac{d}{dx}\left(\frac{1}{x}\right) = 1 + 2\left(-\frac{1}{x^2}\right) = 1 - \frac{2}{x^2}.$$

The slope at $x = 1$ is

$$\left.\frac{dy}{dx}\right|_{x=1} = \left[1 - \frac{2}{x^2}\right]_{x=1} = 1 - 2 = -1.$$

The line through $(1, 3)$ with slope $m = -1$ is

$$\begin{aligned}y - 3 &= (-1)(x - 1) && \text{Point-slope equation} \\ y &= -x + 1 + 3 \\ y &= -x + 4.\end{aligned}$$

The choice of which rules to use in solving a differentiation problem can make a difference in how much work you have to do. Here is an example.

EXAMPLE 13 Choosing Which Rule to Use

Rather than using the Quotient Rule to find the derivative of

$$y = \frac{(x-1)(x^2-2x)}{x^4},$$

expand the numerator and divide by x^4 :

$$y = \frac{(x-1)(x^2-2x)}{x^4} = \frac{x^3 - 3x^2 + 2x}{x^4} = x^{-1} - 3x^{-2} + 2x^{-3}.$$

Then use the Sum and Power Rules:

$$\begin{aligned}\frac{dy}{dx} &= -x^{-2} - 3(-2)x^{-3} + 2(-3)x^{-4} \\ &= -\frac{1}{x^2} + \frac{6}{x^3} - \frac{6}{x^4}.\end{aligned}$$

Second- and Higher-Order Derivatives

If $y = f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function. If f' is also differentiable, then we can differentiate f' to get a new function of x denoted by f'' . So $f'' = (f')'$. The function f'' is called the **second derivative** of f because it is the derivative of the first derivative. Notationally,

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy'}{dx} = y'' = D^2(f)(x) = D_x^2 f(x).$$

The symbol D^2 means the operation of differentiation is performed twice.

If $y = x^6$, then $y' = 6x^5$ and we have

$$y'' = \frac{dy'}{dx} = \frac{d}{dx} (6x^5) = 30x^4.$$

Thus $D^2(x^6) = 30x^4$.

If y'' is differentiable, its derivative, $y''' = dy''/dx = d^3y/dx^3$ is the **third derivative** of y with respect to x . The names continue as you imagine, with

$$y^{(n)} = \frac{d}{dx} y^{(n-1)} = \frac{d^n y}{dx^n} = D^n y$$

denoting the **n th derivative** of y with respect to x for any positive integer n .

We can interpret the second derivative as the rate of change of the slope of the tangent to the graph of $y = f(x)$ at each point. You will see in the next chapter that the second derivative reveals whether the graph bends upward or downward from the tangent line as we move off the point of tangency. In the next section, we interpret both the second and third derivatives in terms of motion along a straight line.

EXAMPLE 14 Finding Higher Derivatives

The first four derivatives of $y = x^3 - 3x^2 + 2$ are

$$\text{First derivative: } y' = 3x^2 - 6x$$

$$\text{Second derivative: } y'' = 6x - 6$$

$$\text{Third derivative: } y''' = 6$$

$$\text{Fourth derivative: } y^{(4)} = 0.$$

The function has derivatives of all orders, the fifth and later derivatives all being zero.

How to Read the Symbols for Derivatives

y'	“y prime”
y''	“y double prime”
$\frac{d^2y}{dx^2}$	“d squared y dx squared”
y'''	“y triple prime”
$y^{(n)}$	“y super n”
$\frac{d^n y}{dx^n}$	“d to the n of y by dx to the n”
D^n	“D to the n”

EXERCISES 3.2

Derivative Calculations

In Exercises 1–12, find the first and second derivatives.

1. $y = -x^2 + 3$
2. $y = x^2 + x + 8$
3. $s = 5t^3 - 3t^5$
4. $w = 3z^7 - 7z^3 + 21z^2$
5. $y = \frac{4x^3}{3} - x$
6. $y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$
7. $w = 3z^{-2} - \frac{1}{z}$
8. $s = -2t^{-1} + \frac{4}{t^2}$
9. $y = 6x^2 - 10x - 5x^{-2}$
10. $y = 4 - 2x - x^{-3}$
11. $r = \frac{1}{3s^2} - \frac{5}{2s}$
12. $r = \frac{12}{\theta} - \frac{4}{\theta^3} + \frac{1}{\theta^4}$

In Exercises 13–16, find y' (a) by applying the Product Rule and (b) by multiplying the factors to produce a sum of simpler terms to differentiate.

13. $y = (3 - x^2)(x^3 - x + 1)$
14. $y = (x - 1)(x^2 + x + 1)$
15. $y = (x^2 + 1)\left(x + 5 + \frac{1}{x}\right)$
16. $y = \left(x + \frac{1}{x}\right)\left(x - \frac{1}{x} + 1\right)$

Find the derivatives of the functions in Exercises 17–28.

17. $y = \frac{2x + 5}{3x - 2}$
18. $z = \frac{2x + 1}{x^2 - 1}$
19. $g(x) = \frac{x^2 - 4}{x + 0.5}$
20. $f(t) = \frac{t^2 - 1}{t^2 + t - 2}$
21. $v = (1 - t)(1 + t^2)^{-1}$
22. $w = (2x - 7)^{-1}(x + 5)$
23. $f(s) = \frac{\sqrt{s} - 1}{\sqrt{s} + 1}$
24. $u = \frac{5x + 1}{2\sqrt{x}}$
25. $v = \frac{1 + x - 4\sqrt{x}}{x}$
26. $r = 2\left(\frac{1}{\sqrt{\theta}} + \sqrt{\theta}\right)$
27. $y = \frac{1}{(x^2 - 1)(x^2 + x + 1)}$
28. $y = \frac{(x + 1)(x + 2)}{(x - 1)(x - 2)}$

Find the derivatives of all orders of the functions in Exercises 29 and 30.

29. $y = \frac{x^4}{2} - \frac{3}{2}x^2 - x$
30. $y = \frac{x^5}{120}$

Find the first and second derivatives of the functions in Exercises 31–38.

31. $y = \frac{x^3 + 7}{x}$
32. $s = \frac{t^2 + 5t - 1}{t^2}$
33. $r = \frac{(\theta - 1)(\theta^2 + \theta + 1)}{\theta^3}$
34. $u = \frac{(x^2 + x)(x^2 - x + 1)}{x^4}$
35. $w = \left(\frac{1 + 3z}{3z}\right)(3 - z)$
36. $w = (z + 1)(z - 1)(z^2 + 1)$

$$37. p = \left(\frac{q^2 + 3}{12q}\right)\left(\frac{q^4 - 1}{q^3}\right) \quad 38. p = \frac{q^2 + 3}{(q - 1)^3 + (q + 1)^3}$$

Using Numerical Values

39. Suppose u and v are functions of x that are differentiable at $x = 0$ and that

$$u(0) = 5, \quad u'(0) = -3, \quad v(0) = -1, \quad v'(0) = 2.$$

Find the values of the following derivatives at $x = 0$.

- a. $\frac{d}{dx}(uv)$
- b. $\frac{d}{dx}\left(\frac{u}{v}\right)$
- c. $\frac{d}{dx}\left(\frac{v}{u}\right)$
- d. $\frac{d}{dx}(7v - 2u)$

40. Suppose u and v are differentiable functions of x and that

$$u(1) = 2, \quad u'(1) = 0, \quad v(1) = 5, \quad v'(1) = -1.$$

Find the values of the following derivatives at $x = 1$.

- a. $\frac{d}{dx}(uv)$
- b. $\frac{d}{dx}\left(\frac{u}{v}\right)$
- c. $\frac{d}{dx}\left(\frac{v}{u}\right)$
- d. $\frac{d}{dx}(7v - 2u)$

Slopes and Tangents

41. a. **Normal to a curve** Find an equation for the line perpendicular to the tangent to the curve $y = x^3 - 4x + 1$ at the point $(2, 1)$.

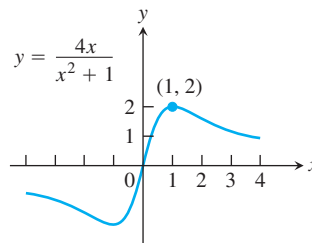
b. **Smallest slope** What is the smallest slope on the curve? At what point on the curve does the curve have this slope?

c. **Tangents having specified slope** Find equations for the tangents to the curve at the points where the slope of the curve is 8.

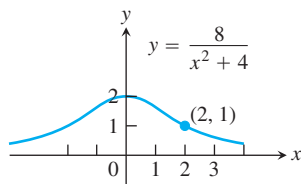
42. a. **Horizontal tangents** Find equations for the horizontal tangents to the curve $y = x^3 - 3x - 2$. Also find equations for the lines that are perpendicular to these tangents at the points of tangency.

b. **Smallest slope** What is the smallest slope on the curve? At what point on the curve does the curve have this slope? Find an equation for the line that is perpendicular to the curve's tangent at this point.

43. Find the tangents to *Newton's serpentine* (graphed here) at the origin and the point $(1, 2)$.



44. Find the tangent to the *Witch of Agnesi* (graphed here) at the point (2, 1).



45. **Quadratic tangent to identity function** The curve $y = ax^2 + bx + c$ passes through the point (1, 2) and is tangent to the line $y = x$ at the origin. Find a , b , and c .
46. **Quadratics having a common tangent** The curves $y = x^2 + ax + b$ and $y = cx - x^2$ have a common tangent line at the point (1, 0). Find a , b , and c .
47. a. Find an equation for the line that is tangent to the curve $y = x^3 - x$ at the point $(-1, 0)$.
- T** b. Graph the curve and tangent line together. The tangent intersects the curve at another point. Use Zoom and Trace to estimate the point's coordinates.
- T** c. Confirm your estimates of the coordinates of the second intersection point by solving the equations for the curve and tangent simultaneously (Solver key).
48. a. Find an equation for the line that is tangent to the curve $y = x^3 - 6x^2 + 5x$ at the origin.
- T** b. Graph the curve and tangent together. The tangent intersects the curve at another point. Use Zoom and Trace to estimate the point's coordinates.
- T** c. Confirm your estimates of the coordinates of the second intersection point by solving the equations for the curve and tangent simultaneously (Solver key).

Theory and Examples

49. The general polynomial of degree n has the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where $a_n \neq 0$. Find $P'(x)$.

50. **The body's reaction to medicine** The reaction of the body to a dose of medicine can sometimes be represented by an equation of the form

$$R = M^2 \left(\frac{C}{2} - \frac{M}{3} \right),$$

where C is a positive constant and M is the amount of medicine absorbed in the blood. If the reaction is a change in blood pressure, R is measured in millimeters of mercury. If the reaction is a change in temperature, R is measured in degrees, and so on.

Find dR/dM . This derivative, as a function of M , is called the sensitivity of the body to the medicine. In Section 4.5, we will see

how to find the amount of medicine to which the body is most sensitive.

51. Suppose that the function v in the Product Rule has a constant value c . What does the Product Rule then say? What does this say about the Constant Multiple Rule?

52. The Reciprocal Rule

- a. The *Reciprocal Rule* says that at any point where the function $v(x)$ is differentiable and different from zero,

$$\frac{d}{dx} \left(\frac{1}{v} \right) = -\frac{1}{v^2} \frac{dv}{dx}.$$

Show that the Reciprocal Rule is a special case of the Quotient Rule.

- b. Show that the Reciprocal Rule and the Product Rule together imply the Quotient Rule.

53. **Generalizing the Product Rule** The Product Rule gives the formula

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

for the derivative of the product uv of two differentiable functions of x .

- a. What is the analogous formula for the derivative of the product uvw of *three* differentiable functions of x ?
- b. What is the formula for the derivative of the product $u_1 u_2 u_3 u_4$ of *four* differentiable functions of x ?
- c. What is the formula for the derivative of a product $u_1 u_2 u_3 \cdots u_n$ of a finite number n of differentiable functions of x ?

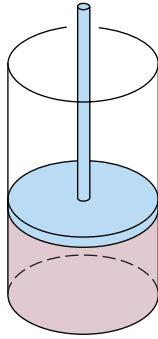
54. Rational Powers

- a. Find $\frac{d}{dx}(x^{3/2})$ by writing $x^{3/2}$ as $x \cdot x^{1/2}$ and using the Product Rule. Express your answer as a rational number times a rational power of x . Work parts (b) and (c) by a similar method.
- b. Find $\frac{d}{dx}(x^{5/2})$.
- c. Find $\frac{d}{dx}(x^{7/2})$.
- d. What patterns do you see in your answers to parts (a), (b), and (c)? Rational powers are one of the topics in Section 3.6.

55. **Cylinder pressure** If gas in a cylinder is maintained at a constant temperature T , the pressure P is related to the volume V by a formula of the form

$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2},$$

in which a , b , n , and R are constants. Find dP/dV . (See accompanying figure.)



- 56. The best quantity to order** One of the formulas for inventory management says that the average weekly cost of ordering, paying for, and holding merchandise is

$$A(q) = \frac{km}{q} + cm + \frac{hq}{2},$$

where q is the quantity you order when things run low (shoes, radios, brooms, or whatever the item might be); k is the cost of placing an order (the same, no matter how often you order); c is the cost of one item (a constant); m is the number of items sold each week (a constant); and h is the weekly holding cost per item (a constant that takes into account things such as space, utilities, insurance, and security). Find dA/dq and d^2A/dq^2 .

3.3

The Derivative as a Rate of Change

In Section 2.1, we initiated the study of average and instantaneous rates of change. In this section, we continue our investigations of applications in which derivatives are used to model the rates at which things change in the world around us. We revisit the study of motion along a line and examine other applications.

It is natural to think of change as change with respect to time, but other variables can be treated in the same way. For example, a physician may want to know how change in dosage affects the body's response to a drug. An economist may want to study how the cost of producing steel varies with the number of tons produced.

Instantaneous Rates of Change

If we interpret the difference quotient $(f(x + h) - f(x))/h$ as the average rate of change in f over the interval from x to $x + h$, we can interpret its limit as $h \rightarrow 0$ as the rate at which f is changing at the point x .

DEFINITION Instantaneous Rate of Change

The **instantaneous rate of change** of f with respect to x at x_0 is the derivative

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided the limit exists.

Thus, instantaneous rates are limits of average rates.

It is conventional to use the word *instantaneous* even when x does not represent time. The word is, however, frequently omitted. When we say *rate of change*, we mean *instantaneous rate of change*.

EXAMPLE 1 How a Circle's Area Changes with Its Diameter

The area A of a circle is related to its diameter by the equation

$$A = \frac{\pi}{4} D^2.$$

How fast does the area change with respect to the diameter when the diameter is 10 m?

Solution The rate of change of the area with respect to the diameter is

$$\frac{dA}{dD} = \frac{\pi}{4} \cdot 2D = \frac{\pi D}{2}.$$

When $D = 10$ m, the area is changing at rate $(\pi/2)10 = 5\pi$ m²/m. ■

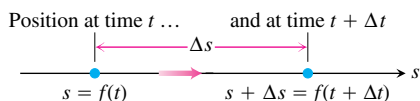


FIGURE 3.12 The positions of a body moving along a coordinate line at time t and shortly later at time $t + \Delta t$.

Motion Along a Line: Displacement, Velocity, Speed, Acceleration, and Jerk

Suppose that an object is moving along a coordinate line (say an s -axis) so that we know its position s on that line as a function of time t :

$$s = f(t).$$

The **displacement** of the object over the time interval from t to $t + \Delta t$ (Figure 3.12) is

$$\Delta s = f(t + \Delta t) - f(t),$$

and the **average velocity** of the object over that time interval is

$$v_{av} = \frac{\text{displacement}}{\text{travel time}} = \frac{\Delta s}{\Delta t} = \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

To find the body's velocity at the exact instant t , we take the limit of the average velocity over the interval from t to $t + \Delta t$ as Δt shrinks to zero. This limit is the derivative of f with respect to t .

DEFINITION Velocity

Velocity (instantaneous velocity) is the derivative of position with respect to time. If a body's position at time t is $s = f(t)$, then the body's velocity at time t is

$$v(t) = \frac{ds}{dt} = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

EXAMPLE 2 Finding the Velocity of a Race Car

Figure 3.13 shows the time-to-distance graph of a 1996 Riley & Scott Mk III-Olds WSC race car. The slope of the secant PQ is the average velocity for the 3-sec interval from $t = 2$ to $t = 5$ sec; in this case, it is about 100 ft/sec or 68 mph.

The slope of the tangent at P is the speedometer reading at $t = 2$ sec, about 57 ft/sec or 39 mph. The acceleration for the period shown is a nearly constant 28.5 ft/sec² during

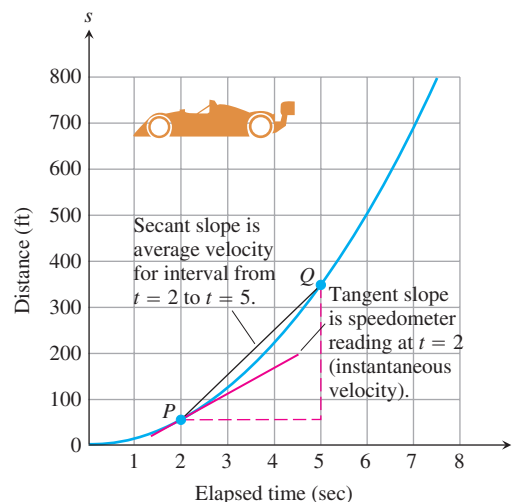


FIGURE 3.13 The time-to-distance graph for Example 2. The slope of the tangent line at P is the instantaneous velocity at $t = 2$ sec.

each second, which is about $0.89g$, where g is the acceleration due to gravity. The race car's top speed is an estimated 190 mph. (Source: *Road and Track*, March 1997.) ■

Besides telling how fast an object is moving, its velocity tells the direction of motion. When the object is moving forward (s increasing), the velocity is positive; when the body is moving backward (s decreasing), the velocity is negative (Figure 3.14).

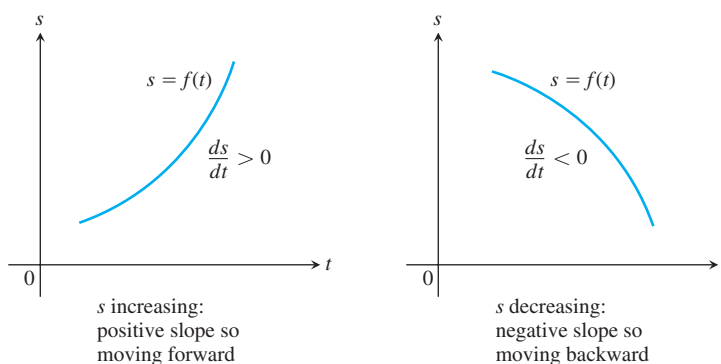


FIGURE 3.14 For motion $s = f(t)$ along a straight line, $v = ds/dt$ is positive when s increases and negative when s decreases.

If we drive to a friend's house and back at 30 mph, say, the speedometer will show 30 on the way over but it will not show -30 on the way back, even though our distance from home is decreasing. The speedometer always shows *speed*, which is the absolute value of velocity. Speed measures the rate of progress regardless of direction.

DEFINITION **Speed**

Speed is the absolute value of velocity.

$$\text{Speed} = |v(t)| = \left| \frac{ds}{dt} \right|$$

EXAMPLE 3 Horizontal Motion

Figure 3.15 shows the velocity $v = f'(t)$ of a particle moving on a coordinate line. The particle moves forward for the first 3 sec, moves backward for the next 2 sec, stands still for a second, and moves forward again. The particle achieves its greatest speed at time $t = 4$, while moving backward. ■

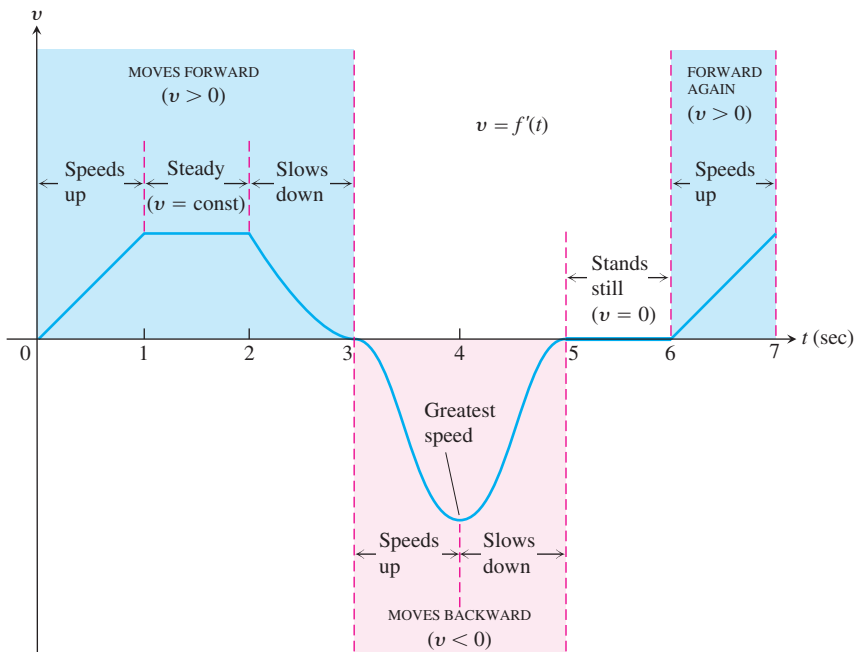


FIGURE 3.15 The velocity graph for Example 3.

HISTORICAL BIOGRAPHY

Bernard Bolzano
(1781–1848)

The rate at which a body’s velocity changes is the body’s *acceleration*. The acceleration measures how quickly the body picks up or loses speed.

A sudden change in acceleration is called a *jerk*. When a ride in a car or a bus is jerky, it is not that the accelerations involved are necessarily large but that the changes in acceleration are abrupt.

DEFINITIONS Acceleration, Jerk

Acceleration is the derivative of velocity with respect to time. If a body's position at time t is $s = f(t)$, then the body's acceleration at time t is

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

Jerk is the derivative of acceleration with respect to time:

$$j(t) = \frac{da}{dt} = \frac{d^3s}{dt^3}.$$

Near the surface of the Earth all bodies fall with the same constant acceleration. Galileo's experiments with free fall (Example 1, Section 2.1) lead to the equation

$$s = \frac{1}{2}gt^2,$$

where s is distance and g is the acceleration due to Earth's gravity. This equation holds in a vacuum, where there is no air resistance, and closely models the fall of dense, heavy objects, such as rocks or steel tools, for the first few seconds of their fall, before air resistance starts to slow them down.

The value of g in the equation $s = (1/2)gt^2$ depends on the units used to measure t and s . With t in seconds (the usual unit), the value of g determined by measurement at sea level is approximately 32 ft/sec² (feet per second squared) in English units, and $g = 9.8$ m/sec² (meters per second squared) in metric units. (These gravitational constants depend on the distance from Earth's center of mass, and are slightly lower on top of Mt. Everest, for example.)

The jerk of the constant acceleration of gravity ($g = 32$ ft/sec²) is zero:

$$j = \frac{d}{dt}(g) = 0.$$

An object does not exhibit jerkiness during free fall.

EXAMPLE 4 Modeling Free Fall

Figure 3.16 shows the free fall of a heavy ball bearing released from rest at time $t = 0$ sec.

- (a) How many meters does the ball fall in the first 2 sec?
 (b) What is its velocity, speed, and acceleration then?

Solution

- (a) The metric free-fall equation is $s = 4.9t^2$. During the first 2 sec, the ball falls

$$s(2) = 4.9(2)^2 = 19.6 \text{ m}.$$

- (b) At any time t , *velocity* is the derivative of position:

$$v(t) = s'(t) = \frac{d}{dt}(4.9t^2) = 9.8t.$$

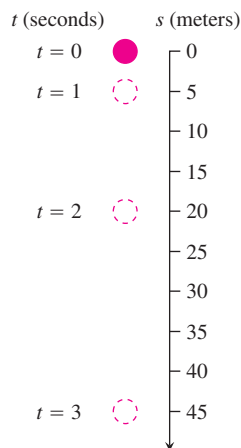


FIGURE 3.16 A ball bearing falling from rest (Example 4).

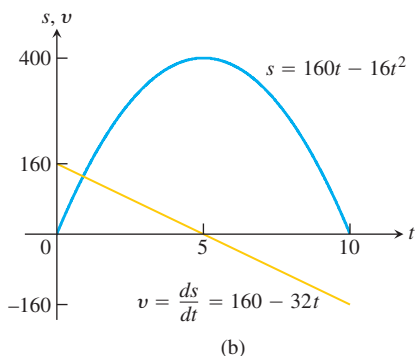
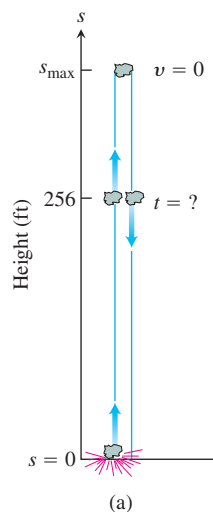


FIGURE 3.17 (a) The rock in Example 5. (b) The graphs of s and v as functions of time; s is largest when $v = ds/dt = 0$. The graph of s is *not* the path of the rock: It is a plot of height versus time. The slope of the plot is the rock's velocity, graphed here as a straight line.

At $t = 2$, the velocity is

$$v(2) = 19.6 \text{ m/sec}$$

in the downward (increasing s) direction. The *speed* at $t = 2$ is

$$\text{Speed} = |v(2)| = 19.6 \text{ m/sec}.$$

The *acceleration* at any time t is

$$a(t) = v'(t) = s''(t) = 9.8 \text{ m/sec}^2.$$

At $t = 2$, the acceleration is 9.8 m/sec^2 . ■

EXAMPLE 5 Modeling Vertical Motion

A dynamite blast blows a heavy rock straight up with a launch velocity of 160 ft/sec (about 109 mph) (Figure 3.17a). It reaches a height of $s = 160t - 16t^2$ ft after t sec.

- How high does the rock go?
- What are the velocity and speed of the rock when it is 256 ft above the ground on the way up? On the way down?
- What is the acceleration of the rock at any time t during its flight (after the blast)?
- When does the rock hit the ground again?

Solution

- In the coordinate system we have chosen, s measures height from the ground up, so the velocity is positive on the way up and negative on the way down. The instant the rock is at its highest point is the one instant during the flight when the velocity is 0. To find the maximum height, all we need to do is to find when $v = 0$ and evaluate s at this time.

At any time t , the velocity is

$$v = \frac{ds}{dt} = \frac{d}{dt}(160t - 16t^2) = 160 - 32t \text{ ft/sec}.$$

The velocity is zero when

$$160 - 32t = 0 \quad \text{or} \quad t = 5 \text{ sec}.$$

The rock's height at $t = 5$ sec is

$$s_{\max} = s(5) = 160(5) - 16(5)^2 = 800 - 400 = 400 \text{ ft}.$$

See Figure 3.17b.

- To find the rock's velocity at 256 ft on the way up and again on the way down, we first find the two values of t for which

$$s(t) = 160t - 16t^2 = 256.$$

To solve this equation, we write

$$16t^2 - 160t + 256 = 0$$

$$16(t^2 - 10t + 16) = 0$$

$$(t - 2)(t - 8) = 0$$

$$t = 2 \text{ sec}, t = 8 \text{ sec}.$$

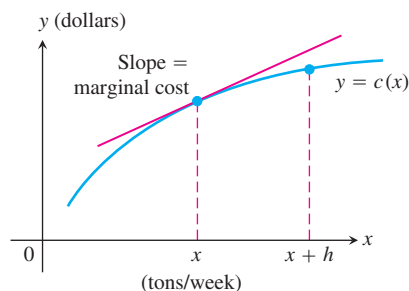


FIGURE 3.18 Weekly steel production: $c(x)$ is the cost of producing x tons per week. The cost of producing an additional h tons is $c(x + h) - c(x)$.

The rock is 256 ft above the ground 2 sec after the explosion and again 8 sec after the explosion. The rock's velocities at these times are

$$v(2) = 160 - 32(2) = 160 - 64 = 96 \text{ ft/sec.}$$

$$v(8) = 160 - 32(8) = 160 - 256 = -96 \text{ ft/sec.}$$

At both instants, the rock's speed is 96 ft/sec. Since $v(2) > 0$, the rock is moving upward (s is increasing) at $t = 2$ sec; it is moving downward (s is decreasing) at $t = 8$ because $v(8) < 0$.

- (c) At any time during its flight following the explosion, the rock's acceleration is a constant

$$a = \frac{dv}{dt} = \frac{d}{dt}(160 - 32t) = -32 \text{ ft/sec}^2.$$

The acceleration is always downward. As the rock rises, it slows down; as it falls, it speeds up.

- (d) The rock hits the ground at the positive time t for which $s = 0$. The equation $160t - 16t^2 = 0$ factors to give $16t(10 - t) = 0$, so it has solutions $t = 0$ and $t = 10$. At $t = 0$, the blast occurred and the rock was thrown upward. It returned to the ground 10 sec later. ■

Derivatives in Economics

Engineers use the terms *velocity* and *acceleration* to refer to the derivatives of functions describing motion. Economists, too, have a specialized vocabulary for rates of change and derivatives. They call them *marginals*.

In a manufacturing operation, the *cost of production* $c(x)$ is a function of x , the number of units produced. The **marginal cost of production** is the rate of change of cost with respect to level of production, so it is dc/dx .

Suppose that $c(x)$ represents the dollars needed to produce x tons of steel in one week. It costs more to produce $x + h$ units per week, and the cost difference, divided by h , is the average cost of producing each additional ton:

$$\frac{c(x + h) - c(x)}{h} = \text{average cost of each of the additional } h \text{ tons of steel produced.}$$

The limit of this ratio as $h \rightarrow 0$ is the *marginal cost* of producing more steel per week when the current weekly production is x tons (Figure 3.18).

$$\frac{dc}{dx} = \lim_{h \rightarrow 0} \frac{c(x + h) - c(x)}{h} = \text{marginal cost of production.}$$

Sometimes the marginal cost of production is loosely defined to be the extra cost of producing one unit:

$$\frac{\Delta c}{\Delta x} = \frac{c(x + 1) - c(x)}{1},$$

which is approximated by the value of dc/dx at x . This approximation is acceptable if the slope of the graph of c does not change quickly near x . Then the difference quotient will be close to its limit dc/dx , which is the rise in the tangent line if $\Delta x = 1$ (Figure 3.19). The approximation works best for large values of x .

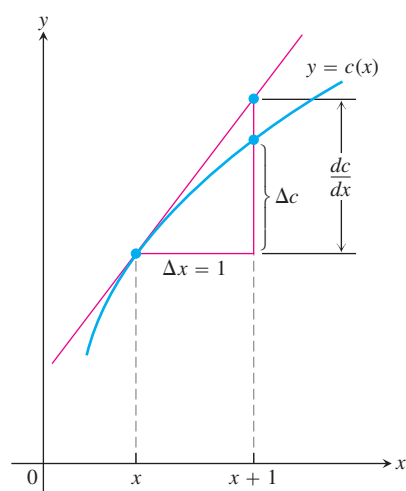


FIGURE 3.19 The marginal cost dc/dx is approximately the extra cost Δc of producing $\Delta x = 1$ more unit.

Economists often represent a total cost function by a cubic polynomial

$$c(x) = \alpha x^3 + \beta x^2 + \gamma x + \delta$$

where δ represents *fixed costs* such as rent, heat, equipment capitalization, and management costs. The other terms represent *variable costs* such as the costs of raw materials, taxes, and labor. Fixed costs are independent of the number of units produced, whereas variable costs depend on the quantity produced. A cubic polynomial is usually complicated enough to capture the cost behavior on a relevant quantity interval.

EXAMPLE 6 Marginal Cost and Marginal Revenue

Suppose that it costs

$$c(x) = x^3 - 6x^2 + 15x$$

dollars to produce x radiators when 8 to 30 radiators are produced and that

$$r(x) = x^3 - 3x^2 + 12x$$

gives the dollar revenue from selling x radiators. Your shop currently produces 10 radiators a day. About how much extra will it cost to produce one more radiator a day, and what is your estimated increase in revenue for selling 11 radiators a day?

Solution The cost of producing one more radiator a day when 10 are produced is about $c'(10)$:

$$c'(x) = \frac{d}{dx}(x^3 - 6x^2 + 15x) = 3x^2 - 12x + 15$$

$$c'(10) = 3(100) - 12(10) + 15 = 195.$$

The additional cost will be about \$195. The marginal revenue is

$$r'(x) = \frac{d}{dx}(x^3 - 3x^2 + 12x) = 3x^2 - 6x + 12.$$

The marginal revenue function estimates the increase in revenue that will result from selling one additional unit. If you currently sell 10 radiators a day, you can expect your revenue to increase by about

$$r'(10) = 3(100) - 6(10) + 12 = \$252$$

if you increase sales to 11 radiators a day. ■

EXAMPLE 7 Marginal Tax Rate

To get some feel for the language of marginal rates, consider marginal tax rates. If your marginal income tax rate is 28% and your income increases by \$1000, you can expect to pay an extra \$280 in taxes. This does not mean that you pay 28% of your entire income in taxes. It just means that at your current income level I , the rate of increase of taxes T with respect to income is $dT/dI = 0.28$. You will pay \$0.28 out of every extra dollar you earn in taxes. Of course, if you earn a lot more, you may land in a higher tax bracket and your marginal rate will increase. ■

Sensitivity to Change

When a small change in x produces a large change in the value of a function $f(x)$, we say that the function is relatively **sensitive** to changes in x . The derivative $f'(x)$ is a measure of this sensitivity.

EXAMPLE 8 Genetic Data and Sensitivity to Change

The Austrian monk Gregor Johann Mendel (1822–1884), working with garden peas and other plants, provided the first scientific explanation of hybridization.

His careful records showed that if p (a number between 0 and 1) is the frequency of the gene for smooth skin in peas (dominant) and $(1 - p)$ is the frequency of the gene for wrinkled skin in peas, then the proportion of smooth-skinned peas in the next generation will be

$$y = 2p(1 - p) + p^2 = 2p - p^2.$$

The graph of y versus p in Figure 3.20a suggests that the value of y is more sensitive to a change in p when p is small than when p is large. Indeed, this fact is borne out by the derivative graph in Figure 3.20b, which shows that dy/dp is close to 2 when p is near 0 and close to 0 when p is near 1.

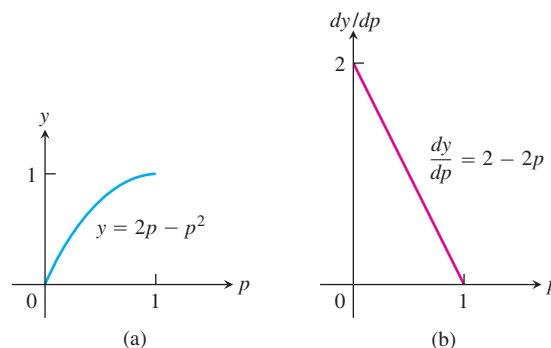


FIGURE 3.20 (a) The graph of $y = 2p - p^2$, describing the proportion of smooth-skinned peas. (b) The graph of dy/dp (Example 8).

The implication for genetics is that introducing a few more dominant genes into a highly recessive population (where the frequency of wrinkled skin peas is small) will have a more dramatic effect on later generations than will a similar increase in a highly dominant population. ■

EXERCISES 3.3

Motion Along a Coordinate Line

Exercises 1–6 give the positions $s = f(t)$ of a body moving on a coordinate line, with s in meters and t in seconds.

- a. Find the body's displacement and average velocity for the given time interval.
- b. Find the body's speed and acceleration at the endpoints of the interval.
- c. When, if ever, during the interval does the body change direction?
 1. $s = t^2 - 3t + 2$, $0 \leq t \leq 2$
 2. $s = 6t - t^2$, $0 \leq t \leq 6$

3. $s = -t^3 + 3t^2 - 3t$, $0 \leq t \leq 3$
4. $s = (t^4/4) - t^3 + t^2$, $0 \leq t \leq 3$
5. $s = \frac{25}{t^2} - \frac{5}{t}$, $1 \leq t \leq 5$
6. $s = \frac{25}{t+5}$, $-4 \leq t \leq 0$
7. **Particle motion** At time t , the position of a body moving along the s -axis is $s = t^3 - 6t^2 + 9t$ m.
 - a. Find the body's acceleration each time the velocity is zero.
 - b. Find the body's speed each time the acceleration is zero.
 - c. Find the total distance traveled by the body from $t = 0$ to $t = 2$.
8. **Particle motion** At time $t \geq 0$, the velocity of a body moving along the s -axis is $v = t^2 - 4t + 3$.
 - a. Find the body's acceleration each time the velocity is zero.
 - b. When is the body moving forward? Backward?
 - c. When is the body's velocity increasing? Decreasing?

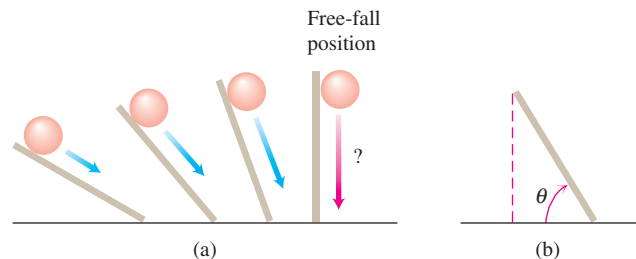
Free-Fall Applications

9. **Free fall on Mars and Jupiter** The equations for free fall at the surfaces of Mars and Jupiter (s in meters, t in seconds) are $s = 1.86t^2$ on Mars and $s = 11.44t^2$ on Jupiter. How long does it take a rock falling from rest to reach a velocity of 27.8 m/sec (about 100 km/h) on each planet?
10. **Lunar projectile motion** A rock thrown vertically upward from the surface of the moon at a velocity of 24 m/sec (about 86 km/h) reaches a height of $s = 24t - 0.8t^2$ meters in t sec.
 - a. Find the rock's velocity and acceleration at time t . (The acceleration in this case is the acceleration of gravity on the moon.)
 - b. How long does it take the rock to reach its highest point?
 - c. How high does the rock go?
 - d. How long does it take the rock to reach half its maximum height?
 - e. How long is the rock aloft?
11. **Finding g on a small airless planet** Explorers on a small airless planet used a spring gun to launch a ball bearing vertically upward from the surface at a launch velocity of 15 m/sec. Because the acceleration of gravity at the planet's surface was g_s m/sec², the explorers expected the ball bearing to reach a height of $s = 15t - (1/2)g_s t^2$ meters t sec later. The ball bearing reached its maximum height 20 sec after being launched. What was the value of g_s ?
12. **Speeding bullet** A 45-caliber bullet fired straight up from the surface of the moon would reach a height of $s = 832t - 2.6t^2$ feet after t sec. On Earth, in the absence of air, its height would be $s = 832t - 16t^2$ ft after t sec. How long will the bullet be aloft in each case? How high will the bullet go?
13. **Free fall from the Tower of Pisa** Had Galileo dropped a cannonball from the Tower of Pisa, 179 ft above the ground, the ball's

height above ground t sec into the fall would have been $s = 179 - 16t^2$.

- a. What would have been the ball's velocity, speed, and acceleration at time t ?
 - b. About how long would it have taken the ball to hit the ground?
 - c. What would have been the ball's velocity at the moment of impact?
14. **Galileo's free-fall formula** Galileo developed a formula for a body's velocity during free fall by rolling balls from rest down increasingly steep inclined planks and looking for a limiting formula that would predict a ball's behavior when the plank was vertical and the ball fell freely; see part (a) of the accompanying figure. He found that, for any given angle of the plank, the ball's velocity t sec into motion was a constant multiple of t . That is, the velocity was given by a formula of the form $v = kt$. The value of the constant k depended on the inclination of the plank.
- In modern notation—part (b) of the figure—with distance in meters and time in seconds, what Galileo determined by experiment was that, for any given angle θ , the ball's velocity t sec into the roll was

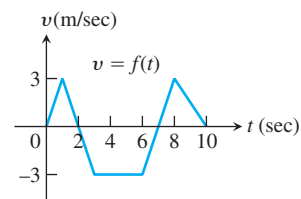
$$v = 9.8(\sin \theta)t \text{ m/sec.}$$



- a. What is the equation for the ball's velocity during free fall?
- b. Building on your work in part (a), what constant acceleration does a freely falling body experience near the surface of Earth?

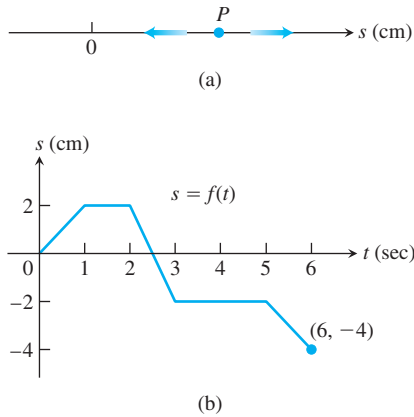
Conclusions About Motion from Graphs

15. The accompanying figure shows the velocity $v = ds/dt = f(t)$ (m/sec) of a body moving along a coordinate line.



- a. When does the body reverse direction?
- b. When (approximately) is the body moving at a constant speed?
- c. Graph the body's speed for $0 \leq t \leq 10$.
- d. Graph the acceleration, where defined.

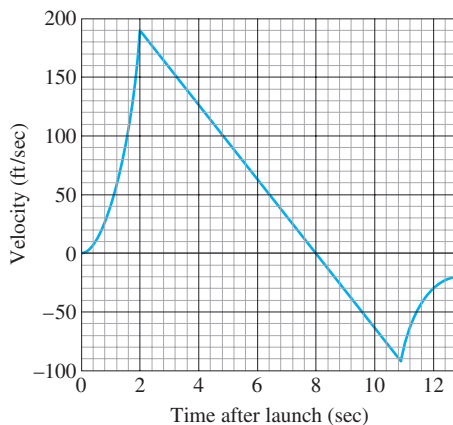
16. A particle P moves on the number line shown in part (a) of the accompanying figure. Part (b) shows the position of P as a function of time t .



- When is P moving to the left? Moving to the right? Standing still?
 - Graph the particle's velocity and speed (where defined).
17. **Launching a rocket** When a model rocket is launched, the propellant burns for a few seconds, accelerating the rocket upward. After burnout, the rocket coasts upward for a while and then begins to fall. A small explosive charge pops out a parachute shortly after the rocket starts down. The parachute slows the rocket to keep it from breaking when it lands.

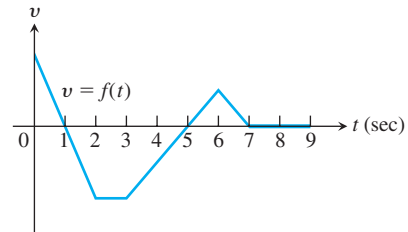
The figure here shows velocity data from the flight of the model rocket. Use the data to answer the following.

- How fast was the rocket climbing when the engine stopped?
- For how many seconds did the engine burn?

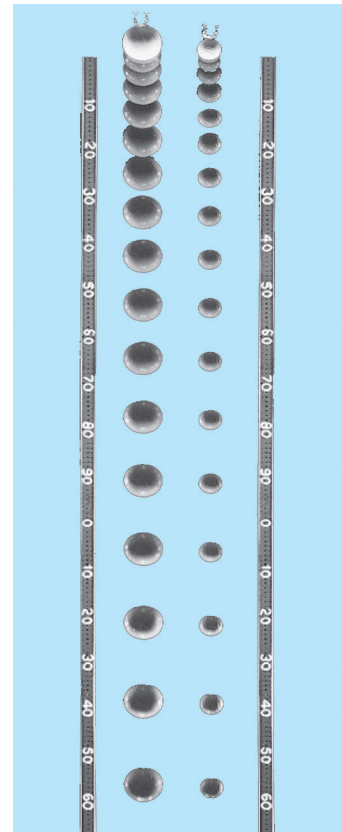


- When did the rocket reach its highest point? What was its velocity then?
- When did the parachute pop out? How fast was the rocket falling then?
- How long did the rocket fall before the parachute opened?

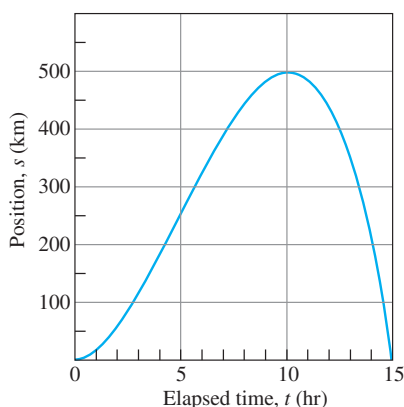
- When was the rocket's acceleration greatest?
 - When was the acceleration constant? What was its value then (to the nearest integer)?
18. The accompanying figure shows the velocity $v = f(t)$ of a particle moving on a coordinate line.



- When does the particle move forward? Move backward? Speed up? Slow down?
 - When is the particle's acceleration positive? Negative? Zero?
 - When does the particle move at its greatest speed?
 - When does the particle stand still for more than an instant?
19. **Two falling balls** The multiframe photograph in the accompanying figure shows two balls falling from rest. The vertical rulers are marked in centimeters. Use the equation $s = 490t^2$ (the free-fall equation for s in centimeters and t in seconds) to answer the following questions.



- a. How long did it take the balls to fall the first 160 cm? What was their average velocity for the period?
- b. How fast were the balls falling when they reached the 160-cm mark? What was their acceleration then?
- c. About how fast was the light flashing (flashes per second)?
20. **A traveling truck** The accompanying graph shows the position s of a truck traveling on a highway. The truck starts at $t = 0$ and returns 15 h later at $t = 15$.
- a. Use the technique described in Section 3.1, Example 3, to graph the truck's velocity $v = ds/dt$ for $0 \leq t \leq 15$. Then repeat the process, with the velocity curve, to graph the truck's acceleration dv/dt .
- b. Suppose that $s = 15t^2 - t^3$. Graph ds/dt and d^2s/dt^2 and compare your graphs with those in part (a).



21. The graphs in Figure 3.21 show the position s , velocity $v = ds/dt$, and acceleration $a = d^2s/dt^2$ of a body moving along a coordinate line as functions of time t . Which graph is which? Give reasons for your answers.

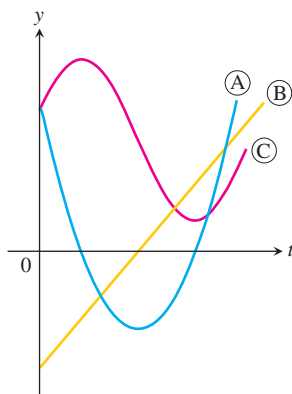


FIGURE 3.21 The graphs for Exercise 21.

22. The graphs in Figure 3.22 show the position s , the velocity $v = ds/dt$, and the acceleration $a = d^2s/dt^2$ of a body moving along the coordinate line as functions of time t . Which graph is which? Give reasons for your answers.

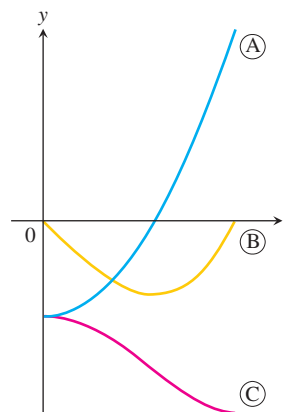


FIGURE 3.22 The graphs for Exercise 22.

Economics

23. **Marginal cost** Suppose that the dollar cost of producing x washing machines is $c(x) = 2000 + 100x - 0.1x^2$.
- a. Find the average cost per machine of producing the first 100 washing machines.
- b. Find the marginal cost when 100 washing machines are produced.
- c. Show that the marginal cost when 100 washing machines are produced is approximately the cost of producing one more washing machine after the first 100 have been made, by calculating the latter cost directly.
24. **Marginal revenue** Suppose that the revenue from selling x washing machines is

$$r(x) = 20,000 \left(1 - \frac{1}{x} \right)$$

dollars.

- a. Find the marginal revenue when 100 machines are produced.
- b. Use the function $r'(x)$ to estimate the increase in revenue that will result from increasing production from 100 machines a week to 101 machines a week.
- c. Find the limit of $r'(x)$ as $x \rightarrow \infty$. How would you interpret this number?

Additional Applications

25. **Bacterium population** When a bactericide was added to a nutrient broth in which bacteria were growing, the bacterium population continued to grow for a while, but then stopped growing and began to decline. The size of the population at time t (hours) was $b = 10^6 + 10^4t - 10^3t^2$. Find the growth rates at
- a. $t = 0$ hours.
- b. $t = 5$ hours.
- c. $t = 10$ hours.

- 26. Draining a tank** The number of gallons of water in a tank t minutes after the tank has started to drain is $Q(t) = 200(30 - t)^2$. How fast is the water running out at the end of 10 min? What is the average rate at which the water flows out during the first 10 min?

T 27. Draining a tank It takes 12 hours to drain a storage tank by opening the valve at the bottom. The depth y of fluid in the tank t hours after the valve is opened is given by the formula

$$y = 6\left(1 - \frac{t}{12}\right)^2 \text{ m.}$$

- Find the rate dy/dt (m/h) at which the tank is draining at time t .
 - When is the fluid level in the tank falling fastest? Slowest? What are the values of dy/dt at these times?
 - Graph y and dy/dt together and discuss the behavior of y in relation to the signs and values of dy/dt .
- 28. Inflating a balloon** The volume $V = (4/3)\pi r^3$ of a spherical balloon changes with the radius.
- At what rate (ft^3/ft) does the volume change with respect to the radius when $r = 2$ ft?
 - By approximately how much does the volume increase when the radius changes from 2 to 2.2 ft?
- 29. Airplane takeoff** Suppose that the distance an aircraft travels along a runway before takeoff is given by $D = (10/9)t^2$, where D is measured in meters from the starting point and t is measured in seconds from the time the brakes are released. The aircraft will become airborne when its speed reaches 200 km/h. How long will it take to become airborne, and what distance will it travel in that time?
- 30. Volcanic lava fountains** Although the November 1959 Kilauea Iki eruption on the island of Hawaii began with a line of fountains along the wall of the crater, activity was later confined to a single vent in the crater's floor, which at one point shot lava 1900 ft straight into the air (a world record). What was the lava's exit velocity in feet per second? In miles per hour? (*Hint:* If v_0 is the exit velocity of a particle of lava, its height t sec later will be $s = v_0 t - 16t^2$ ft. Begin by finding the time at which $ds/dt = 0$. Neglect air resistance.)

T Exercises 31–34 give the position function $s = f(t)$ of a body moving along the s -axis as a function of time t . Graph f together with the velocity function $v(t) = ds/dt = f'(t)$ and the acceleration function $a(t) = d^2s/dt^2 = f''(t)$. Comment on the body's behavior in relation to the signs and values of v and a . Include in your commentary such topics as the following:

- When is the body momentarily at rest?
 - When does it move to the left (down) or to the right (up)?
 - When does it change direction?
 - When does it speed up and slow down?
 - When is it moving fastest (highest speed)? Slowest?
 - When is it farthest from the axis origin?
- 31.** $s = 200t - 16t^2$, $0 \leq t \leq 12.5$ (a heavy object fired straight up from Earth's surface at 200 ft/sec)
- 32.** $s = t^2 - 3t + 2$, $0 \leq t \leq 5$
- 33.** $s = t^3 - 6t^2 + 7t$, $0 \leq t \leq 4$
- 34.** $s = 4 - 7t + 6t^2 - t^3$, $0 \leq t \leq 4$
- 35. Thoroughbred racing** A racehorse is running a 10-furlong race. (A furlong is 220 yards, although we will use furlongs and seconds as our units in this exercise.) As the horse passes each furlong marker (F), a steward records the time elapsed (t) since the beginning of the race, as shown in the table:

F	0	1	2	3	4	5	6	7	8	9	10
t	0	20	33	46	59	73	86	100	112	124	135

- How long does it take the horse to finish the race?
- What is the average speed of the horse over the first 5 furlongs?
- What is the approximate speed of the horse as it passes the 3-furlong marker?
- During which portion of the race is the horse running the fastest?
- During which portion of the race is the horse accelerating the fastest?

3.4

Derivatives of Trigonometric Functions

Many of the phenomena we want information about are approximately periodic (electromagnetic fields, heart rhythms, tides, weather). The derivatives of sines and cosines play a key role in describing periodic changes. This section shows how to differentiate the six basic trigonometric functions.

Derivative of the Sine Function

To calculate the derivative of $f(x) = \sin x$, for x measured in radians, we combine the limits in Example 5a and Theorem 7 in Section 2.4 with the angle sum identity for the sine:

$$\sin(x + h) = \sin x \cos h + \cos x \sin h.$$

If $f(x) = \sin x$, then

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} && \text{Derivative definition} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(\sin x \cos h + \cos x \sin h) - \sin x}{h} && \text{Sine angle sum identity} \\
 &= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h} \\
 &= \lim_{h \rightarrow 0} \left(\sin x \cdot \frac{\cos h - 1}{h} \right) + \lim_{h \rightarrow 0} \left(\cos x \cdot \frac{\sin h}{h} \right) \\
 &= \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\
 &= \sin x \cdot 0 + \cos x \cdot 1 \\
 &= \cos x. && \text{Example 5(a) and Theorem 7, Section 2.4}
 \end{aligned}$$

The derivative of the sine function is the cosine function:

$$\frac{d}{dx}(\sin x) = \cos x.$$

EXAMPLE 1 Derivatives Involving the Sine

(a) $y = x^2 - \sin x$:

$$\begin{aligned}
 \frac{dy}{dx} &= 2x - \frac{d}{dx}(\sin x) && \text{Difference Rule} \\
 &= 2x - \cos x.
 \end{aligned}$$

(b) $y = x^2 \sin x$:

$$\begin{aligned}
 \frac{dy}{dx} &= x^2 \frac{d}{dx}(\sin x) + 2x \sin x && \text{Product Rule} \\
 &= x^2 \cos x + 2x \sin x.
 \end{aligned}$$

(c) $y = \frac{\sin x}{x}$:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{x \cdot \frac{d}{dx}(\sin x) - \sin x \cdot 1}{x^2} && \text{Quotient Rule} \\
 &= \frac{x \cos x - \sin x}{x^2}.
 \end{aligned}$$

Derivative of the Cosine Function

With the help of the angle sum formula for the cosine,

$$\cos(x+h) = \cos x \cos h - \sin x \sin h,$$

we have

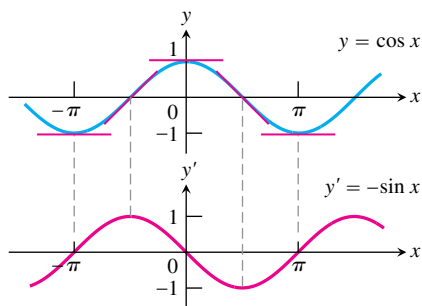


FIGURE 3.23 The curve $y' = -\sin x$ as the graph of the slopes of the tangents to the curve $y = \cos x$.

$$\begin{aligned}
 \frac{d}{dx}(\cos x) &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h} && \text{Derivative definition} \\
 &= \lim_{h \rightarrow 0} \frac{(\cos x \cos h - \sin x \sin h) - \cos x}{h} && \text{Cosine angle sum identity} \\
 &= \lim_{h \rightarrow 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h} \\
 &= \lim_{h \rightarrow 0} \cos x \cdot \frac{\cos h - 1}{h} - \lim_{h \rightarrow 0} \sin x \cdot \frac{\sin h}{h} \\
 &= \cos x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} - \sin x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\
 &= \cos x \cdot 0 - \sin x \cdot 1 \\
 &= -\sin x.
 \end{aligned}$$

Example 5(a) and
Theorem 7, Section 2.4

The derivative of the cosine function is the negative of the sine function:

$$\frac{d}{dx}(\cos x) = -\sin x$$

Figure 3.23 shows a way to visualize this result.

EXAMPLE 2 Derivatives Involving the Cosine

(a) $y = 5x + \cos x$:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{d}{dx}(5x) + \frac{d}{dx}(\cos x) && \text{Sum Rule} \\
 &= 5 - \sin x.
 \end{aligned}$$

(b) $y = \sin x \cos x$:

$$\begin{aligned}
 \frac{dy}{dx} &= \sin x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(\sin x) && \text{Product Rule} \\
 &= \sin x(-\sin x) + \cos x(\cos x) \\
 &= \cos^2 x - \sin^2 x.
 \end{aligned}$$

(c) $y = \frac{\cos x}{1 - \sin x}$:

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{(1 - \sin x) \frac{d}{dx}(\cos x) - \cos x \frac{d}{dx}(1 - \sin x)}{(1 - \sin x)^2} && \text{Quotient Rule} \\
 &= \frac{(1 - \sin x)(-\sin x) - \cos x(0 - \cos x)}{(1 - \sin x)^2} \\
 &= \frac{1 - \sin x}{(1 - \sin x)^2} && \sin^2 x + \cos^2 x = 1 \\
 &= \frac{1}{1 - \sin x}.
 \end{aligned}$$

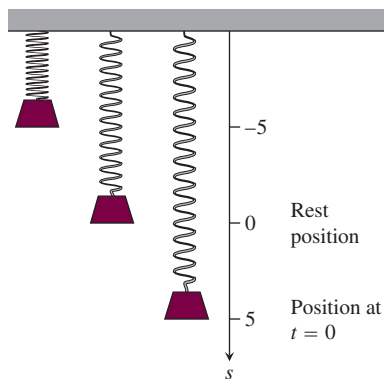


FIGURE 3.24 A body hanging from a vertical spring and then displaced oscillates above and below its rest position. Its motion is described by trigonometric functions (Example 3).

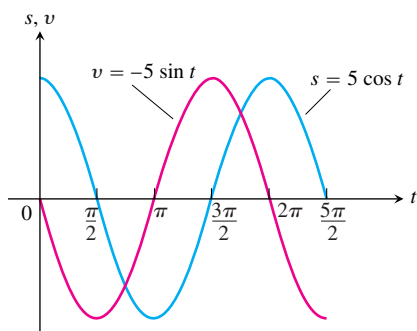


FIGURE 3.25 The graphs of the position and velocity of the body in Example 3.

Simple Harmonic Motion

The motion of a body bobbing freely up and down on the end of a spring or bungee cord is an example of *simple harmonic motion*. The next example describes a case in which there are no opposing forces such as friction or buoyancy to slow the motion down.

EXAMPLE 3 Motion on a Spring

A body hanging from a spring (Figure 3.24) is stretched 5 units beyond its rest position and released at time $t = 0$ to bob up and down. Its position at any later time t is

$$s = 5 \cos t.$$

What are its velocity and acceleration at time t ?

Solution We have

$$\text{Position:} \quad s = 5 \cos t$$

$$\text{Velocity:} \quad v = \frac{ds}{dt} = \frac{d}{dt}(5 \cos t) = -5 \sin t$$

$$\text{Acceleration:} \quad a = \frac{dv}{dt} = \frac{d}{dt}(-5 \sin t) = -5 \cos t.$$

Notice how much we can learn from these equations:

1. As time passes, the weight moves down and up between $s = -5$ and $s = 5$ on the s -axis. The amplitude of the motion is 5. The period of the motion is 2π .
2. The velocity $v = -5 \sin t$ attains its greatest magnitude, 5, when $\cos t = 0$, as the graphs show in Figure 3.25. Hence, the speed of the weight, $|v| = 5|\sin t|$, is greatest when $\cos t = 0$, that is, when $s = 0$ (the rest position). The speed of the weight is zero when $\sin t = 0$. This occurs when $s = 5 \cos t = \pm 5$, at the endpoints of the interval of motion.
3. The acceleration value is always the exact opposite of the position value. When the weight is above the rest position, gravity is pulling it back down; when the weight is below the rest position, the spring is pulling it back up.
4. The acceleration, $a = -5 \cos t$, is zero only at the rest position, where $\cos t = 0$ and the force of gravity and the force from the spring offset each other. When the weight is anywhere else, the two forces are unequal and acceleration is nonzero. The acceleration is greatest in magnitude at the points farthest from the rest position, where $\cos t = \pm 1$. ■

EXAMPLE 4 Jerk

The jerk of the simple harmonic motion in Example 3 is

$$j = \frac{da}{dt} = \frac{d}{dt}(-5 \cos t) = 5 \sin t.$$

It has its greatest magnitude when $\sin t = \pm 1$, not at the extremes of the displacement but at the rest position, where the acceleration changes direction and sign. ■

Derivatives of the Other Basic Trigonometric Functions

Because $\sin x$ and $\cos x$ are differentiable functions of x , the related functions

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}, \quad \sec x = \frac{1}{\cos x}, \quad \text{and} \quad \csc x = \frac{1}{\sin x}$$

are differentiable at every value of x at which they are defined. Their derivatives, calculated from the Quotient Rule, are given by the following formulas. Notice the negative signs in the derivative formulas for the cofunctions.

Derivatives of the Other Trigonometric Functions

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

To show a typical calculation, we derive the derivative of the tangent function. The other derivations are left to Exercise 50.

EXAMPLE 5

Find $d(\tan x)/dx$.

Solution

$$\begin{aligned}\frac{d}{dx}(\tan x) &= \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} && \text{Quotient Rule} \\ &= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

EXAMPLE 6

Find y'' if $y = \sec x$.

Solution

$$\begin{aligned}y &= \sec x \\ y' &= \sec x \tan x \\ y'' &= \frac{d}{dx}(\sec x \tan x) \\ &= \sec x \frac{d}{dx}(\tan x) + \tan x \frac{d}{dx}(\sec x) && \text{Product Rule} \\ &= \sec x(\sec^2 x) + \tan x(\sec x \tan x) \\ &= \sec^3 x + \sec x \tan^2 x\end{aligned}$$

The differentiability of the trigonometric functions throughout their domains gives another proof of their continuity at every point in their domains (Theorem 1, Section 3.1). So we can calculate limits of algebraic combinations and composites of trigonometric functions by direct substitution.

EXAMPLE 7 Finding a Trigonometric Limit

$$\lim_{x \rightarrow 0} \frac{\sqrt{2 + \sec x}}{\cos(\pi - \tan x)} = \frac{\sqrt{2 + \sec 0}}{\cos(\pi - \tan 0)} = \frac{\sqrt{2 + 1}}{\cos(\pi - 0)} = \frac{\sqrt{3}}{-1} = -\sqrt{3} \quad \blacksquare$$

EXERCISES 3.4

Derivatives

In Exercises 1–12, find dy/dx .

1. $y = -10x + 3 \cos x$
2. $y = \frac{3}{x} + 5 \sin x$
3. $y = \csc x - 4\sqrt{x} + 7$
4. $y = x^2 \cot x - \frac{1}{x^2}$
5. $y = (\sec x + \tan x)(\sec x - \tan x)$
6. $y = (\sin x + \cos x) \sec x$
7. $y = \frac{\cot x}{1 + \cot x}$
8. $y = \frac{\cos x}{1 + \sin x}$
9. $y = \frac{4}{\cos x} + \frac{1}{\tan x}$
10. $y = \frac{\cos x}{x} + \frac{x}{\cos x}$
11. $y = x^2 \sin x + 2x \cos x - 2 \sin x$
12. $y = x^2 \cos x - 2x \sin x - 2 \cos x$

In Exercises 13–16, find ds/dt .

13. $s = \tan t - t$
14. $s = t^2 - \sec t + 1$
15. $s = \frac{1 + \csc t}{1 - \csc t}$
16. $s = \frac{\sin t}{1 - \cos t}$

In Exercises 17–20, find $dr/d\theta$.

17. $r = 4 - \theta^2 \sin \theta$
18. $r = \theta \sin \theta + \cos \theta$
19. $r = \sec \theta \csc \theta$
20. $r = (1 + \sec \theta) \sin \theta$

In Exercises 21–24, find dp/dq .

21. $p = 5 + \frac{1}{\cot q}$
22. $p = (1 + \csc q) \cos q$
23. $p = \frac{\sin q + \cos q}{\cos q}$
24. $p = \frac{\tan q}{1 + \tan q}$

25. Find y'' if

- a. $y = \csc x$
- b. $y = \sec x$

26. Find $y^{(4)} = d^4 y/dx^4$ if

- a. $y = -2 \sin x$
- b. $y = 9 \cos x$

Tangent Lines

In Exercises 27–30, graph the curves over the given intervals, together with their tangents at the given values of x . Label each curve and tangent with its equation.

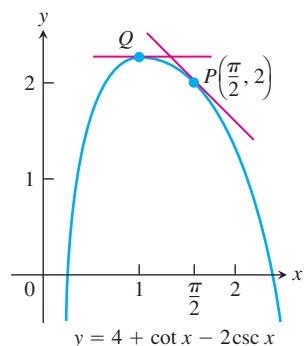
27. $y = \sin x, \quad -3\pi/2 \leq x \leq 2\pi$
 $x = -\pi, 0, 3\pi/2$
28. $y = \tan x, \quad -\pi/2 < x < \pi/2$
 $x = -\pi/3, 0, \pi/3$
29. $y = \sec x, \quad -\pi/2 < x < \pi/2$
 $x = -\pi/3, \pi/4$
30. $y = 1 + \cos x, \quad -3\pi/2 \leq x \leq 2\pi$
 $x = -\pi/3, 3\pi/2$

T Do the graphs of the functions in Exercises 31–34 have any horizontal tangents in the interval $0 \leq x \leq 2\pi$? If so, where? If not, why not? Visualize your findings by graphing the functions with a grapher.

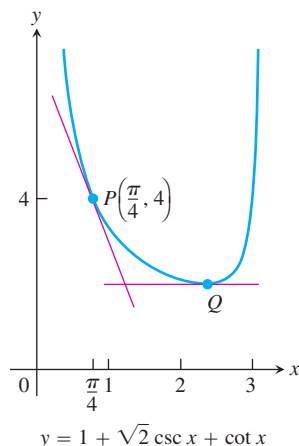
31. $y = x + \sin x$
32. $y = 2x + \sin x$
33. $y = x - \cot x$
34. $y = x + 2 \cos x$
35. Find all points on the curve $y = \tan x, -\pi/2 < x < \pi/2$, where the tangent line is parallel to the line $y = 2x$. Sketch the curve and tangent(s) together, labeling each with its equation.
36. Find all points on the curve $y = \cot x, 0 < x < \pi$, where the tangent line is parallel to the line $y = -x$. Sketch the curve and tangent(s) together, labeling each with its equation.

In Exercises 37 and 38, find an equation for (a) the tangent to the curve at P and (b) the horizontal tangent to the curve at Q .

37.



38.



Trigonometric Limits

Find the limits in Exercises 39–44.

39. $\lim_{x \rightarrow 2} \sin\left(\frac{1}{x} - \frac{1}{2}\right)$

40. $\lim_{x \rightarrow -\pi/6} \sqrt{1 + \cos(\pi \csc x)}$

41. $\lim_{x \rightarrow 0} \sec\left[\cos x + \pi \tan\left(\frac{\pi}{4 \sec x}\right) - 1\right]$

42. $\lim_{x \rightarrow 0} \sin\left(\frac{\pi + \tan x}{\tan x - 2 \sec x}\right)$

43. $\lim_{t \rightarrow 0} \tan\left(1 - \frac{\sin t}{t}\right)$

44. $\lim_{\theta \rightarrow 0} \cos\left(\frac{\pi \theta}{\sin \theta}\right)$

Simple Harmonic Motion

The equations in Exercises 45 and 46 give the position $s = f(t)$ of a body moving on a coordinate line (s in meters, t in seconds). Find the body's velocity, speed, acceleration, and jerk at time $t = \pi/4$ sec.

45. $s = 2 - 2 \sin t$

46. $s = \sin t + \cos t$

Theory and Examples

47. Is there a value of c that will make

$$f(x) = \begin{cases} \frac{\sin^2 3x}{x^2}, & x \neq 0 \\ c, & x = 0 \end{cases}$$

continuous at $x = 0$? Give reasons for your answer.

48. Is there a value of b that will make

$$g(x) = \begin{cases} x + b, & x < 0 \\ \cos x, & x \geq 0 \end{cases}$$

continuous at $x = 0$? Differentiable at $x = 0$? Give reasons for your answers.

49. Find $d^{999}/dx^{999}(\cos x)$.

50. Derive the formula for the derivative with respect to x of

a. $\sec x$. b. $\csc x$. c. $\cot x$.

T 51. Graph $y = \cos x$ for $-\pi \leq x \leq 2\pi$. On the same screen, graph

$$y = \frac{\sin(x+h) - \sin x}{h}$$

for $h = 1, 0.5, 0.3$, and 0.1 . Then, in a new window, try $h = -1, -0.5$, and -0.3 . What happens as $h \rightarrow 0^+$? As $h \rightarrow 0^-$? What phenomenon is being illustrated here?

T 52. Graph $y = -\sin x$ for $-\pi \leq x \leq 2\pi$. On the same screen, graph

$$y = \frac{\cos(x+h) - \cos x}{h}$$

for $h = 1, 0.5, 0.3$, and 0.1 . Then, in a new window, try $h = -1, -0.5$, and -0.3 . What happens as $h \rightarrow 0^+$? As $h \rightarrow 0^-$? What phenomenon is being illustrated here?

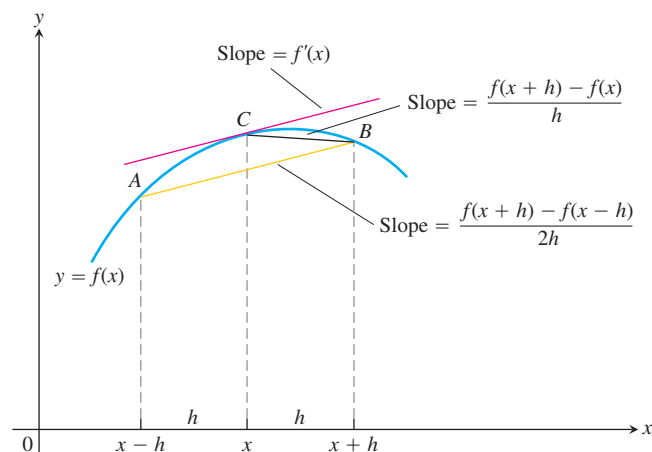
T 53. **Centered difference quotients** The *centered difference quotient*

$$\frac{f(x+h) - f(x-h)}{2h}$$

is used to approximate $f'(x)$ in numerical work because (1) its limit as $h \rightarrow 0$ equals $f'(x)$ when $f'(x)$ exists, and (2) it usually gives a better approximation of $f'(x)$ for a given value of h than Fermat's difference quotient

$$\frac{f(x+h) - f(x)}{h}.$$

See the accompanying figure.



- a. To see how rapidly the centered difference quotient for $f(x) = \sin x$ converges to $f'(x) = \cos x$, graph $y = \cos x$ together with

$$y = \frac{\sin(x+h) - \sin(x-h)}{2h}$$

over the interval $[-\pi, 2\pi]$ for $h = 1, 0.5$, and 0.3 . Compare the results with those obtained in Exercise 51 for the same values of h .

- b. To see how rapidly the centered difference quotient for $f(x) = \cos x$ converges to $f'(x) = -\sin x$, graph $y = -\sin x$ together with

$$y = \frac{\cos(x+h) - \cos(x-h)}{2h}$$

over the interval $[-\pi, 2\pi]$ for $h = 1, 0.5$, and 0.3 . Compare the results with those obtained in Exercise 52 for the same values of h .

- 54. A caution about centered difference quotients** (Continuation of Exercise 53.) The quotient

$$\frac{f(x+h) - f(x-h)}{2h}$$

may have a limit as $h \rightarrow 0$ when f has no derivative at x . As a case in point, take $f(x) = |x|$ and calculate

$$\lim_{h \rightarrow 0} \frac{|0+h| - |0-h|}{2h}.$$

As you will see, the limit exists even though $f(x) = |x|$ has no derivative at $x = 0$. *Moral:* Before using a centered difference quotient, be sure the derivative exists.

- T 55. Slopes on the graph of the tangent function** Graph $y = \tan x$ and its derivative together on $(-\pi/2, \pi/2)$. Does the graph of the tangent function appear to have a smallest slope? a largest slope? Is the slope ever negative? Give reasons for your answers.

- T 56. Slopes on the graph of the cotangent function** Graph $y = \cot x$ and its derivative together for $0 < x < \pi$. Does the graph of the cotangent function appear to have a smallest slope? A largest slope? Is the slope ever positive? Give reasons for your answers.

- T 57. Exploring $(\sin kx)/x$** Graph $y = (\sin x)/x$, $y = (\sin 2x)/x$, and $y = (\sin 4x)/x$ together over the interval $-2 \leq x \leq 2$. Where does each graph appear to cross the y -axis? Do the graphs really intersect the axis? What would you expect the graphs of $y = (\sin 5x)/x$ and $y = (\sin(-3x))/x$ to do as $x \rightarrow 0$? Why? What about the graph of $y = (\sin kx)/x$ for other values of k ? Give reasons for your answers.

- T 58. Radians versus degrees: degree mode derivatives** What happens to the derivatives of $\sin x$ and $\cos x$ if x is measured in degrees instead of radians? To find out, take the following steps.

- a. With your graphing calculator or computer grapher in *degree mode*, graph

$$f(h) = \frac{\sin h}{h}$$

and estimate $\lim_{h \rightarrow 0} f(h)$. Compare your estimate with $\pi/180$. Is there any reason to believe the limit *should* be $\pi/180$?

- b. With your grapher still in degree mode, estimate

$$\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}.$$

- c. Now go back to the derivation of the formula for the derivative of $\sin x$ in the text and carry out the steps of the derivation using degree-mode limits. What formula do you obtain for the derivative?
- d. Work through the derivation of the formula for the derivative of $\cos x$ using degree-mode limits. What formula do you obtain for the derivative?
- e. The disadvantages of the degree-mode formulas become apparent as you start taking derivatives of higher order. Try it. What are the second and third degree-mode derivatives of $\sin x$ and $\cos x$?

3.5

The Chain Rule and Parametric Equations

We know how to differentiate $y = f(u) = \sin u$ and $u = g(x) = x^2 - 4$, but how do we differentiate a composite like $F(x) = f(g(x)) = \sin(x^2 - 4)$? The differentiation formulas we have studied so far do not tell us how to calculate $F'(x)$. So how do we find the derivative of $F = f \circ g$? The answer is, with the Chain Rule, which says that the derivative of the composite of two differentiable functions is the product of their derivatives evaluated at appropriate points. The Chain Rule is one of the most important and widely used rules of differentiation. This section describes the rule and how to use it. We then apply the rule to describe curves in the plane and their tangent lines in another way.